

Метод матем. индукции.

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$n=1 \quad 1 = \frac{(1+1) \cdot 1}{2} \quad \checkmark$$

$$n=2 \quad 3 = \frac{2 \cdot 3}{2} \quad \checkmark$$

$$n=k \quad \frac{k(k+1)}{2} = 1 + 2 + \dots + k$$

$$n=k+1 \quad 1 + 2 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$n=2 \quad 5 = \frac{2 \cdot 3 \cdot 5}{6} = 5$$

$$n=k \text{ - верно } \Rightarrow n=k+1 \quad 1^2 + 2^2 + \dots + k^2 + (k+1)^2 =$$

$$= \frac{k(k+1)(2k+1)}{6} + \frac{(k+1)^2}{6} = \frac{(k+1)}{6} [6k + 6 + k(2k+1)] =$$

$$= \frac{k+1}{6} (6k + 6 + 2k^2 + k) = \frac{(k+1)(k+2)(2k+3)}{6}$$

$$(1+x_1)(1+x_2) \dots (1+x_n) \geq 1 + x_1 + \dots + x_n$$

$$n=2 \quad \begin{array}{c} (1+x_1)(1+y_1) \\ \parallel \\ 1 + x + y + xy \end{array} \geq 1 + x + y$$

$$x_1, \dots, x_n \geq 0$$

$$n=k+1 \quad (1-x_1) \dots (1+x_k)(1+x_{k+1}) \geq$$

$$2^{2^n} + 1 \text{ делится на } 7 \Rightarrow 2^{2^n} + 4 \div 10$$

$$n=2 \quad 16+4=20 \div 10$$

$$n=3 \quad 256+4=260 \div 10$$

$$n=k \quad 2^{2^k} + 4 = 10m \div 10$$

$$n=k+1 \quad 2^{2^{k+1}} + 4 = 2^{2^k \cdot 2} + 4 = (2^{2^k})^2 + 16 + 8 \cdot 2^{2^k} - 8 \cdot 2^{2^k} - 16 + 4 = (2^{2^k} + 4) - 8(2^{2^k} + 4) + 32 - 12 = 100m^2 - 80m + 20 \div 10$$

$$1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(4n^2-1)}{3}$$

$$n=1 \quad 1 = \frac{1 \cdot 3}{3} \checkmark$$

$$k(2k-1)(2k+1)$$

$$n=2 \quad 1+9 = \frac{2 \cdot 15}{3} \checkmark$$

$$n=k \quad 1^2 + 3^2 + \dots + k^2 = \frac{k(4k^2-1)}{3}$$

$$n=k+1 \quad \frac{1^2 + 3^2 + \dots + k^2 + (k+1)^2}{1^2 + 3^2 + \dots + (2k-1)^2 + (2k+1)^2} = \frac{\frac{k(4k^2-1)}{3} + (k+1)^2}{\frac{(k+1)(4(k+1)^2-1)}{3}}$$

$$= \frac{k(4k^2-1)}{3} + (2k+1)^2 = \frac{k(2k-1)(2k+1)}{3} + 3(2k+1)^2$$

$$= (2k+1)(6k+3+8k^2-k) = (2k+1)(4k^2+5k+3)$$

$$= \frac{(2k+1)(4k^2+5k+3)}{4k^3+4k^2+8k^2+8k+3k+3} = \frac{(2k+1)(4k^2+5k+3)}{4k^3+8k^2+3k+3}$$

$$= \frac{(k+1)(4(k+1)^2-1)}{3} \checkmark$$

$$\bullet 1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$$

$$n=1 \quad 1=1 \quad \checkmark \quad n=3 \quad 1+8+27=36 \quad \checkmark$$

$$n=k \quad 1^3 + 2^3 + \dots + k^3 = (1 + 2 + \dots + k)^2$$

$$n=k+1 \quad 1^3 + 2^3 + \dots + k^3 + (k+1)^3 = (1 + 2 + \dots + k)^2 + (k+1)^3$$

$$\frac{k^2(1+k)^2}{4} + \frac{4(k+1)^3}{4} = \frac{(k+1)^2(4k+4+k^2)}{4} =$$

$$= \frac{(k+1)^2(k+2)^2}{4} = \left(\frac{(k+1)(k+1)+1}{2} \right)^2 = (1 + 2 + \dots + (k+1))^2$$

$$\bullet 1 \cdot 2 + 2 \cdot 5 + \dots + n(3n-1) = n^2(n+1)$$

$$n=1 \quad 2=2 \quad \checkmark \quad n=2 \quad 12=4 \cdot 3 \quad \checkmark$$

$$n=k \quad 1 \cdot 2 + 2 \cdot 5 + \dots + k(3k-1) = k^2(k+1)$$

$$n=k+1 \quad 1 \cdot 2 + 2 \cdot 5 + \dots + k(3k-1) + (k+1)(3(k+1)-1) =$$

$$= k^2(k+1) + (k+1)(3(k+1)-1) =$$

$$= k^2(k+1) + (k+1)(3k+2) = (k+1)(k^2+3k+2) =$$

$$= (k+1)^2(k+1+1)$$

$$\bullet \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

$$n=2 \quad \frac{1}{3} + \frac{1}{15} = \frac{2}{5}$$

$$n=k \quad \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}$$

$$n=k+1 \quad \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} =$$

$$= \frac{k}{2k+1} + \frac{1}{(2(k+1)+1)(2(k+1)-1)} = \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} =$$

$$= \frac{k(2k+3)+1}{(2k+1)(2k+3)} = \frac{2k^2+3k+1}{4k^2+8k+3} = \frac{(k+1)(2k+1)}{(2k+1)(2k+3)} =$$

$$= \frac{k+1}{2k+1+1}$$

$$\bullet \quad 5 \cdot 2^{3n-2} + 3^{3n-1} \div 19$$

$$n=1 \quad 5 \cdot 2 + 9 = 19$$

$$n=k \quad 5 \cdot 2^{3k-2} + 3^{3k-1} = m \div 19$$

$$n=k+1 \quad 5 \cdot 2^{3(k+1)-2} + 3^{3(k+1)-1} = 5 \cdot 2^{3k+1} + 3^{3k+2} = 5 \cdot 8 \cdot 2^{3k-2} +$$

$$+ 27 \cdot 3^{3k-1} = 8(5 \cdot 2^{3k-2} + 3^{3k-1}) + 19 \cdot 3^{3k-1} =$$

$$= 8m + 19 \cdot 3^{3k-1} \div 19$$

$$\bullet \quad n(2n^2-3n+1) \div 6$$

$$n=2 \quad 2(8-6+1) = 6 \div 6$$

$$n=k \quad k(2k^2-3k+1) \div 6$$

$$n=k+1 \quad (k+1)(2(k+1)^2-3(k+1)+1) = (k+1)(2k^2+4k+2-3k-3+1) =$$

$$= (k+1)(2k^2+k) = 2k^3+2k^2+k = k(2k^2+3k+1) =$$

$$= \underbrace{k(2k-3k+1)}_{n=k} + \underbrace{6k^2}_{\div 6}$$

$$\bullet \quad n^3 - n \div 7$$

$$n^7 - n : 7$$

$$n=2 \quad 2^7 - 2 = 126 : 7$$

$$n=k \quad k^7 - k : 7$$

$$\begin{aligned} n=k+1 \quad (k+1)^7 - k &= k^7 + 7k^6 + 21k^5 + 35k^4 + 35k^3 + 7k^2 + k + 1 - k - 1 \\ &= 7(k^6 + 3k^5 + 5k^4 + 5k^3 + 3k^2 + k) + (k^7 - k) : 7 \end{aligned}$$

$$\bullet \quad \arctan \frac{1}{2} + \arctan \frac{1}{8} + \dots + \arctan \frac{1}{2n^2} = \arctan \frac{n}{n+1}$$

$$n=2 \quad \arctan \frac{1}{8} + \arctan \frac{1}{2} = \arctan \frac{2}{3}$$

$$\arctan x + \arctan y = \arctan \frac{x+y}{1-xy}$$

$$n=k \quad \arctan \frac{1}{2} + \arctan \frac{1}{8} + \dots + \arctan \frac{1}{2k^2} = \arctan \frac{k}{k+1}$$

$$n=k+1 \quad \arctan \frac{k}{k+1} + \arctan \frac{1}{2(k+1)^2} =$$

$$= \arctan \left(\frac{\frac{k}{k+1} + \frac{1}{2(k+1)^2}}{1 - \frac{k}{2(k+1)^3}} \right) = \arctan \frac{(k+1)(2k^2 + 2k + 1)}{2(k+1)^3 - k}$$

$$= \arctan \left(\frac{2k^2(k+1) + 2k(k+1) + (k+1)}{2k^2(k+2) + 2k(k+2) + (k+2)} \right) =$$

$$= \arctan \frac{k+1}{k+2}$$

$$\bullet x_1, \dots, x_n > 0 \quad x_1 + x_2 + \dots + x_n \leq \frac{1}{2}$$

$$(1-x_1)(1-x_2)\dots(1-x_n) \geq \frac{1}{2}$$

$$n=2 \quad x_1, x_2 > 0 \quad x_1 + x_2 \leq \frac{1}{2}$$

$$(1-x_1)(1-x_2) = 1 - x_1 - x_2 + x_1x_2 = \underbrace{1 - (x_1 + x_2)}_{\geq \frac{1}{2}} + \underbrace{x_1x_2}_{\geq 0} \geq \frac{1}{2}$$

$$n=k \quad (1-x_1)\dots(1-x_k) \geq \frac{1}{2} \text{, wenn } x_1 + \dots + x_k \leq \frac{1}{2}$$

$$n=k+1 \quad (1-x_1)\dots(1-x_{k+1}) \geq \frac{1}{2} \Rightarrow (1-x_1)\dots(1-x_k) \geq \frac{1}{2}$$

$$(1-x_1)\dots(1-x_{k+1}) = (1-x_1)\dots(1-x_k) \underbrace{(1-x_{k+1})}_{\geq \frac{1}{2}} \geq \frac{1}{2}$$

$$(1-x_1)\dots(1-x_k) \geq \frac{1}{2}$$

$$\bullet x_1, \dots, x_n = 1, \text{ for } x_i > 0 \quad \sum_{i=1}^n x_i \geq 1$$

$$n=2 \quad x_1 \cdot x_2 = 1, \quad x_1 = \frac{1}{x_2} \quad x_1 + x_2 = x_2 + \frac{1}{x_2} = \frac{x_2^2 + 1}{x_2} \geq 2$$

$$x_2^2 + 1 \geq 2x_2 \quad (x_2 - 1)^2 \geq 0 \quad - \text{beweisen}$$

$$n=k \quad x_1 \cdot \dots \cdot x_k = 1 \quad \sum_{i=1}^k x_i \geq k$$

$$n=k+1 \quad x_1 \cdot \dots \cdot x_k \cdot x_{k+1} = 1 \quad x_1 + x_2 + \dots + x_k + x_{k+1} \geq k+1$$

$$x_i \cdot x_j = \tilde{x}_i \Rightarrow x_1 \cdot \dots \cdot \tilde{x}_i \cdot x_j \cdot \dots \cdot x_{k+1} = x_1 \cdot \dots \cdot x_{k+1} = 1$$

$$x_1 + x_2 + \dots + \tilde{x}_i + x_j + \dots + x_{k+1} \geq k+1$$

$$x_1 + x_2 + \dots + x_k$$

$$x_i \cdot x_j \leq x_i + x_j - 1$$

$$x_1 + x_2 + \dots + x_{k+1} - 1 \geq x_1 + x_2 + \dots + x_i \cdot x_j + \dots + x_{k+1} \geq k$$

$$x_i \cdot x_j \leq x_i + x_j - 1 \quad (x_i - 1)(x_j - 1) \leq 0$$

$$\cdot \frac{1}{3} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2n-1}{2n} < \frac{1}{\sqrt{3n+1}} \quad n > 1$$

$$n=2 \quad \frac{3}{8} < \frac{1}{\sqrt{4}} \quad \frac{3}{64} < \frac{9}{63} \quad \checkmark$$

$$n=k \quad \frac{2k-1}{2k} < \frac{1}{\sqrt{3k+1}} \quad \checkmark$$

$$n=k+1 \quad \frac{2k-1}{2k} \cdot \frac{2(k+1)-1}{2(k+1)} < \frac{1}{\sqrt{3k+1}}$$

$$\frac{2k-1}{2k} \cdot \frac{2k+1}{2(k+1)} = \frac{2(k+1)(k-1)}{2k(k+1)} = \frac{(2k-1)(2k+1)}{4k(k+1)} < \frac{1}{\sqrt{3k+1}}$$

$$\frac{4k^2-1}{4k(k+1)} < \frac{1}{\sqrt{3k+1}} \quad \left(\frac{4k^2-1}{16k^2(k+1)^2} \right) < \frac{1}{3k+1}$$

$$(4k^2-1)(3k+1) < 16k^2(1+k)^2 \quad k^2+2k+1$$

$$12k^3+4k^2-3k-1 < 16k^4+32k^3+16k^2$$

$$-16k^4-30k^3-12k^2-3k-1 < 0$$

$$\bullet (2n)! > \frac{4n}{n+1} (n!)^2 \quad n > 1$$

$$n=2 \quad 24 > \frac{4}{3} \cdot 4$$

$$n=3 \quad 6! > \frac{12}{4} (3!)^2 \quad 2 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 > 3(3 \cdot 3 \cdot 2 \cdot 2)$$

$$n=k \quad (2k)! > \frac{4k}{k+1} (k!)^2$$

$$n=k+1 \quad (2k+2)! > \frac{4(k+1)(k+1)!}{k+2}$$

$$(2n)! = n! (n+1) \cdot \dots \cdot (2n)$$

$$(n+1) \cdot \dots \cdot (2n) > \frac{4n}{n+1} n! = \frac{4n}{n+1}$$

$$\bullet \quad n > 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2^{n-1}} > \frac{n}{2} \quad n > 1$$

$$n=2 \quad 2 > 1 + \frac{1}{2} > 1$$

$$n=k \quad k > 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2^{k-1}} > \frac{k}{2}$$

$$n=k+1 \quad k+1 > \underbrace{1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2^{k-1}}}_{> \frac{k}{2}} + \frac{1}{2^k} + \frac{1}{2^{k+1}} + \dots + \frac{1}{2^{k+1}} > \frac{k+1}{2}$$

$$1 + \dots + \frac{1}{2^{k-1}} + \dots + \frac{1}{2^{k+1}} > \frac{k}{2} + \frac{1}{2^k} + \dots + \frac{1}{2^{k+1}} > \frac{k+1}{2}$$

$$\frac{1}{2^k} + \frac{1}{2^{k+1}} + \dots + \frac{1}{2^{k+1}} > \frac{1}{2} \quad \xrightarrow{\text{2 k-nya keses. uz kom}} \frac{1}{2^{k+1} - 1}$$

$$\frac{1}{2^k} + \frac{1}{2^{k+1}} + \dots + \frac{1}{2^{k+n-1}} > 2^4 \cdot \frac{1}{2^{k+1}} = \frac{2^4}{2 \cdot 2^k} > \frac{2^4}{2 \cdot 2^k} = \frac{1}{2}$$

$$\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} > \frac{13}{24}, n > 1$$

$$n=2 \quad \frac{1}{3} + \frac{1}{4} = \frac{7}{12} > \frac{13}{24} \quad \checkmark$$

$$n=3 \quad \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{22}{60} > \frac{13}{24} \quad \checkmark$$

$$n=4 \quad \frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{2k} > \frac{13}{24} \quad \checkmark$$

$$n=k+1 \quad \frac{1}{(k+1)+1} + \frac{1}{(k+1)+2} + \dots + \frac{1}{2(k+1)} > \frac{13}{24}$$

$$\Rightarrow \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+2} \Rightarrow$$

$$= \frac{1}{k+1} + \underbrace{\frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k}}_E + \frac{1}{2k+1} + \frac{1}{2k+2} \Rightarrow$$

$$\Rightarrow \frac{1}{k+1} + E + \frac{1}{2k+1} + \frac{1}{2k+2} = E + \frac{1}{2(k+1)(2k+1)} > \frac{13}{24}$$

$$\frac{1}{n+1} + \dots + \frac{1}{3n+1} > 1 \quad n > 1$$

$$n=2 \quad \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{7} > 1$$

$$n=k \quad \frac{1}{k+1} + \dots + \frac{1}{3k+1} > 1$$

$$n=k+1 \quad \frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{3k+1} + \frac{1}{3k+2} + \frac{1}{3k+3} + \frac{1}{3k+4} > 1$$

$$\bullet \quad 2\sqrt{n} > 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

$$n=2 \quad 2\sqrt{2} > \frac{\sqrt{24}}{\sqrt{2}} > \sqrt{2} \quad \checkmark$$

$$h=3 \quad 2\sqrt{3} > 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} > \sqrt{3} \quad \checkmark$$

$$n=k \quad 2\sqrt{k} > 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} > \sqrt{4} \quad \checkmark$$

$$h=k+1 \quad \sqrt{k+1} > 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k+1}} > \sqrt{k+1}$$

$$2\sqrt{k+1} > 2\sqrt{k} + \frac{1}{\sqrt{k+1}} > 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} > \sqrt{k} + \frac{1}{\sqrt{k+1}} > \sqrt{k+1}$$

$$\bullet \quad 2!4! \dots (2n)! > [(n+1)!]^n$$

$$n=2 \quad 2! \cdot 4! > (3!)^2 \quad \checkmark$$

$$n=k \quad 2! \cdot 4! \cdot \dots \cdot (2k)! > ((k+1)!)^4$$

$$n=k+1 \quad 2! \cdot 4! \cdots (2k)!(2k+2)! > [k+1!]^k (k+2)! > (k+2!)^{k+1}$$

$$(k+1)!^4 (2k+2)! > (k+1)!^4 (k+2)(k+2)!$$

$$\frac{(k+2)! (k+3)(k+4) \dots (k+k) (2k+1)(2k+2)}{(k+3) \dots (2k+2)} \geq (k+2)! (k+2)^k$$

$(k+3), (2k+2)$

$$\begin{array}{l} (k+3) \dots (2k+2) \\ (k+2) \dots (2k+1) \\ k+1 \dots (2k) \end{array}$$

$$\bullet \quad n^{n+1} > (n+1)^n \quad n \geq 3$$

$$n=3 \quad 3^4 > 4^3 \quad 81 > 64$$

$$n=k \quad k^{k+1} > (k+1)^k$$

$$n=k+1 \quad (k+1)^{k+2} > (k+2)^{k+1} \quad |k+1| |k+1|^{k+1} > |k+2|^k |k+2|$$

$$\frac{(k+1)^{k+2}}{(k+2)^{k+1}} > 1 \Rightarrow \frac{(k+1)^k}{k^{k+1}} < 1$$

$$\frac{(k+1)^{k+2}}{(k+2)^{k+1}} > \frac{(k+1)^{k+2}}{(k+2)^{k+1}} \cdot \frac{\cancel{k^{k+1}}}{\cancel{k^{k+1}}} = \frac{(k+1)^k}{k^{k+1}} \cdot \frac{(k+1)^{2k+2}}{(k^2+2k)^{k+1}}$$

$$= \frac{(k^2+2k+1)^{k+1}}{(k^2+2k)^{k+1}} = \left(\frac{k^2+2k+1}{k^2+2k} \right)^{k+1} > 1$$

$$\bullet \quad 1+2+2^2+\dots+2^{n-1} = 2^n - 1$$

$$n=1 \quad 1 = 1$$

$$n=2 \quad 1+2 = 3$$

$$n=3 \quad 1+2+4 = 7$$

$$n=k \quad 1+2+\dots+2^{k-1} = 2^k - 1$$

$$n=k+1 \quad 1+2+\dots+2^{k-1} + 2^k = 2 \cdot 2^k - 1 \Rightarrow$$

$$\begin{aligned} 2 \cdot 2^k - 1 + 2^k &= \\ &= 2^k \cdot 2 - 1 = 2^{k+1} - 1 \end{aligned}$$

11 1. M. №1-14, №27, 28, 30.

12. c $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap B = A \bar{B}$

$$A \cap (B \cup C) = A \bar{B} \bar{C} = A \bar{B} \cdot A \bar{C} = A \bar{B} (A \cap B) \cap (A \cap C)$$

• $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$A \cap (B \cap C) =$$

№1 a) $A \subseteq A_2 \Rightarrow \forall b \in A_1 ; \forall c \in A_2 \Rightarrow b \in A_2 \Rightarrow$

$$\Rightarrow A_1 = A_2 = A \quad \underline{\forall x \in A}$$

б) если $A \subseteq B$ и $B \subseteq C$, то $A \subseteq C$

$$C = \{x \mid x \in B\}, B = \{y \mid y \in A\} \Rightarrow A \subseteq C$$

в) $A \cap B \subseteq A \subseteq A \cup B$

$$A = \{x \mid x \in A \cap B\} \Rightarrow x \in A \cup B$$

г) $A \cap B \subseteq B \subseteq A \cup B$

$$A \cap B = \{x \mid x \in A, \underline{x \in B}\}$$

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

д) $A \cap B \subseteq A$

$$A \cap B = \{x \mid x \in A, \text{ и } x \in B\}$$

№2 $x^2 - 7x + 6 = 0$

$$B = \{1, 6\}$$

$$x_1 = 1, x_2 = 6$$

$$A \in \{x_1, x_2\}$$

$$\Rightarrow A = B$$

$$\bullet 2^n n! < (n+1)^n, n > 1$$

$$n=2 \quad 8 < 9$$

$$n=3 \quad 48 < 64$$

$$n=k \quad 2^k \cdot k! < (k+1)^k$$

$$n=k+1 \quad 2^{k+1} \cdot k! (k+1) < (k+2)^{k+1} = 2(k+1)^k (k+2)^{k+1}$$

$$2 \cdot 2^k \cdot k! (k+1)$$

$$2(k+1)^{k+1} < (k+2)^{k+1}$$

$$\frac{2(k+1)^{k+1} \cdot (k+2)^{k+1}}{(k+2)^{k+1}} = \frac{2(k+2)^{k+1}}{\left(\frac{k+2}{k+1}\right)^{k+1}} =$$

$$= \frac{2(k+2)^{k+1}}{\left(1 + \frac{1}{k+1}\right)^{k+1}} >> 2(k+2)^{k+1}$$

$$\left(1 + \frac{1}{k+1}\right)^{k+1} >> 2$$

$$\left(1 + \frac{1}{k+1}\right)^{k+1} > 2 \quad \text{no sumang}$$

$$\sqrt{3} \quad \emptyset \neq \{\emptyset\}$$

$$|\emptyset| = 0 \quad |\{\emptyset\}| = 1$$

$$\sqrt{4} \quad \underbrace{\{\{1, 2\}, \{2, 3\}\}}_A \neq \underbrace{\{1, 2, 3\}}_B$$

$$|A| = 2 \neq |B| = 3$$

$$\sqrt{5} \quad a) \quad \emptyset \subseteq A \subseteq U$$

$$A \subseteq$$

$$b) \text{ even } A \subseteq \emptyset \Rightarrow A = \emptyset; U \subseteq A \Rightarrow A = U$$

$$b) A \cup \emptyset = A, A \cap \emptyset = \emptyset, A \cup U = U, A \cap U = A$$

$$A = \{x \mid x \in A\}$$

$$\Rightarrow A \cup \emptyset = A$$

$$|\emptyset| = 0$$

$$A \cap \emptyset = \emptyset$$

$$U = \{x \mid \forall x \quad x \cup U = U$$

$$\sqrt{6} \quad \emptyset$$

$$\sqrt{7} \quad A \cap B = \emptyset, A \cap C = \emptyset, (A \cap B) \cap C = \emptyset$$

$$x \in A \cap B, x \notin C, x \in (A \cap B) \cap C$$

Rem

$$\sqrt{8} \quad \alpha(x) = \beta(x) \cdot \gamma(x)$$

$\chi(x)$ - un-ko всех корней

$$x \in X(B) \Leftrightarrow x(x) = 0 \Leftrightarrow B(x) = 0 \cdot x(x) = 0, \text{ and}$$

$$x \notin B(x) = 0, x(x) = 0 \Leftrightarrow x \in X(B) \vee X(x)$$

$$X(x) = X(B) \cup X(x)$$

now

$$A \subseteq B \Leftrightarrow A \cup B = B \Leftrightarrow A \cap B = A \Leftrightarrow A \cap B = \emptyset \Leftrightarrow (A \cup B) = U$$

$$A \cup B = B \Leftrightarrow \cancel{A \supset B} \Rightarrow \cancel{A \subseteq B} \Rightarrow \cancel{A \cap B = \emptyset} \Rightarrow$$

now

$$a) A \cup A = A \cap A = A$$

$$x \in A \quad x \in A \cup A, x \in A \cap A, x \in A$$

$$b) A \cap B = B \cap A$$

$$A \cap B = \{x : x \in A \wedge x \in B\}$$

$$B \cap A = \{x : x \in B \wedge x \in A\}$$

$$c) A \cup B = B \cup A$$

$$A \cup B = \{x : x \in A \vee x \in B\}$$

$$B \cup A = \{x : x \in B \vee x \in A\}$$

$$d) A \cap (B \cap C) = (A \cap B) \cap C$$

$$\text{prob. 11} \quad \begin{cases} x \in A \\ x \in B \cap C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in B \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow \begin{cases} x \in A \cap B \\ x \in C \end{cases} \Rightarrow x \in (A \cap B) \cap C$$

$$\text{pr.} \quad \begin{cases} x \in A \cap B \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in B \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in A \cap C \\ x \in B \end{cases} \Rightarrow x \in (A \cap C) \cap B$$

$$g) A \cup (B \cap C) = (A \cup B) \cap C$$

$$1.7' \begin{cases} x \in A \\ \{x \in B\} \\ \{x \in C\} \end{cases} \Rightarrow x \in A$$

$$e) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\begin{cases} x \in A \\ \{x \in (B \cup C)\} \end{cases} \Rightarrow \begin{cases} x \in A \\ \{x \in B\} \\ \{x \in C\} \end{cases} \Rightarrow \begin{cases} x \in A \\ \{x \in B\} \\ \{x \in A\} \\ \{x \in C\} \end{cases} \Rightarrow (A \cap B) \cup (A \cap C)$$

$$a) A \cup B \cap C = (A \cup B) \cap (A \cup C)$$

$$\begin{cases} x \in A \\ \{x \in (B \cap C)\} \end{cases} \Rightarrow \begin{cases} x \in A \\ \{x \in B\} \\ \{x \in C\} \end{cases} \Rightarrow \begin{cases} x \in A \\ \{x \in B\} \\ \{x \in A\} \\ \{x \in C\} \end{cases} \Rightarrow \begin{cases} x \in A \cup B \\ \{x \in A \cup C\} \end{cases} \Rightarrow (A \cup B) \cap (A \cup C)$$

$$3) (A \cap B) \cup (A \cap C) = (A \cap C) \cup (A \cap B) \cup (B \cap C) \cup (B \cap D)$$

$$\begin{cases} x \in (A \cap B) \\ \{x \in (A \cap C)\} \end{cases} \Rightarrow \begin{cases} x \in A \\ \{x \in B\} \\ \{x \in C\} \\ \{x \in D\} \end{cases} \Rightarrow \begin{cases} x \in A \\ \{x \in C\} \\ \{x \in A\} \\ \{x \in D\} \\ \{x \in B\} \\ \{x \in C\} \\ \{x \in B\} \\ \{x \in D\} \end{cases} \Rightarrow (A \cap C) \cup (A \cap B) \cup (B \cap C) \cup (B \cap D)$$

$$N12 a) \neg(A \cap B) = \neg A \cup \neg B$$

$$x \notin (A \cap B) \Rightarrow \begin{cases} x \notin A \\ x \notin B \end{cases} \Rightarrow \neg A \cup \neg B$$

$$A \cap B =$$

$$8) \neg(A \cup B) = (\neg A) \cap (\neg B)$$

$$A \setminus B = A \bar{B}$$

$$x \notin A \cup B \Rightarrow \begin{cases} x \notin A \\ x \notin B \end{cases} \Rightarrow \bar{A} \cdot \bar{B}$$

$$6) A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

$$\begin{aligned} \frac{8}{4} \begin{cases} x \in A \\ x \notin (B \cup C) \end{cases} &\Rightarrow \begin{cases} x \in A \\ x \notin B \\ x \notin C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \notin B \\ x \notin C \end{cases} \Rightarrow (A \setminus B) \cap (A \setminus C) \\ x \notin (B \cup C) &= \overline{B \cup C} = \bar{B} \cap \bar{C} \end{aligned}$$

$$2) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\frac{8}{4} \begin{cases} x \in A \\ x \in (B \cup C) \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in B \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in B \\ x \in C \end{cases} \Rightarrow (A \cap B) \cup (A \cap C)$$

$$9) A \setminus (A \cap B) = A \cap \bar{B}$$

$$\begin{aligned} \begin{cases} x \in A \\ x \notin (A \cap B) \end{cases} &\Rightarrow \begin{cases} x \in A \\ x \notin A \\ x \in B \end{cases} \Rightarrow \begin{cases} x \in A \\ x \notin A \\ x \in B \end{cases} \Rightarrow (A \cap \bar{A}) \cup (A \cap B) \Rightarrow \\ &A \cap \bar{B} \end{aligned}$$

$$d) A \setminus B = A \setminus (A \cap B)$$

$$A \bar{B} = A \setminus (A \cap B) = A \setminus (A \cap \bar{B}) = A \bar{B}$$

$$20) A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C) = (A \cap B) \cap \bar{C}$$

$$\begin{aligned} \begin{cases} x \in A \\ x \in (B \setminus C) \end{cases} &\Rightarrow \begin{cases} x \in A \\ x \in B \\ x \notin C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in B \\ x \notin C \end{cases} \Rightarrow (A \cap B) \cap \bar{C} \\ (A \cap B) \cap \bar{C} &= (A \cap B) \setminus (A \cap C) = A \cap B \cap \bar{C} \end{aligned}$$

$$A \cap (B \setminus C) = (A \cap B) \cap \bar{C} \quad A \setminus (B \cap C) = A \cap \bar{B} \cap \bar{C}$$

$$3) (A \setminus B) \setminus C = (A \cdot \bar{B}) \cdot \bar{C} = A\bar{B}\bar{C} \neq (A \cap \bar{B}) \cap \bar{C}$$

$$u) A \cup B = A \cup (B \setminus A) = A \cup (B \cap \bar{A}) = (A \cup B) \cap (A \cup \bar{A}) = A \cup B$$

$$k) \bar{\bar{A}} = A \quad x \in \bar{A} \Rightarrow x \notin A \Rightarrow x \in A$$

$$l) A \cup \bar{A} = U \quad \begin{cases} x \in A \\ x \notin A \end{cases} = A \cup \bar{A} = U$$

$$m) A \cap \bar{A} = \emptyset \quad \begin{cases} x \in A \\ x \notin A \end{cases} \Rightarrow \emptyset$$

$$u) (A \cdot B) \cup (A \cdot \bar{B}) = A \cup A\bar{B} \cup B \cup BA = A \cup B \cup \bar{B} = A \cup B$$

$$o) (\bar{A} \cup B) \cap A = (A \cap \bar{A}) \cup (A \cap B) = A \cap B$$

$$n) A \cap (B \setminus A) = A \cap B \cap \bar{A} = \emptyset$$

$$p) (A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$$

$$\begin{cases} x \in A \cup B \\ x \notin C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \notin C \end{cases} \cup \begin{cases} x \in B \\ x \notin C \end{cases} \Rightarrow (A \setminus C) \cup (B \setminus C)$$

$$c) A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C) \quad A \cap \bar{B} \cup C$$

$$\begin{cases} x \in A \\ x \notin B \cap C \end{cases} \Rightarrow \begin{cases} x \in A \\ x \notin B \\ x \notin C \end{cases} \cup \begin{cases} x \in A \\ x \notin C \end{cases} = (A \setminus B) \cup (A \setminus C)$$

$$m) A \setminus (B \cup C) = A \cdot \overline{(B \cup C)} = A \cdot \bar{B} \cdot \bar{C} = (A \cdot \bar{B}) \cdot (A \cdot \bar{C}) = (A \setminus B) \cap (A \setminus C)$$

$$\begin{cases} A \cap X = B \\ A \cup X = C \end{cases}$$

$$X = (C \setminus A) \cup (A \cap X) = (C \setminus A) \cup B$$



N 28

$$\begin{cases} A \cap X = B \\ X \cap A = C \end{cases}$$

$$B \subseteq A, A \cap C = \emptyset$$



$$A \cap X = B$$



$$X \cap A = C$$

$$X = (A \setminus B) \cup C$$

N 30

$$\begin{cases} A \setminus X = B \\ A \cup X = C \end{cases}$$

$$B \subseteq A \subseteq C$$



$$A \setminus X = B$$



$$A \cup X = C$$

$$X = C \setminus B$$

N 13

$$a1 \quad A \cup B \subseteq C \Leftrightarrow A \subseteq C \cup B \subseteq C$$

$$\text{Пусть } x \in A \cup B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases}$$

$$\text{если } x \in A \Rightarrow A \subseteq C \Rightarrow x \in C$$

$$\text{если } x \in B \Rightarrow B \subseteq C \Rightarrow x \in C$$

$$\Rightarrow A \cup B \subseteq C$$

$$\text{b) } A \subseteq B \cap C \Leftrightarrow A \subseteq B \text{ u } A \subseteq C$$

$$\text{Prüfung } x \in B \cap C \Rightarrow \begin{cases} x \in B \\ x \in C \end{cases}$$

$$\begin{cases} x \in B \\ x \in C \end{cases} \Rightarrow x \in A$$

$$12 \text{ a) } \overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$x \in \overline{A \cap B} \Leftrightarrow x \notin (A \cap B) \Leftrightarrow \begin{cases} x \notin A \\ x \notin B \end{cases} \Leftrightarrow \begin{cases} x \in \bar{A} \\ x \in \bar{B} \end{cases} \Leftrightarrow \text{my. 17}$$

$$x \in (\bar{A} \cup \bar{B}) \Rightarrow \begin{cases} x \in \bar{A} \\ x \in \bar{B} \end{cases} \Rightarrow$$

$$\text{b) } \overline{A \cup B} = \bar{A} \cdot \bar{B}$$

$$x \in \overline{A \cup B} \Leftrightarrow x \notin (A \cup B) \Leftrightarrow \begin{cases} x \notin A \\ x \notin B \end{cases}$$

$$x \in A \cup B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases}$$

$$x \notin A \cup B \Leftrightarrow \begin{cases} x \notin A \\ x \notin B \end{cases} \Leftrightarrow \bar{A} \cdot \bar{B}$$

$$\text{b) } A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cap \overline{B \cup C} = A \cap \bar{B} \cap \bar{C}$$

$$1) A \cap (A \cap B) = A \cap (\bar{A} \cdot \bar{B}) = A \cap (\bar{A} \cup B) = A \cap B$$

N/13

$$a) A \cup B \subseteq C \Leftrightarrow A \subseteq C \cup B \subseteq C$$

$$x \in C \quad x \in A \cup B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} - x \in C \Rightarrow \begin{cases} x \in A \Rightarrow x \in C \\ x \in B \Rightarrow x \in C \Rightarrow B \subseteq C \end{cases}$$

$$b) A \subseteq B \cap C \Leftrightarrow A \subseteq B \cup A \subseteq C$$

$$x \in A \Rightarrow x \in B \cap C \Rightarrow \begin{cases} x \in B \Rightarrow A \subseteq B \\ x \in C \Rightarrow A \subseteq C \end{cases}$$

$$b) A \cap B \subseteq C \Leftrightarrow A \subseteq \bar{B} \cup A \subseteq C \quad A \subseteq \bar{B} \cup C$$

$$x \in A \cap B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow x \in C$$

$$\text{Прямое } x \in A \Rightarrow \begin{cases} x \in B \cap A \\ x \in A \cap \bar{B} \end{cases} \Rightarrow \begin{cases} x \in C \Rightarrow x \in \bar{B} \cup C \\ x \in \bar{B} \end{cases}$$

$$\text{Прямое } A \subseteq \bar{B} \cup C \Rightarrow \overset{\text{лемма}}{x \in A \Rightarrow} \begin{cases} x \notin B \\ x \in C \end{cases}$$

$$x \in A \cap B \Rightarrow \begin{cases} x \in B \\ x \in A \end{cases} = \begin{cases} x \in B \\ x \notin B \end{cases} \Rightarrow \begin{cases} x \in B \Rightarrow x \in C \\ x \in C \end{cases}$$

$$2) A \subseteq B \cup C \Leftrightarrow A \cap \bar{B} \subseteq C$$

$$x \in A \Rightarrow x \in B \cup C \Rightarrow \begin{cases} x \in B \\ x \in C \end{cases}$$

$$\text{Прямое } x \in A \cap \bar{B} \Rightarrow \begin{cases} x \in A \\ x \notin B \end{cases} \Rightarrow \begin{cases} \begin{cases} x \in B \\ x \in C \end{cases} \\ \text{лемма} \\ x \notin B \end{cases} \Rightarrow \begin{cases} x \in C \\ x \notin B \end{cases} \Rightarrow x \in C$$

Pycoms $x \in A \cap B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow x \in C \Rightarrow \begin{cases} A \subseteq C \\ B \subseteq C \end{cases}$

$\Rightarrow x \in A \Rightarrow x \in B \Rightarrow \begin{cases} x \in A \cap B \\ x \in B \end{cases} \Rightarrow \begin{cases} x \in C \\ x \in B \end{cases} \Rightarrow A \subseteq B \cup C$

g) $(A \cap B) \cup B = A \Leftrightarrow B \subseteq A$

$A \cap B \cup B = (A \cup B) \cap (B \cup B) = A \cup B = A$

$B \subseteq A$

e) $(A \cap B) \cup C \subseteq A \cap (B \cup C) \Leftrightarrow C \subseteq A$

$A \cap (B \cup C) \subseteq A \cap B \cup C \Leftrightarrow A \cap B \cup A \cap C$

Pycoms $x \in C \Rightarrow x \in A \Rightarrow$

m) $A \subseteq B \Rightarrow A \cup B \subseteq B \cup C$

$\forall x \in A \Rightarrow x \in B$

Pycoms $x \in A \cup C \Rightarrow \begin{cases} x \in A \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in B \\ x \in C \end{cases} \Rightarrow x \in B \cup C$

3) $A \subseteq B \Rightarrow A \cap C \subseteq B \cap C$

$\forall x \in A \Rightarrow x \in B$

$\exists x \in A \cap C \Rightarrow \begin{cases} x \in A \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in B \\ x \in C \end{cases}$

u) $A \subseteq B \Rightarrow (A \cap C) \subseteq (B \cap C)$

$x \in A \Rightarrow x \in B$

$\exists x \in A \cap C \Rightarrow \begin{cases} x \in A \\ x \in C \end{cases} \Rightarrow \begin{cases} x \in B \\ x \in C \end{cases}$

$$\text{Ans a) } A \Delta B = B \Delta A$$

$$A \Delta B = A\bar{B} \vee \bar{A}B \quad B \Delta A = \bar{A}B \vee A\bar{B}$$

$$\text{b) } A \Delta (B \Delta C) = A \Delta (B\bar{C} \vee \bar{B}C) = A(\bar{B}\bar{C} \vee \bar{B}C \vee B\bar{C} \vee B\bar{C}) =$$

$$(\bar{A} \vee B)(\bar{A} \vee C)$$

$$= \bar{A}\bar{B}\bar{C} \vee \bar{A}\bar{B}C \vee A\bar{B}\bar{C} \vee A\bar{B}C$$

$$(A \Delta B) \Delta C = (A\bar{B} \vee \bar{A}B) \Delta C = (\bar{A}\bar{B} \vee \bar{A}B \vee A\bar{B} \vee AB) \Delta C =$$

$$= \bar{A}\bar{B}\bar{C} \vee \bar{A}\bar{B}C \vee A\bar{B}\bar{C} \vee A\bar{B}C$$

$$\text{b) } A(B \Delta C) = A(\bar{B}\bar{C} \vee \bar{B}C) = A\bar{B}\bar{C} \vee A\bar{B}C$$

$$(AB) \Delta (AC) = AB\bar{A}C \vee (AB) \Delta C = AB\bar{A} \vee C \vee A\bar{B} \vee B\bar{A}C =$$

$$= AB\bar{C} \vee A\bar{B}C$$

$$\text{c) } A \Delta (A \Delta B) = A \Delta (A\bar{B} \vee \bar{A}B) = \bar{A}(A\bar{B} \vee \bar{A}B) \vee$$

$$\vee A(\bar{A}\bar{B} \vee \bar{A}B) = \bar{A}B \vee A(\bar{A} \vee B)(\bar{A} \vee B) = \bar{A}B \vee AB = B$$

$$\text{g) } (A \Delta B) \Delta (AB) = (\bar{A}B \vee \bar{A}B) \Delta AB = AB(\bar{A}B \vee \bar{A}B) \vee$$

$$\vee \bar{A}B(\bar{A}B \vee \bar{A}B) = AB(\bar{A} \vee B)(\bar{A} \vee B) \vee (\bar{A} \vee B)(\bar{A}B \vee \bar{A}B) =$$

$$= AB \vee \bar{A}B \vee \bar{A}B = A \vee B$$

$$\text{e) } A \vee B = A\bar{B} ; A \Delta (AB) = \bar{A} \cdot AB \vee A(\bar{A}B) =$$

$$= A(\bar{A} \vee B) = A\bar{B}$$

$$\text{h) } A \Delta \emptyset = \text{Pythagoras } A\bar{A} = \emptyset \Rightarrow A \Delta A\bar{A} =$$

$$= \bar{A} \cdot A\bar{A} \vee A(A \cdot \bar{A}) = A(\bar{A} \vee A) = A$$

$$\text{i) } A \Delta A = A \cdot \bar{A} \vee \bar{A} \cdot A = \emptyset$$

$$\text{j) } A \Delta U = \text{Pythagoras } U = 1 \Rightarrow A \Delta 1 = \bar{A} \cdot 1 \vee A \cdot 0 = \bar{A}$$

$$x1) A \Delta B, \vee(A \Delta B) = (\bar{A}B \vee A\bar{B}) \vee (A\bar{B} \vee \bar{A}B) = A\bar{B} \vee B\bar{A}$$

$$N13 x1) A \subseteq B \Rightarrow (C \setminus B) \subseteq (C \setminus A)$$

$$x \in A \Rightarrow x \in B$$

$$x \in C \setminus B \Rightarrow \begin{cases} x \in C \\ x \notin B \end{cases} \Rightarrow \begin{cases} x \in C \\ x \notin A \end{cases} \Rightarrow x \in C \setminus A$$

$$1) A \subseteq B \Rightarrow \neg B \subseteq \neg A$$

$$x \in A \Rightarrow x \in B$$

$$\forall x \in B \Rightarrow x \in A$$

$$x \in A \cup B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Leftrightarrow \begin{cases} x \in A \\ x \in B \\ x \in A \cap B \\ x \in A \setminus B \\ x \in B \setminus A \end{cases} \Leftrightarrow$$

$$u) A \cup B \Rightarrow A \cap B \Rightarrow A = B \Leftrightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Leftrightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Leftrightarrow A \cap B = A \cap B \Rightarrow A = B$$

$$\text{Пример } A = B = A \Rightarrow A \cup B = A \cup A = A \Rightarrow A \cap B = A \cap A = A \Rightarrow A = B$$

$$u) A = \neg B \Leftrightarrow A \cap B = \emptyset \text{ и } A \cup B = U$$

$$\text{Пример } A = \neg B = A \Rightarrow \neg B \cap B = \emptyset \text{ и } \bar{B} \cup B = U$$

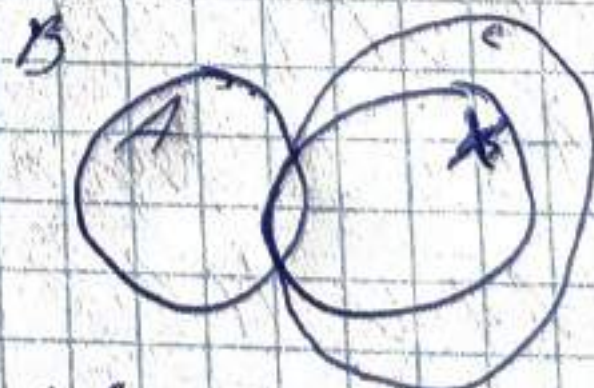
$$x \in A \Rightarrow x \in \bar{B} \quad x \in A \cap B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow \begin{cases} x \in \bar{B} \\ x \in B \end{cases} \Rightarrow \emptyset$$

$$x \in A \cup B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow \begin{cases} x \in \bar{B} \\ x \in B \end{cases} \Rightarrow U$$

$$\sqrt{14, 27, 28, 30, 32}$$

$$\sqrt{32} \quad 91 \quad \begin{cases} A \cup X = B \cap X \\ A \cap X = C \cup X \end{cases}$$

$$\begin{cases} x \in A \\ x \in X \end{cases} \Leftrightarrow \begin{cases} x \in B \\ x \in X \end{cases} \Rightarrow A \subseteq X$$





$$A = \{x \mid x \subseteq A\}$$

$$x = A$$

$$C \subseteq A$$

N6 ~~$x \in A \Rightarrow x \subseteq A$~~ $A \times B = \{(x, y) : x \in A, y \in B\}$

$$\{(x, y) : x, y \in \mathbb{R}, y = \sin x\}$$

a) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

$$x, y \in U.M. \Leftrightarrow \begin{cases} x \in A \cup B \\ y \in C \end{cases} \Leftrightarrow \begin{cases} x \in A \\ x \in B \\ y \in C \end{cases} \Leftrightarrow \begin{cases} (x \in A \wedge y \in C) \vee (x \in B \wedge y \in C) \end{cases}$$

$$\Leftrightarrow x, y \in (A \times C) \cup (B \times C)$$

b) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

$$x, y \in U.M. \Leftrightarrow \begin{cases} x \in A \\ y \in B \cup C \end{cases} \Leftrightarrow \begin{cases} x \in A \\ y \in B \\ y \in C \end{cases} \Leftrightarrow (A \times B) \cup (A \times C)$$

b) $(A \cup B) \times (C \cup D) = (A \times C) \cup (A \times D) \cup (B \times C) \cup (B \times D)$

$$x, y \in U.M. \Leftrightarrow \begin{cases} x \in A \cup B \\ y \in C \cup D \end{cases} \Leftrightarrow \begin{cases} x \in A \\ x \in B \\ y \in C \\ y \in D \end{cases} \Leftrightarrow \begin{cases} (x \in A \wedge y \in C) \vee (x \in A \wedge y \in D) \vee (x \in B \wedge y \in C) \vee (x \in B \wedge y \in D) \end{cases}$$

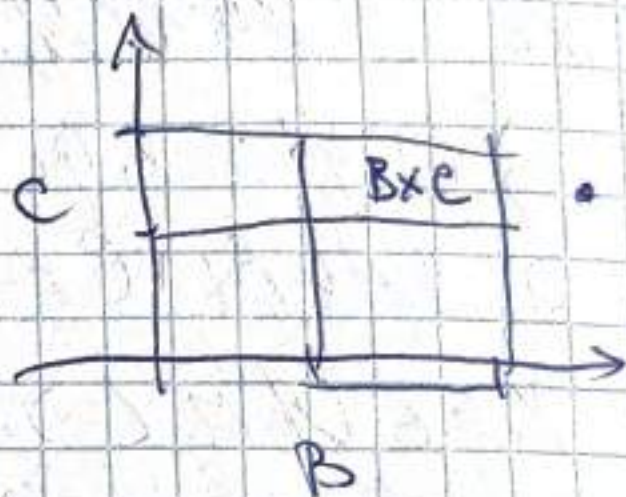
$$\Leftrightarrow x, y \in (A \times C) \cup (A \times D) \cup (B \times C) \cup (B \times D)$$

$$a) (A \setminus B) \times C = (A \times C) \setminus (B \times C)$$

$$x, y \in \mathcal{U} \text{ s.t. } (x, y) \in (A \setminus B) \times C \Leftrightarrow \begin{cases} x \in A \setminus B \\ y \in C \end{cases} \Leftrightarrow \begin{cases} \begin{cases} x \in A \\ x \notin B \end{cases} \\ y \in C \end{cases} \Leftrightarrow \begin{cases} \begin{cases} x \in A \\ y \in C \end{cases} \\ \begin{cases} x \notin B \\ y \in C \end{cases} \end{cases} \Leftrightarrow (A \times C) \setminus (B \times C)$$

$$\Leftrightarrow (A \times C) \setminus (B \times C)$$

$$\Leftrightarrow (A \times C) \setminus (B \times C)$$



$$g) A \times (B \setminus C) = (A \times B) \setminus (A \times C)$$

$$x, y \in \mathcal{U} \text{ s.t. } (x, y) \in A \times (B \setminus C) \Leftrightarrow \begin{cases} x \in A \\ y \in B \setminus C \end{cases} \Leftrightarrow \begin{cases} x \in A \\ \begin{cases} y \in B \\ y \notin C \end{cases} \end{cases} \Leftrightarrow \begin{cases} \begin{cases} x \in A \\ y \in B \end{cases} \\ \begin{cases} x \in A \\ y \notin C \end{cases} \end{cases} \Leftrightarrow (A \times B) \setminus (A \times C)$$

$$e) A \times B = (A \times D) \cap (C \times B), \text{ s.t. } A \subseteq C \text{ and } B \subseteq D$$

$$\begin{cases} x \in A \Rightarrow x \in C \\ y \in B \Rightarrow y \in D \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in C \\ y \in B \\ y \in D \end{cases} \Rightarrow (A \times B) \cap (A \times B)$$

$$x, y \in \mathcal{U} \text{ s.t. } \begin{cases} x \in A \\ x \in C \\ x \in C \\ y \in B \end{cases} \Rightarrow x \in A, A \subseteq C \Rightarrow x \in C \Rightarrow y \in B, B \subseteq D \Rightarrow x \in B$$

$$\begin{cases} x \in A \\ x \in C \\ x \in B \\ x \in D \end{cases} \Leftrightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow A \times B$$

$$2) U^2 \setminus (A \times B) = \{ (U \setminus A) \times B \} \cup \{ U \times (U \setminus B) \}$$



$$x, y \in \text{pr. ?} \Leftrightarrow \begin{cases} x \in U \setminus A \\ y \in U \\ x \in U \\ y \in U \setminus B \end{cases} \Leftrightarrow \begin{cases} x \notin A \\ y \in U \\ x \in U \\ y \notin B \end{cases} \Leftrightarrow \begin{matrix} x \notin A \\ x, y \notin A \times U \\ xy \in U \times B \end{matrix}$$

$$\begin{cases} x \in U \\ y \in U \\ x \notin A \\ x \in U \\ y \in U \\ y \notin B \end{cases} \rightarrow \begin{cases} x \notin A \\ y \notin B \\ x \in A \\ y \in B \\ x \in A \\ x \notin B \\ x \notin A \\ x \in B \end{cases} \rightarrow (x, y) \notin A \times B \Rightarrow (x, y) \in U^2 \setminus (A \times B)$$

bcz ~~not in~~ $A \times B$

$$3) \bigcup_{k \in K} A_k \times \bigcup_{t \in T} B_t = \bigcup_{(k,t) \in K \times T} (A_k \times B_t)$$

$$(A_1 \cup A_2 \cup A_3) \times (B_1 \cup B_2 \cup B_3)$$

N_8 δ_p p_R R^{-1} $R \circ R$ $R \circ R^{-1}$ $R^{-1} \circ R$
dim. map dim. map

$$R_1 R_2 = \{ (x, y) : \exists z : (x, z) \in R_1, (z, y) \in R_2 \}$$

$$R_2(R_1)$$

$$a) R = \{(x, y) : x, y \in \mathbb{N}, y \leq x\}$$

$$\delta_R = \rho_R = \mathbb{N}$$

$$R^{-1} = \{(x, y) : x, y \in \mathbb{N}, x \leq y\}$$

$$R \circ R^{-1} = \{(x, y) : x, y \in \mathbb{N}, \exists z \in \mathbb{N} : (x, z) \in R, (z, y) \in R^{-1}\}$$

$$z : x \text{ und } z \leq x$$

$$z : y \text{ und } z \leq y$$

$$\Rightarrow R \circ R^{-1} = \mathbb{N}^2$$

$$R \circ R = \{(x, y) : x, y \in \mathbb{N}, \exists z \in \mathbb{N}, (x, z) \in R, (z, y) \in R\} = \mathbb{N}$$

$$z : x$$

$$z \leq x$$

$$y : z$$

$$y \leq z$$

$\Downarrow$

$$y \leq x$$

$$y : x$$

$$R^{-1} \circ R = \{(x, y) : x, y \in \mathbb{N}, \exists z \in \mathbb{N}, (x, z) \in R^{-1}, (z, y) \in R\} = \mathbb{N}^2$$

$$\sqrt{8}, 32$$

$$x : z$$

$$x \leq z$$

$$y : z$$

$$y \leq z$$

$$8 \quad R = \{ (x, y) : x, y \in \mathbb{N}, x : y \}$$

$$\tilde{S}_R = P_R = \mathbb{N}$$

$$R^{-1} = \{ (x, y) : x, y \in \mathbb{N}, y : x \}$$

$$R \circ R = \{ (x, y) : x, y \in \mathbb{N}, \exists z \in \mathbb{N}, (x, z) \in R, (y, z) \in R \} = \mathbb{N}^2$$

$\begin{matrix} x : z & y : z \\ x = kz & y = mz \end{matrix}$

$$R \circ R^{-1} = \{ (x, y) : x, y \in \mathbb{N}, \exists z \in \mathbb{N}, (x, z) \in R, (y, z) \in R^{-1} \} = R$$

$\begin{matrix} x : z & z : y \\ x = kz & z = my \end{matrix}$
 $x = km y$

$$R^{-1} \circ R = \{ (x, y) : x, y \in \mathbb{N}, \exists z \in \mathbb{N}, (x, z) \in R^{-1}, (z, y) \in R \} = \mathbb{N}^2$$

$\begin{matrix} z : x & z : y \\ z = kx & z = my \end{matrix}$
 $z = kmxy$

$$6) \quad R = \{ (x, y) : x, y \in \mathbb{D} \text{ u } x + y \leq 0 \}$$

$$\tilde{S}_R = P_R = \mathbb{D}$$

$$R^{-1} = \{ (x, y) : x, y \in \mathbb{D} \text{ u } y + x \leq 0 \} \text{ монотонно возрастающая}$$

$$R \circ R = \{ (x, y) : x, y \in \mathbb{D}, \exists z \in \mathbb{D} : (x, z) \in R, (z, y) \in R \} = \mathbb{D}^2$$

$\begin{matrix} x + z \leq 0 & y + z \leq 0 \end{matrix}$

$$R \circ R^{-1} = \{ (x, y) : x, y \in \mathbb{D}, \exists z \in \mathbb{D}, (x, z) \in R, (z, y) \in R^{-1} \} = \mathbb{D}^2$$

$\begin{matrix} x + z \leq 0 & y + z \leq 0 \end{matrix}$

$$R^{-1} \circ R = \{ (x, y) : x, y \in \mathbb{D}, \exists z \in \mathbb{D}, (x, z) \in R^{-1}, (z, y) \in R \} = \mathbb{D}^2$$

$\begin{matrix} z + x \leq 0 & z + y \leq 0 \end{matrix}$

$$2) R = \{(x, y) : x, y \in D, 2x \geq 3y\}$$

$$\delta_R = \rho_R = D$$

$$R^{-1} = \{(x, y) : x, y \in D, 2y \geq 3x\}$$

$$2x \geq 3z \geq \frac{3}{2}y$$

$$4x \geq 3y$$

$$R \circ R = \{(x, y) : x, y \in D, \exists z \in D \underset{2x \geq 3z}{(x, z) \in R}, \underset{2z \geq 3y}{(z, y) \in R}\} = ?$$

$$R^{-1} \circ R = \{(x, y) : x, y \in D, \exists z \in D \underset{2z \geq 3x}{(x, z) \in R^{-1}}, \underset{2z \geq 3y}{(z, y) \in R}\} = D^2$$

$$R \circ R^{-1} = \{(x, y) : x, y \in D, \exists z \in D \underset{2x \geq 3z}{(x, z) \in R}, \underset{2y \geq 3z}{(z, y) \in R^{-1}}\} = D^2$$

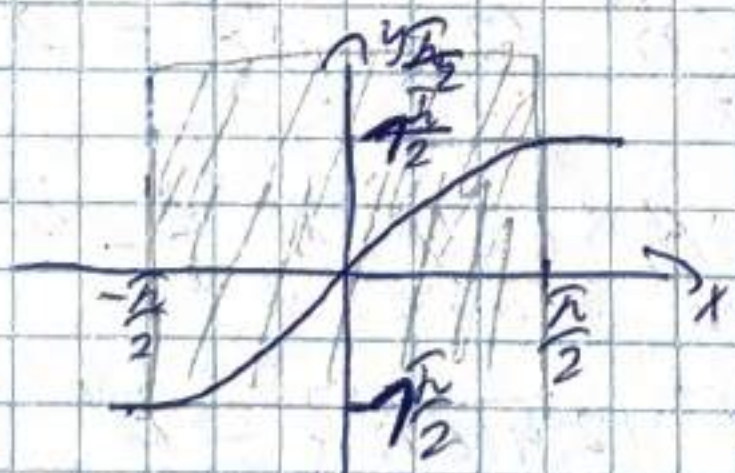
$$g) R = \{(x, y) : x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}], y \geq \sin x\}$$

$$\delta_R = [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\rho_R = [-1, \frac{\pi}{2}]$$

$$R^{-1} = \{(x, y) : x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}], x \geq \sin y\}$$

$$R \circ R = \{(x, y) : x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}], \exists z \in [-\frac{\pi}{2}, \frac{\pi}{2}], z \geq \sin x, y \geq \sin z\}$$



$$R \circ R^{-1} = \{(x, y) : x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}],$$

$$x \geq \sin x, z \geq \sin y\} = D^2$$

$$R^{-1} \circ R = \{(x, y) : x, y \in [-\frac{\pi}{2}, \frac{\pi}{2}],$$

$$x \geq \sin z, y \geq \sin z\}$$

$$= [-1, \frac{\pi}{2}] \times [-1, \frac{\pi}{2}] = [-1, \frac{\pi}{2}]^2$$

$$\sqrt{32} \quad \delta 1 \{ A \setminus X = X \setminus B$$

$$\{ X \setminus A = C \setminus X$$

$$\{ A \bar{X} \cup B \bar{X} \cup \bar{A} B X \cup \bar{B} X = \emptyset$$

$$\{ X \bar{A} \cup \bar{A} X \cup C \bar{X} \cup C \bar{X} A = \emptyset$$

$$\{ A \bar{X} \cap X \bar{B} = \emptyset$$

$$\{ X \bar{A} \cap C \bar{X} = \emptyset$$

$$\{ A \bar{X} \cup B X = \emptyset$$

$$\{ \bar{A} X \cup C \bar{X} = \emptyset$$

$$\begin{cases} A \bar{X} = \emptyset \\ \bar{B} X = \emptyset \\ \bar{A} X = \emptyset \\ C \bar{X} = \emptyset \end{cases}$$

$$\begin{cases} A \in X \\ X \in B \\ X \in A \\ X \subseteq C \end{cases} \quad X \neq A : \quad \begin{matrix} C \subseteq A \subseteq B \\ \parallel \\ X \end{matrix}$$

$$\delta 1 \{ A X = B \bar{X}$$

$$\{ C \cup X = X \bar{A}$$

$$\{ \bar{A} \bar{B} X \cup A X \cup \bar{A} B \bar{X} \cup B \bar{X} = \emptyset$$

$$\{ C A \cup C \bar{X} \cup A X = \emptyset$$

$$\{ A X \cap B \bar{X} = \emptyset$$

$$\{ (C \cup X) \cap \bar{A} X = \emptyset$$

$$\{ A X \cup B \bar{X} = \emptyset$$

$$\{ \bar{C} A X \cup \bar{C} \bar{X} \cup C \bar{X} \cup A X = \emptyset$$

$$A X = \emptyset$$

$$X \in \bar{A}$$

$$(B \cup X) \in X \Rightarrow (B \cup C) \in \bar{A}$$

$$B \bar{X} = \emptyset$$

$$B \in X$$

$$X \in \bar{A}$$

$$C \bar{X} = \emptyset$$

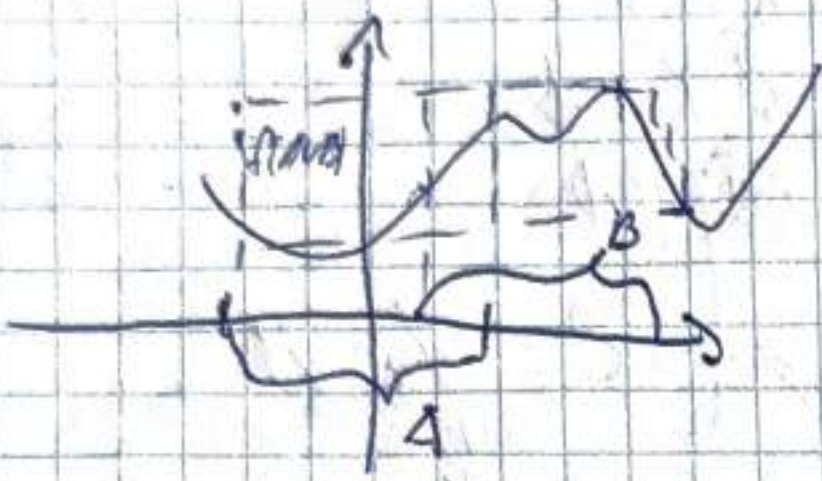
$$C \in X$$

$$A \bar{X} = \emptyset$$

$$\bar{A} \in X \in \bar{A}$$

N 31

9) $f(A \cup B) = f(A) \cup f(B)$



$$x \in A \cup B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow \begin{cases} f(x) \in f(A) \\ f(x) \in f(B) \end{cases} \Rightarrow f(x) \in f(A) \cup f(B)$$

$$\Rightarrow f(A \cup B) \subseteq f(A) \cup f(B)$$

$$x \in A \cup B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow \begin{cases} f(x) \in f(A) \\ f(x) \in f(B) \end{cases} \Rightarrow f(A \cup B) \subseteq f(A) \cup f(B)$$

$$\Downarrow$$

$$f(x) = f(A \cup B)$$

допущено

$$f(x) \subseteq f(A \cup B) \Rightarrow \begin{cases} f(x) \in f(A) \\ f(x) \in f(B) \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in B \\ x \in B \end{cases} \Rightarrow \begin{cases} x \in A \cup B \\ x \in A \cup B \end{cases} \Rightarrow$$

не монотон.

$$\Rightarrow f(x) \in f(A \cup B) \Rightarrow f(A) \cup f(B) \subseteq f(A \cup B)$$

8) $f(\bigcup_{i \in I} A_i) = \bigcup_{i \in I} f(A_i)$

N 32 9) $f(A \cap B) \subseteq f(A) \cap f(B)$

$$x \in A \cap B \Rightarrow \begin{cases} x \in A \\ x \in B \end{cases} \Rightarrow \begin{cases} f(x) \in f(A) \\ f(x) \in f(B) \end{cases} \Rightarrow f(x) \in f(A) \cap f(B) = f(A \cap B)$$

$$\Downarrow$$

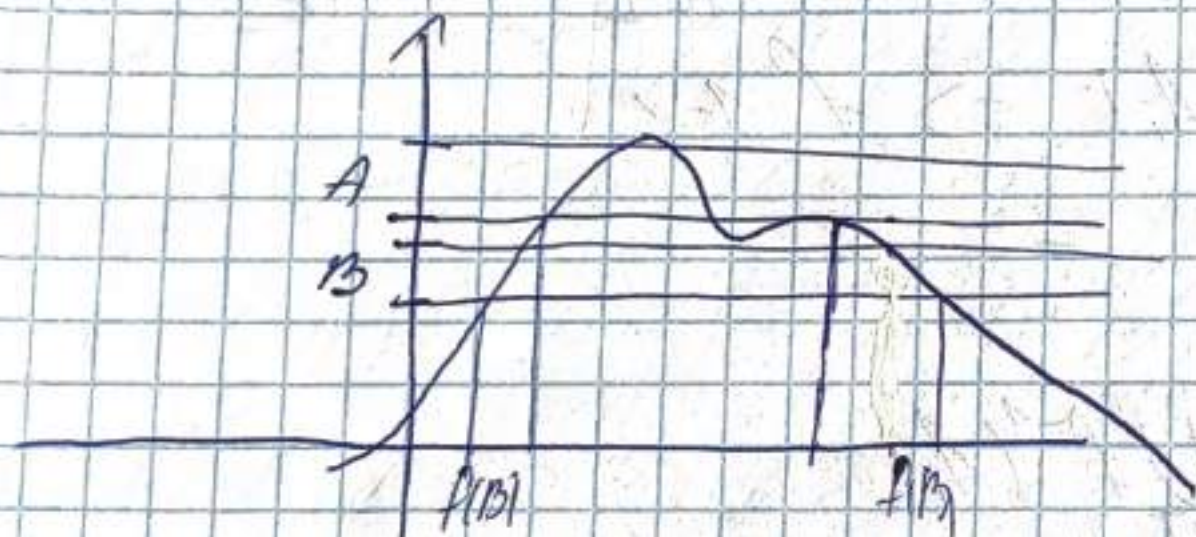
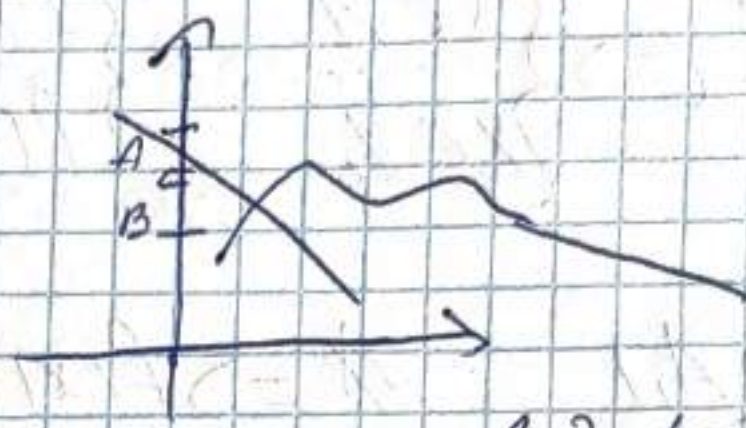
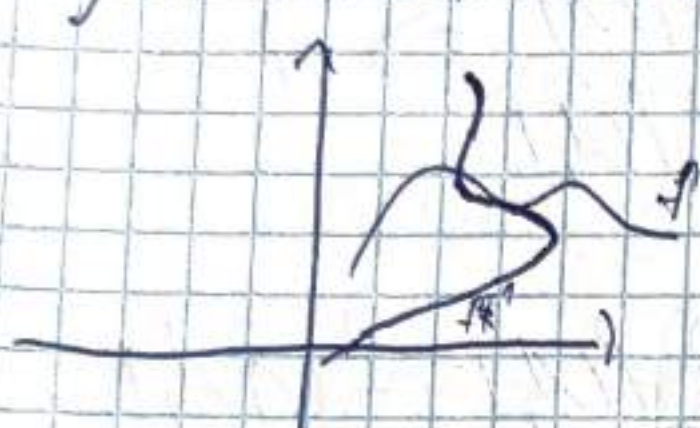
$$f(x) \in f(A \cap B)$$

оп. монотон. / yes.

$$f(A \cap B) = f(A) \cap f(B) \Rightarrow f - \text{на } \text{взаимног. comb}$$

$$\sqrt{38} \quad a) f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

$$y = f^{-1}(x) \in f^{-1}(A \cup B)$$



- 1-2) man ung pab
- 2) Hevab. man ung
- 3) g-bd no (bavamlan u mequm
- u a) gpyrny
- 8) geyam. ipomf
- 6) g-bd unomamb

$$x \in f^{-1}(A \cup B) \Leftrightarrow f(x) \in A \cup B$$

$$\Leftrightarrow \begin{cases} f(x) \in A \\ f(x) \in B \end{cases} \Leftrightarrow \begin{cases} x \in f^{-1}(A) \\ x \in f^{-1}(B) \end{cases} \Rightarrow \bigcup_{i \in I} f(A_i) \subseteq f\left(\bigcup_{i \in I} A_i\right)$$

$$\sqrt{31} \quad 8) f\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f(A_i)$$

$$x \in \bigcup_{i \in I} A_i \Leftrightarrow \begin{cases} x \in A \\ x \in A_i \end{cases} \Rightarrow \begin{cases} f(x) \in f(A) \\ f(x) \in f(A_i) \end{cases} \Rightarrow f\left(\bigcup_{i \in I} A_i\right) \subseteq \bigcup_{i \in I} f(A_i)$$

$$\downarrow$$

$$f(x) \subseteq f\left(\bigcup_{i \in I} A_i\right)$$

$$\bigcup_{i \in I} f(A_i) \subseteq \bigcup_{i \in I} f(A_i)$$

$$f(x) \subseteq \bigcup_{i \in I} f(A_i) \Rightarrow \begin{cases} f(x) \in f(A) \\ f(x) \in f(A_i) \end{cases} \Rightarrow \begin{cases} x \in A \\ x \in A_i \end{cases} \Rightarrow f(x) \in f\left(\bigcup_{i \in I} A_i\right)$$

$$\text{N32 } \delta_1 \quad f\left(\bigcap_{i \in I} A_i\right) \subseteq \bigcap_{i \in I} f(A_i)$$

$$x \in \bigcap_{i \in I} A_i \Leftrightarrow \begin{cases} x \in A_i \\ x \in A_i \end{cases} \Rightarrow \begin{cases} f(x) \in f(A_i) \\ f(x) \in f(A_i) \end{cases} \Rightarrow \star$$

$$f(x) \in f\left(\bigcap_{i \in I} A_i\right) \quad \star \Rightarrow f(x) \in \bigcap_{i \in I} f(A_i) = f\left(\bigcap_{i \in I} A_i\right)$$

$$f\left(\bigcap_{i \in I} A_i\right) = \bigcap_{i \in I} f(A_i) \Rightarrow f \text{ 1-1}$$

N38

$$\delta_1 \quad f^{-1}\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f^{-1}(A_i)$$

$$x \in f^{-1}\left(\bigcup_{i \in I} A_i\right) \Rightarrow f(x) \in \bigcup_{i \in I} A_i \Rightarrow$$

$$\Rightarrow \begin{cases} f(x) \in A_i \\ f(x) \in A_i \end{cases} \Rightarrow \begin{cases} x \in f^{-1}(A_i) \\ x \in f^{-1}(A_i) \end{cases} \Rightarrow f^{-1}(A_i) \cup \dots \cup f^{-1}(A_i)$$

$$N38 \quad b) f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

$$\begin{aligned} & \text{~~x \in f^{-1}(A \cap B)~~ } x \in f^{-1}(A \cap B) \Leftrightarrow \exists y \in A, y \in B: f(x) = y \Rightarrow \\ & \Rightarrow \begin{cases} y \in A, f(x) = y \Leftrightarrow x \in f^{-1}(A) \\ y \in B, f(x) = y \Leftrightarrow x \in f^{-1}(B) \end{cases} \Leftrightarrow f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B) \end{aligned}$$

$$g) f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B)$$

$$\begin{aligned} & x \in f^{-1}(A \setminus B) \Leftrightarrow \exists y \in A, y \notin B: y = f(x) \Rightarrow \\ & \Rightarrow \begin{cases} y \in A, f(x) = y \Leftrightarrow x \in f^{-1}(A) \\ y \notin B, f(x) = y \Leftrightarrow x \notin f^{-1}(B) \end{cases} \exists y_2 \in B: f(x) \neq y_2 \Leftrightarrow \\ & \text{~~x \in f^{-1}(A), x \notin f^{-1}(B)~~ } x \in f^{-1}(A), x \notin f^{-1}(B) \end{aligned}$$

$$x \in f^{-1}(A \setminus B) \Leftrightarrow f(x) \in A \setminus B$$

$$\Leftrightarrow \begin{cases} f(x) \in A \\ f(x) \notin B \end{cases} \Leftrightarrow \begin{cases} x \in f^{-1}(A) \\ x \notin f^{-1}(B) \end{cases} \Rightarrow f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B)$$

N1 100 - всего

70 - роз

75 - розы

80 - розы

85 - розы

30) мин. 10 роз

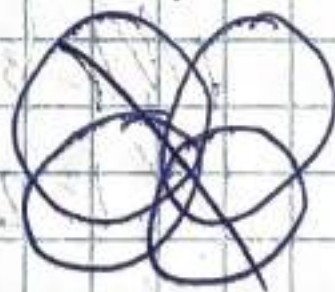
25

20

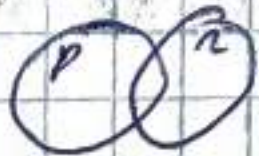
15

$$100 - 90 = 10$$

розы



$$70 + 75 - 100 = 45$$



$$80 + 75 - 100 = 55$$

$$N_2 \quad A = \{x \in \mathbb{N}, 1000 \in \mathbb{N} : x \bmod 3 = 1\}$$

$$B = \{y \in \mathbb{N}, 1000 \in \mathbb{N} : y \bmod 4 = 3\}$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$A = \{x \in \mathbb{N}, 1000 \in \mathbb{N} : x = 3n + 1\}$$

$$x = 3n + 1 \quad n = 0, \dots, 333 \Rightarrow |A| = 334$$

$$y = 4m + 3 \quad m = 0, \dots, 249 \Rightarrow |B| = 250$$

$$3n + 1 = 4m + 3 \quad z = 19 \quad z \in A \cap B$$

$$z = 7 + 12k \quad k = 0, 82 \Rightarrow |A \cap B| = 83$$

$$7 + 12k \leq 1000$$

$$|A \cup B| = 334 + 250 - 83 = 584 - 83 = 501$$

$$\sqrt{3} \quad N_1 = \{x \in \mathbb{N}, x \in [1, \dots, 1000], x : 30\}$$

$$N_2 = \{y \in \mathbb{N}, y \in [1, \dots, 1000], y : 42\}$$

$$N_3 = \{z \in \mathbb{N}, z \in [1, \dots, 1000], z : 105\}$$

$$|N_1 \cup N_2 \cup N_3| = |N_1| + |N_2| + |N_3| - |N_1 \cap N_2| - |N_1 \cap N_3| - |N_2 \cap N_3| + |N_1 \cap N_2 \cap N_3|$$

$$x = 1000 : 30 \quad |N_1| = 33$$

$$y = 1000 : 42 \quad |N_2| = 23$$

$$z = 1000 : 105 \quad |N_3| = 9$$

$$|N_1 \cap N_2| = 1000 : 210 = 4$$

$$|N_2 \cap N_3| = 1000 : 207 = 4$$

$$|N_1 \cap N_2 \cap N_3| = 1000 : 210 = 4$$

$$|N_1 \cap N_2 \cap N_3| = 1000 : 210 = 4$$

$$33 + 23 + 9 - 4 - 4 - 4 + 4 =$$

$$33 + 23 + 9 = 65 - 8 = 57$$

$$30 \cdot 42 \cdot 105 =$$

$$5 \cdot 2 \cdot 3 \cdot 7 \cdot 3 \cdot 7 \cdot 3 \cdot 5 = 210$$

№4 $[0, 1]$ $[0, 1]$ $[0, 1]$ $[0, 1]$

$A = \left\{ \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^k}, \dots \right\}$

$B = \left\{ 0, 1, \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^k}, \dots \right\}$

$f(x) = y$ — биекция

$f(x) = \begin{cases} y, & x \notin A \\ 0, & x = \frac{1}{2} \\ 1, & x = \frac{1}{4} \\ \frac{1}{2^{k-2}}, & x = \frac{1}{2^k}, k \geq 3 \end{cases}$

№5 $[a, b]$ $[c, d]$

$\frac{x-a}{b-a} = \frac{y-c}{d-c}$

биекция

$\frac{d-c}{b-a} (x-a) + c = y$

№6

$x \in (7, 8)$

$y \in (3, +\infty)$

$y = \frac{1}{x-7} + 2$

№7

$U = \{1, \dots, 1000 \in \mathbb{N}\}$

$A = B, C \subseteq U$

$|A \cap B|$

$|A \cup B \cup C|$

$A = \{x \bmod 3 = 1\}$

$|A \cup B|$

$B = \{y \bmod 5 = 2\}$

$|B \cup C|$

$C = \{z \bmod 7 = 4\}$

$|A \cap C|$

$$|A| = \begin{cases} 3n+1 \leq 1000 \\ n \leq 333 \end{cases} \quad |A| = 334$$

$$|B| = \begin{cases} 5n+2 \leq 1000 \\ 5n \leq 998 \\ n \leq 199 \end{cases} \quad |B| = 200$$

$$|C| = \begin{cases} 7n+4 \leq 1000 \\ 7n \leq 996 \\ n \leq 142 \end{cases} \quad |C| = 143$$

$$|A \cap B| = \begin{cases} x \bmod 3 = 1 \\ x \bmod 5 = 2 \end{cases} \quad \begin{matrix} 7+15k = 67 \\ 22+3n \leq 1000 \end{matrix}$$

$$|A \cup B| = |A| + |B| - |A \cap B| = 534 - 67 = 467$$

$$|B \cap C| : |B \cap C| = \begin{cases} x \bmod 5 = 2 \\ x \bmod 7 = 4 \end{cases} \quad \begin{matrix} 32+35k \leq 1000 \\ = 28 \end{matrix}$$

$$|B \cup C| = |B| + |C| - |B \cap C| = 343 - 28 = 315$$

$$|A \cap C| = \begin{cases} x \bmod 3 = 1 \\ x \bmod 7 \neq 4 \end{cases} = \overline{A \cap C} \quad \begin{matrix} 0, 4, 7, 10 \\ 11, \dots \end{matrix}$$

~~$|A \cap C| = \begin{cases} x \bmod 3 = 1 \\ x \bmod 7 = 4 \end{cases}$~~

$$4+21n \leq 1000$$

$$n \leq 47$$

$$|A \cap C| = 48$$

$$|A \cap C| = 334 - 48 = 286$$

$$\begin{cases} x \bmod 3 = 1 & 1, 4, 7, 10, 13, 16, 19, 22, 25 \\ x \bmod 5 = 2 & 2, 7, 12, 17, 22, 27, 32, 37, 42 \\ x \bmod 7 = 4 & 4, 11, 18, 25, 32, 39, 46 \end{cases}$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$|A \cap C| = \begin{cases} x \bmod 3 = 1 & 1, 4, 7 \\ x \bmod 7 \neq 4 & = 48 \end{cases}$$

$$|B \cap C| = \begin{cases} x \bmod 5 = 2 & 2, 7, 12, 17, 22, 27, 32 \\ x \bmod 7 = 4 & 4, 11, 18, 25, 32 \end{cases}$$

$$32+35n \leq 1000 \quad 28$$

$$|A \cup B \cup C| = 334 + 200 + 947 - 67 - 48 - 28 + 9 = 544$$

$$|A \cap B \cap C| = \begin{cases} x \bmod 3 = 1 & 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, \dots, 67 \\ x \bmod 5 = 2 & 2, 7, 12, 17, 22, 27, 32, 37, 42, 47, 52, 57, 62, 67, 72, 77, 82 \\ x \bmod 7 = 4 & 4, 11, 18, 25, 32, 39, 46, 53, 60, 67 \end{cases}$$

$$67 + 105n \leq 1000$$

$$n \leq 8$$

$$|A \cap B \cap C| = 9$$

$$41, 42, 45, 61, 79 - 81, 82 - 85$$

$$\sqrt{4n} \quad x_n = \frac{n}{n+1}$$

$$\lim_{n \rightarrow \infty} x_n \frac{n}{n+1} = 1$$

$$\forall \varepsilon > 0 \exists N = N(\varepsilon) : \forall n > N |x_n - 1| < \varepsilon$$

ε	0.1	0.01	0.001
N	9	99	999

$$|\frac{n}{n+1} - 1| < \frac{1}{10}$$

$$\frac{1}{n+1} < \frac{1}{100}$$

$$\frac{1}{n+1} < \frac{1}{10}$$

$$n+1 > 100$$

$$n+1 > 10$$

$$n > 99$$

$$n > 9$$

$$\sqrt{46} \quad \lim_{n \rightarrow \infty} \frac{10000n}{n^2+1} = 0$$

$$\sqrt{47} \quad \lim_{n \rightarrow \infty} |\sqrt{n+1} - \sqrt{n}| = (\infty - \infty) = \lim_{n \rightarrow \infty} \frac{n+1-n}{\sqrt{n+1} + \sqrt{n}} = 0$$

$$\sqrt{48} \quad \lim_{n \rightarrow \infty} \frac{\sqrt[3]{n^2} \sin n!}{n+1} = \lim_{n \rightarrow \infty} \frac{\sin n!}{\sqrt[3]{n^2} + \sqrt[3]{n^2}} = 0$$

$$\sqrt{49} \quad \lim_{n \rightarrow \infty} \frac{1-2^{n+1}+3^{n+1}}{1-2^{n+1}+3^{n+1}} = \lim_{n \rightarrow \infty} \frac{3^n \left(1+\frac{2}{3}\right)^{n+1}}{3^n \left(1+\frac{2}{3}\right)^{n+1}} = \frac{1}{3}$$

$$\sqrt{50} \quad \lim_{n \rightarrow \infty} \frac{1+a+a^2+\dots+a^n}{1+b+b^2+\dots+b^n} = \left| \frac{|a| < 1}{|b| < 1} \right| = \lim_{n \rightarrow \infty} \frac{(1-a^{n+1})(1-b)}{(1-a)(1-b^{n+1})} =$$

$$\sum_n = \frac{b(1-a^{n+1})}{1-a} = \frac{1-b}{1-a}$$

$$\sum_{\infty} = \frac{b}{1-a}$$

$$\sqrt{51} \quad \lim_{n \rightarrow \infty} \left| \frac{1}{n^2} + \frac{2}{n^2} + \dots + \frac{n-1}{n^2} \right| = \lim_{n \rightarrow \infty} \frac{1+n-1}{2} \cdot n \cdot \frac{1}{n^2} = \frac{1}{2}$$

$$\sqrt{52} \quad \lim_{n \rightarrow \infty} \left| \frac{1}{n} - \frac{2}{n} + \frac{3}{n} - \dots + \frac{(-1)^{n-1} \cdot n}{n} \right| = \frac{1}{2}$$

$$n\text{-term} \quad X_n = \left| \frac{1-2}{n} + \frac{3-4}{n} + \dots + \frac{n-1-n}{n} \right| = \left| -\frac{1}{n} \cdot \frac{n}{2} \right| = \frac{1}{2}$$

$$n\text{-term} \quad X_n = \left| \frac{1-2}{n} + \frac{3-4}{n} + \dots + \frac{n-2-(n-1)}{n} + 1 \right| =$$

$$= \left| -\frac{1}{n} \cdot \frac{n-1}{2} + 1 \right| = \left| \frac{-n+1+2n}{2n} \right| = \left| \frac{n+1}{2n} \right| \Rightarrow \frac{1}{2}$$

$$\sqrt{53} \quad \lim_{n \rightarrow \infty} \left| \frac{1}{n^3} + \frac{2^2}{n^3} + \dots + \frac{(n-1)^2}{n^3} \right| = \lim_{n \rightarrow \infty} \frac{1}{n^3} (1^2 + 2^2 + \dots + (n-1)^2) =$$

$$= \lim_{n \rightarrow \infty} \frac{(n-1)n(2(n-1)+1)}{6n^3} = \frac{1}{3}$$

$$\sqrt{54} \quad \lim_{n \rightarrow \infty} \left[\frac{1^2}{n^3} + \frac{3^2}{n^3} + \dots + \frac{(2n-1)^2}{n^3} \right] = \lim_{n \rightarrow \infty} \frac{1}{n^3} (1^2 + 2^2 + 3^2 + \dots + (2n)^2 - (2^2 + 4^2 + \dots + (2n)^2)) =$$

$$= \lim_{n \rightarrow \infty} \left\{ \frac{1}{n^3} \left(\frac{2n(2n+1)(4n+1)}{6} - \frac{n(n+1)(2n+1)}{6} \right) \right\} =$$

$$= \frac{46}{6} - \frac{8}{6} - \frac{8}{6} = \frac{4}{3}$$

√55

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \dots + \frac{2n-1}{2^n} \right) = 3$$

$x_n =$

$$\frac{x_n}{2} = \frac{1}{2^2} + \frac{3}{2^3} + \dots + \frac{2n-3}{2^n} + \frac{2n-1}{2^{n+1}}$$

$$\frac{x}{2} = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{n-1}} + \frac{2n-1}{2^{n+1}} \rightarrow 0$$

$$\frac{x}{2} = 1$$

$$\frac{x_n}{2} \rightarrow \frac{3}{2}$$

√56

$$\lim_{n \rightarrow \infty} \left[\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} \right] = \left| \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1} \right| =$$

$$= \lim_{n \rightarrow \infty} \left[1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1} \right] = 1$$

√57

$$\lim_{n \rightarrow \infty} (\sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} \dots \sqrt{2}) = 2 \left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n} \right) = 2$$

√58

$$\lim_{n \rightarrow \infty} \frac{n}{2^n} = 0$$

$$0 < \frac{n}{(n+1)^n} = \frac{n}{1 + n + \frac{n(n-1)}{2} + \frac{n(n-1)(n-2)}{3!} + \dots} < \frac{2n}{n(n+1)} \rightarrow 0$$

$$\sqrt{59} \quad \lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$$

$$0 < \frac{2}{1} \cdot \frac{2}{2} \cdot \underbrace{\frac{2}{3} \cdots \frac{2}{n}}_{n-2} < 2 \cdot \left(\frac{2}{3}\right)^{n-2} \rightarrow 0$$

$$\sqrt{60} \quad \lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0 \quad |a| > 1 \quad q = \frac{1}{a} < 1$$

$$\lim_{n \rightarrow \infty} \frac{n^k}{a^n} = \lim_{n \rightarrow \infty} n^k \cdot q^n$$

$$\frac{n^{[k]}}{a^n} < \frac{n^k}{a^n} < \frac{n^{[k]+1}}{a^n}$$

g-bö nem kez (yvel)

$$0 < \lim_{n \rightarrow \infty} \frac{n^k}{(1+(a-1))^n} = \lim_{n \rightarrow \infty} \frac{n^k}{1 + n(a-1) + \frac{n(n-1)(a-1)^2}{2} + \dots + \frac{n(n-1)\dots(n-k)}{(k+1)! (a-1)^k}$$

$$\leq \lim_{n \rightarrow \infty} \frac{n^k}{\frac{n(n-1)\dots(n-k)}{(k+1)!} (a-1)^{k+1}} = \lim_{n \rightarrow \infty} \frac{(k+1)!}{n(a-1)^{k+1}} = 0$$

$$\sqrt{62} \quad \lim_{n \rightarrow \infty} n q^n = |q| < 1 \Rightarrow a = \frac{1}{q}, |a| > 1 \Rightarrow \lim_{n \rightarrow \infty} \frac{n^k}{a^n} = |k=1| = 0$$

$$\sqrt{61} \quad \lim_{n \rightarrow \infty} \frac{a^n}{n!} =$$

$$\sqrt{63} \quad \lim_{n \rightarrow \infty} \sqrt[n]{n} = \lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1 = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln n} =$$

$$n = (1 + (\sqrt[n]{n} - 1))^n = 1 + n(\sqrt[n]{n} - 1) + \frac{n(n-1)}{2} (\sqrt[n]{n} - 1)^2 + \dots >$$

$$> \frac{n(n-1)}{2} (\sqrt[n]{n} - 1)^2$$

$$(\sqrt[n]{n} - 1)^2 < \frac{2}{n(n-1)} \quad \text{or} \quad |\sqrt[n]{n} - 1| < \sqrt{\frac{2}{n-1}} \rightarrow 0$$

$$\Downarrow$$

$$\sqrt[n]{n} \rightarrow 1$$

$$\sqrt{66} \quad \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n!}} = 0$$

$$\sqrt[n]{n} = 1; \quad n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

$$\sqrt[n]{n!} = \underbrace{\sqrt[n]{2\pi n}}_{\rightarrow 1} \cdot \underbrace{\left(\frac{n}{e}\right)^n}_{\rightarrow \infty}$$

$$\sqrt{77} \quad X_n = p_0 + \frac{p_1}{10} + \frac{p_2}{100} + \dots + \frac{p_n}{10^n} \quad \left(\begin{array}{l} n = (1, 2, \dots) \\ p_i, i = 0, 1, 2, \dots \end{array} \right)$$

3, 1; 3, 12; 3, 124; 3, 1248;

$$9 \geq p_k \geq 0 \quad \forall k \geq 1 \quad p_0 - \text{целое}$$

монотонность, ограниченность

$$X_{n+1} - X_n = \frac{p_{n+1}}{10^{n+1}} \geq 0 \Rightarrow \{X_n\} - \text{неубыв.}$$

$$\forall n \quad X_n < p_0 + 1 \Rightarrow \text{имеет предел}$$

$$\sqrt{78} \quad x_n = \frac{10}{1} \cdot \frac{11}{3} \cdots \left(\frac{n+9}{2n-1} \right) \sim \frac{1}{2} \sim \frac{9-n}{2n+1}$$

$$x_{n+1} - x_n = \underbrace{\frac{10}{1} \cdot \frac{11}{3} \cdots \frac{n+9}{2n-1}}_{>0} \left(\frac{n+10}{2n+1} - 1 \right) = \underbrace{\frac{10}{1} \cdot \frac{11}{3} \cdots \frac{n+9}{2n-1}}_{>0} \cdot \frac{9-n}{2n+1}$$

for $n=9$

$$x_{n+1} - x_n < 0$$

$$x_n > x_{n+1} \Rightarrow \{x_n\} \text{ - убыв}$$

$$x_n > 0 \quad \forall n \quad x_n \rightarrow 0$$

82

$$x_n = a_0 + a_1 q + \dots + a_n q^n \quad |a_k| < M, |q| < 1$$

$$\{x_n\} \text{ с.с.} (\Rightarrow) \{x_n\} \text{ - убыв}$$

$$\{x_n\} \text{ - убыв} (\Rightarrow) \forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall n > N, \forall p > 0 \Rightarrow$$

$$|x_{n+p} - x_n| < \varepsilon$$

$$|x_{n+p} - x_n| = |a_{n+1} q^{n+1} + a_{n+2} q^{n+2} + \dots + a_{n+p} q^{n+p}| <$$

$$< M |q^{n+1} + \dots + q^{n+p}| = M \frac{2^{n+1} (1 - q^p)}{2-2} < \varepsilon$$

$$q^{n+1} \frac{\varepsilon (1-q)}{M(1-q^2)} < \frac{\varepsilon (1-q)}{M \cdot 2}$$

$$n <$$

Nu2

$$a) x_n = \frac{(-1)^{n+1}}{n}$$

$$\lim_{n \rightarrow \infty} \frac{(-1)^{n+1}}{n} = 0$$

$$|x_n| = \frac{1}{n} \quad \frac{1}{n} < \varepsilon$$

$$N > \frac{1}{\varepsilon}$$

ε
 N

0,1
10

0,01
100

0,001
1000

$$N > 10$$

$$b) x_n = \frac{2n}{n^3+1} < \frac{2}{n^2}$$

$$\lim_{n \rightarrow \infty} x_n = 0$$

$$\frac{2}{n^2} < \varepsilon \quad n > \sqrt{\frac{2}{\varepsilon}}$$

ε
 N

0,1

0,01

0,001

$$N > \sqrt{20}$$

$$c) x_n = \frac{1}{n!} \leq \frac{1}{2^{n-1}}$$

$$\frac{1}{2^{n-1}} < \varepsilon$$

$$2^{n-1} > \frac{1}{\varepsilon}$$

ε
 N

0,1

0,01

0,001

$$d) x_n = (-1)^n \cdot 0,999^n$$

ε
 N

0,1

0,01

0,001

N45

a) $\lim_{n \rightarrow \infty} x_n = \infty$

$n > N, |x_n| > \varepsilon$

b) $\lim_{n \rightarrow \infty} x_n = -\infty$

$n > N, x_n < -\varepsilon$

c) $\lim_{n \rightarrow \infty} x_n = +\infty$

$x_n > \varepsilon$

N61

$\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$

Kann 2191

$0 < \left| \frac{a^n}{n!} \right| = \frac{|a|}{1} \cdot \frac{|a|}{2} \cdot \dots \cdot \frac{|a|}{k} \cdot \frac{|a|}{k+1} \cdot \frac{|a|}{k+2} \cdot \dots \cdot \frac{|a|}{n} <$

$< \frac{|a|^k}{k!} \cdot \frac{|a|^{n-k}}{(k+1)^{n-k}}$

$< |a|^k \cdot \left(\frac{1}{2} \right)^{n-k} = \frac{(2|a|)^k}{2^n}$ K-Verhalten
wird

$\lim_{n \rightarrow \infty} \frac{(2|a|)^k}{2^n} = 0$

N64

$x_n = \frac{\cos 1!}{1 \cdot 2} + \frac{\cos 2!}{2 \cdot 3} + \dots + \frac{\cos n!}{n(n+1)}$

$\forall \varepsilon > 0 \exists N: \forall n > N$

$|x_{n+p} - x_n| < \varepsilon$

$|x_{n+p} - x_n| = \left| \frac{\cos(n+1)!}{(n+1)(n+2)} + \dots + \frac{\cos(n+p)!}{(n+p)(n+p+1)} \right| < \frac{1}{(n+1)(n+2)} + \dots + \frac{1}{(n+p)(n+p+1)}$

$= \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+2} - \frac{1}{n+3} + \dots + \frac{1}{n+p} - \frac{1}{n+p+1} = \frac{1}{n+1} - \frac{1}{n+p+1} = \frac{p}{(n+1)(n+p+1)}$

Wegen $p = n+1 \Rightarrow \frac{1}{2(n+1)} < \varepsilon \Rightarrow n+1 > \frac{1}{2\varepsilon} \Rightarrow n > \frac{1}{2\varepsilon} - 1$

№36

$$X_n = \frac{n^2}{2^n} \rightarrow 0$$

$$\Delta X_n = \frac{\sqrt{n}}{100+n} \rightarrow 0$$

$$b) X_n = \frac{1000^n}{n!} \rightarrow 0$$

$$\frac{1}{2}; 1; \frac{9}{8}; \frac{16}{16}; \frac{25}{32}; \frac{36}{64}$$

$\frac{9}{8}$ - макс. чл. ряда.

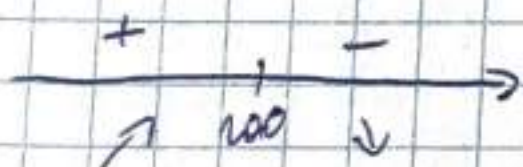
$$\frac{1}{101}; \frac{\sqrt{2}}{102}; \frac{\sqrt{3}}{103} \Rightarrow \text{Sup} = X_{100} = \frac{10}{200} = 0.05$$

$$\Delta X_n' = \frac{\frac{100+n}{25n} - \sqrt{n}}{(100+n)^2} = 0$$

$$\frac{100+n-2n}{25n} = \frac{100-n}{25n}$$

$$100+n = 2n$$

$$n = 100$$



$$b) X_{1000} = \frac{1000^{1000}}{1000!} = \frac{1000^{999}}{999!} = X_{999}$$

$$\frac{1000}{1}; \frac{1000000}{2}; \frac{1000000000}{8}$$

№99

$$X_n = n^2 - 9n - 100 \rightarrow \infty$$

$$X_4 = 16 - 36 - 100 = -120$$

$$X_5 = 25 - 45 - 100 = -120$$

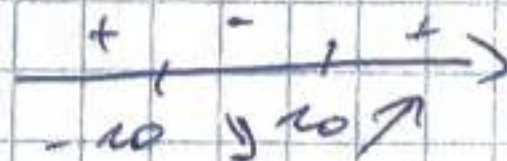
№100

$$X_n = n + \frac{100}{n} \rightarrow \infty$$

$$101; 52; 36 \frac{1}{3}; 29; 25$$

$$X_{10} = 20$$

$$X_n' = 1 - \frac{100}{n^2} = \frac{n^2 - 100}{n^2} = 0$$



N 101 $x_n = 1 - \frac{1}{n} \rightarrow 1$

$\inf x_n, \sup x_n, \lim_{n \rightarrow \infty} x_n, \overline{\lim}_{n \rightarrow \infty} x_n$

$\inf x_n = 0$

$\overline{\lim}_{n \rightarrow \infty} x_n = 1$

$\sup x_n = 1$

$\lim_{n \rightarrow \infty} x_n = 1$

$0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$

N 101, 1

$x_n = (1-n)^{n-1} \left(2 + \frac{3}{n} \right)$

$\inf x_n = -3.5$

$\lim_{n \rightarrow \infty} x_n = 2$

$\sup x_n = 5$

$\lim_{n \rightarrow \infty} x_n = -2$

$5 | -3.5 | 3 | -2.25 | 2.6 | 2.5 |$

N 102

$x_n = \frac{(1-n)^n}{n} + \frac{1+(1-n)^n}{2}$

$\sup x_n = 1.5$

$\overline{\lim}_{n \rightarrow \infty} x_n = 1$

$\inf x_n = -1$

$\lim_{n \rightarrow \infty} x_n = 0$

$0 \quad 0 \quad \frac{1}{2} \quad \frac{2}{3}$

n-remains $x_n = \frac{1}{n} + 1 \rightarrow 1$

n-remains $x_n = -\frac{1}{n} + 0 \rightarrow 0$

N 103

$x_n = 1 + \frac{n}{n+1} \cos \frac{n\pi}{2}$

$\sup x_n =$

$\overline{\lim}_{n \rightarrow \infty} x_n = 2$

$\inf x_n =$

$\lim_{n \rightarrow \infty} x_n = 0$

n-remains $x_n = 1$

n-remains $x_n = 1 + \frac{n}{n+1} \rightarrow 2$

$2 - \frac{n}{n+1} \rightarrow 0$

$$1) \quad 1 + \frac{4}{5} \quad 1 + \frac{8}{9}$$

$$2) \quad 1 - \frac{2}{3} \quad 1 - \frac{6}{15} \rightarrow 0$$

105-120 important

151-164

183-188

$$N104 \quad X_n = 1 + 2(-1)^{n+1} + 3(-1)^{\frac{n(n-1)}{2}}$$

$$\inf X_n = -4$$

$$\sup X_n = 6$$

$$\lim_{n \rightarrow \infty} X_n = 0$$

$$\lim_{n \rightarrow \infty} X_n = -4$$

$$1+2+3=6$$

$$1-2+3=2$$

$$1+2-3=0$$

$$1-2-3=-4$$

$$1$$

$$2$$

$$3$$

$$4$$

$$1+2+3=6$$

$$1-2-3=-4$$

$$1+2-3=0$$

$$1-2+3=2$$

$$N164 \quad (2x)! \quad x=0, \frac{1}{2}, 1;$$

$$x = \frac{n}{2}, n \in 0, 1, \dots, \infty$$

$$y = \log_2 \log_3 (\log_n x) \quad x > 0$$

$$\log_n x > 0 = \log_n 1 \quad x > 1$$

$$\log_2 0 \Rightarrow \log_3 \log_n x > 0 \cdot \log_3 0$$

$$\log_n x > 1 \quad x > n$$

$$y = \sqrt{\ln \lg x}$$

$$\ln \lg x \geq 0 = \ln 1$$

$$\lg x \geq 1 \quad x \in \left[\frac{n}{2} + \pi n; \frac{n}{2} + \pi n \right]$$

$$\sqrt{9; n 2x} + \sqrt{9; n 3x}$$

№ 79 E_x

E_y

y

$$[-1, 2]$$

$$[0, 4]$$

$$x^2$$

$$[10, 1000]$$

$$[1, 3]$$

$$\lg x$$

$$(-\infty; +\infty)$$

$$[0, 1]$$

$$\frac{1}{\pi} \arctg x$$

$$[-1, 0] \cup [0, 1]$$

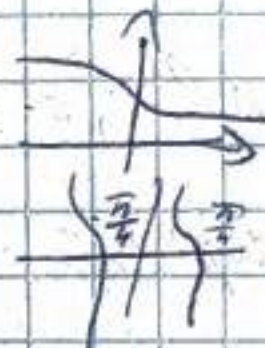
$$(-\infty; -1] \cup [1; +\infty)$$

$$\arcsin \frac{\pi x}{4}$$

$$[-2, -1] \cup [1, 2]$$

$$[1, 2]$$

$$|x|$$



№ 79 $x_n = (1 - \frac{1}{2})(1 - \frac{1}{4}) \dots (1 - \frac{1}{2^n})$

$$x_{n+1} = x_n \cdot (1 - \frac{1}{2^{n+1}})$$

$$x_{n+1} - x_n = (1 - \frac{1}{2})(1 - \frac{1}{4}) \dots (1 - \frac{1}{2^n}) (1 - \frac{1}{2^{n+1}} - 1)$$

$$x_n > x_{n+1} \Rightarrow \{x_n\} - \text{убыв}$$

$$x_n > 0 \quad \forall n \quad x_n \rightarrow 1$$

№ 80

$$x_n = (1 + \frac{1}{2})(1 + \frac{1}{4}) \dots (1 + \frac{1}{2^n})$$

$$x_{n+1} = x_n \cdot (1 + \frac{1}{2^{n+1}}) > x_n - \text{возраст}$$

$$x_n < e^{\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n}} < e^{\frac{1}{2} \cdot \frac{1}{1 - \frac{1}{2}}} = e - \text{ср. верх}$$

с. 112

$$\sqrt{81} \quad x_1 = \sqrt{2}, \quad x_2 = \sqrt{2+\sqrt{2}}, \quad x_n = \sqrt{2+\sqrt{2+\dots+\sqrt{2}}}$$

$$x_{n+1} = \sqrt{2+x_n} \quad x_{n+1} > x_n \Rightarrow \{x_n\} - \text{возр.}$$

$$x_n = \sqrt{2+\sqrt{2+\dots+\sqrt{2}}} < \sqrt{2+\sqrt{2+\dots+\sqrt{4}}}$$

$$x_n < 2 - \text{огр. сверху} \Rightarrow \text{возр. + огр. сверху} = \text{сход.}$$

$$\lim_{n \rightarrow \infty} \sqrt{2+x_n} =$$

$$\sqrt{83} \quad x_n = \frac{\sin 1}{2} + \frac{\sin 2}{2^2} + \dots + \frac{\sin n}{2^n}$$

$$|x_{n+p} - x_n| = \left| \frac{\sin(n+p)}{2^{n+1}} + \dots + \frac{\sin(n+p)}{2^{n+p}} \right| = \frac{1}{2^{n+1}} + \dots + \frac{1}{2^{n+p}} =$$

$$= \frac{1}{2^{n+1}} \left(1 + \frac{1}{2} + \dots + \frac{1}{2^{p-1}} \right) = \frac{1}{2^{n+1}} < \varepsilon \quad 2^n > \frac{1}{\varepsilon}$$

$$\sqrt{85} \quad x_n = 4 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}$$

$$|x_{n+p} - x_n| = \left| \frac{1}{(n+p)^2} + \dots + \frac{1}{(n+p)^2} \right| = \frac{1}{n} \left(\frac{1}{n+1} + \dots + \frac{1}{n+p} \right) =$$

$$= \frac{1}{n} - \frac{1}{n+p} = \frac{n+p-n}{n(n+p)} = \frac{p}{n(n+p)} < \varepsilon$$

$$\sqrt{86} \quad x_n = \frac{(n-1)}{n+1} \cdot \cos \frac{\pi n}{3}$$

$$n \text{ нечетное} \rightarrow 1 - \frac{1}{6}, -\frac{1}{6}, -\frac{3}{10}$$

$$\sup x_n = 1$$

$$\lim_{n \rightarrow \infty} x_n = 1$$

$$\inf x_n = -\frac{1}{2}$$

$$\lim_{n \rightarrow \infty} x_n = \frac{1}{2}$$

$$n \text{ четное} \quad 0, \frac{1}{2}, \frac{1}{3}$$

$$\sqrt{106} \quad x_n = (1-n)^n$$

$$\sup x_n = \infty$$

$$\inf x_n = -\infty$$

$$\lim_{n \rightarrow \infty} x_n = \infty$$

$$\lim_{n \rightarrow \infty} x_n = -\infty$$

$$n \text{ нечетн} \rightarrow \infty$$

$$n \text{ четн} \rightarrow -\infty$$

$$\sqrt{107} \quad x_n = -n[1 + (1-n)^n]$$

$$\sup x_n = -1$$

$$\inf x_n = -\infty$$

$$\lim_{n \rightarrow \infty} x_n = -\infty$$

$$\lim_{n \rightarrow \infty} x_n = -\infty$$

1	2	3	4	5
-1	-6	-3	-10	-5

$$\sqrt{108} \quad x_n = n^{1-n}$$

$$\sup x_n = +\infty$$

$$\inf x_n = 0$$

$$\lim_{n \rightarrow \infty} x_n = +\infty$$

$$\lim_{n \rightarrow \infty} x_n = +\infty$$

1	2	3	4	5
1	2	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$

$$\sqrt{109} \quad x_n = 1 + n \sin \frac{n\pi}{2}$$

$$\sup x_n = +\infty$$

$$\inf x_n = -\infty$$

$$\lim_{n \rightarrow \infty} x_n = +\infty$$

$$\lim_{n \rightarrow \infty} x_n = -\infty$$

$$n \text{ четн} \rightarrow 1$$

$$n \text{ нечетн} \rightarrow$$

1	3	5	7	9
2	-2	6	-6	10

$$\sqrt{110} \quad x_n = \frac{1}{n-10.2}$$

$$\sup x_n = x_n = \frac{10}{8} = 1.25$$

$$\inf x_n = x_n = -5$$

$$\lim_{n \rightarrow \infty} x_n = 0$$

$$\lim_{n \rightarrow \infty} x_n = 0$$

1	2	3	10	11
$\frac{1}{12}$	$\frac{10}{82}$	$\frac{10}{72}$	$\frac{10}{2}$	$\frac{10}{8}$
12	13			
$\frac{10}{18}$	$\frac{10}{28}$			

$$\sqrt{112} \quad x_n = \left(1 + \frac{1}{n}\right)^n \cdot (-1)^n + \sin \frac{\pi n}{4}$$

$$\lim_{n \rightarrow \infty} \overline{x_n} = e+1$$

$$\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ -e + \frac{\sqrt{2}}{2} & e+1 & -e + \frac{\sqrt{2}}{2} & e & -e - \frac{\sqrt{2}}{2} & e-1 & \end{array}$$

$$\lim_{n \rightarrow \infty} \underline{x_n} = -e - \frac{\sqrt{2}}{2}$$

$$\sqrt{113} \quad x_n = \frac{n}{n+1} \sin^2 \frac{\pi n}{4}$$

$$\lim_{n \rightarrow \infty} \overline{x_n} = 1$$

$$\lim_{n \rightarrow \infty} \underline{x_n} = 0$$

$$\sqrt{114} \quad x_n = \sqrt[n]{1 + 2^{n-1}}$$

$$\lim_{n \rightarrow \infty} \underline{x_n} = 1$$

$$\lim_{n \rightarrow \infty} \overline{x_n} = 2$$

?

$$\sqrt{115} \quad x_n = \cos^n \frac{\pi n}{3}$$

$$\lim_{n \rightarrow \infty} \overline{x_n} = 1$$

$$\lim_{n \rightarrow \infty} \underline{x_n} = 0$$

$$\sqrt{116} \quad \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{7}{8}, \dots, \frac{1}{2^n}, \frac{2^n-1}{2^n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0, \quad \lim_{n \rightarrow \infty} \frac{2^n-1}{2^n} = 1$$

$$\sqrt{117} \quad 1, 1 + \frac{1}{2}, \frac{1}{3}, 1 + \frac{1}{3}, \dots, \frac{1}{4}, 1 + \frac{1}{4}, \frac{1}{5} + \frac{1}{4}, \dots, \frac{1}{n+1} + \frac{1}{n}, \frac{1}{n+1}$$

$$0, 1, \frac{1}{2}, \frac{1}{3}, \dots$$

$$\sqrt{118}$$

$$\sqrt{118} \quad \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \quad [1, 0]$$

$$\sqrt{119} \quad x_n = 3(1 - \frac{1}{n}) + 2(-1)^n$$

$$3+2=5 \quad 3-2=1$$

$$\sqrt{120} \quad x_n = \frac{1}{2} [(a+b) + (-1)^n (a-b)]$$

a, b

$$\sqrt{151} \quad y = \frac{x^2}{x+1} \quad (-\infty; -1) \cup (-1; +\infty)$$

$$\sqrt{152} \quad y = \sqrt{3x - x^3}$$

$$3x - x^3 \geq 0 \quad \begin{array}{c} + \\ - \end{array} \begin{array}{c} + \\ - \end{array} \begin{array}{c} + \\ - \end{array} \quad x \in (-\infty; -\sqrt{3}] \cup [0; +\sqrt{3}]$$

$$\sqrt{153} \quad y = (x-2) \sqrt{\frac{x+1}{x-1}} \quad \frac{x+1}{x-1} \geq 0 \quad \begin{array}{c} + \\ - \end{array} \begin{array}{c} + \\ - \end{array} \begin{array}{c} + \\ - \end{array} \quad (-\infty; -1] \cup [1; +\infty)$$

$$\sqrt{154} \quad y = \log(x^2 - 4) \quad x^2 - 4 > 0 \quad \begin{array}{c} + \\ - \end{array} \begin{array}{c} + \\ - \end{array} \begin{array}{c} + \\ - \end{array} \quad (-\infty; -2) \cup (2; +\infty)$$

$$\sqrt{155} \quad y = \log(x+2) + \log x - 2$$

$$\begin{cases} x+2 > 0 \\ x-2 > 0 \end{cases} \quad x > 2$$

$$\sqrt{156} \quad y = \sqrt{\sin(5x)}$$

$$\begin{cases} \sin 5x \geq 0 \\ x \geq 0 \end{cases} \quad 0 \leq x \leq \pi^2 \quad \begin{array}{l} 2\pi n \leq 5x \leq \pi + 2\pi n \\ 4\pi^2 n^2 \leq x \leq (\pi + 2\pi n)^2 \end{array}$$

$$\sqrt{157} \quad y = \sqrt{\cos x^2} \quad -\frac{\pi}{2} + 2\pi n \leq x^2 \leq \frac{\pi}{2} + 2\pi n, n \in \mathbb{Z}$$

$$\sqrt{158} \quad y = |n| \sin \frac{\pi}{x} \quad \sin \frac{\pi}{x} > 0 \quad 2\pi n < \frac{\pi}{x} < \pi + 2\pi n, n \in \mathbb{Z}$$

$$2\pi n < \frac{1}{2\pi} < x < \frac{1}{1+2\pi}$$

$$\sqrt{159} \quad y = \frac{\sqrt{x}}{\sin \pi x} \quad x > 0 \quad x > 0, x \neq n, n \in \mathbb{Z}$$

$$\sqrt{159} \quad y = \arcsin \frac{2x}{1+x} \quad -1 \leq \frac{2x}{1+x} \leq 1$$

$$-\frac{1}{2} \leq \frac{x}{1+x} \leq \frac{1}{2}$$

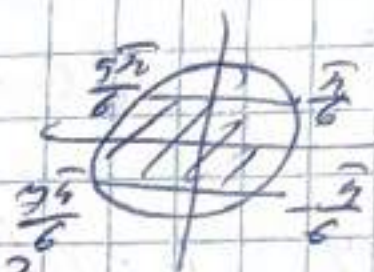
$$-\frac{1}{3} \leq x \leq 1$$

$$-\frac{3}{2} \leq -\frac{1}{1+x} \leq 0$$

$$\sqrt{160} \quad y = \arccos(2 \sin x)$$

$$-1 \leq 2 \sin x \leq 1 \quad x \in \left(-\frac{\pi}{6} + \pi n, \frac{\pi}{6} + \pi n\right), n \in \mathbb{Z}$$

$$-\frac{1}{2} \leq \sin x \leq \frac{1}{2}$$



$\sqrt{161}$

$$y = \lg[\cos(\lg x)]$$

$$\cos(\lg x) > 0 \quad -\frac{\pi}{2} + \pi n < \lg x < \frac{\pi}{2} + \pi n, n \in \mathbb{Z}$$

$$\lg x > 0 \quad 10^{(-\frac{\pi}{2} + \pi n)} < x < 10^{(\frac{\pi}{2} + \pi n)}, n \in \mathbb{Z}$$

$\sqrt{162}$

$$y = (x + |x|) \sqrt{x \sin^2 \pi x} \quad \begin{matrix} x \sin^2 \pi x \geq 0 \\ x \geq 0 \end{matrix}$$

$$\sqrt{163} \quad y = \operatorname{ctg} \pi x + \arccos(2^x)$$

$$-1 \leq 2^x \leq 1$$

$$\operatorname{ctg} \pi x \quad x \neq 0$$

$$x \leq 0$$

$$x < 0, x \neq -n, n \in \mathbb{Z}$$

$$\sqrt{164} \quad y = \arcsin(1-x) + \lg(\lg x)$$

$$-1 \leq 1-x \leq 1$$

$$\lg x > 0$$

$$0 \leq x \leq 2$$

$$x > 1$$

$$x \in (1; 2]$$

$$\sqrt{183} \quad y = a + (b-a)x$$

$$y \in [a, b] \text{ при } b > a, \quad y \in [b, a] \text{ при } a > b$$

$$\sqrt{184} \quad y = \frac{1}{1-x} \quad y \in (1, +\infty)$$

$$\sqrt{185} \quad y = \frac{x}{2x-1} \quad y \in (-\infty; 0) \cup (1, +\infty)$$

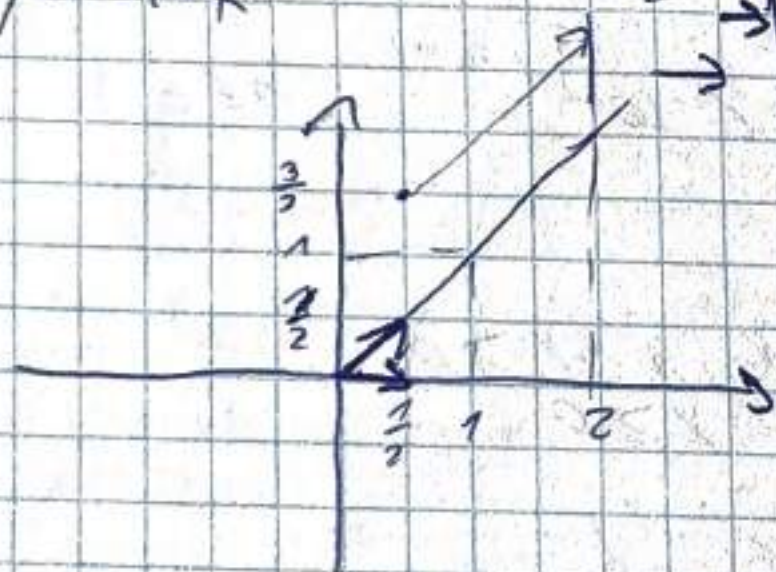
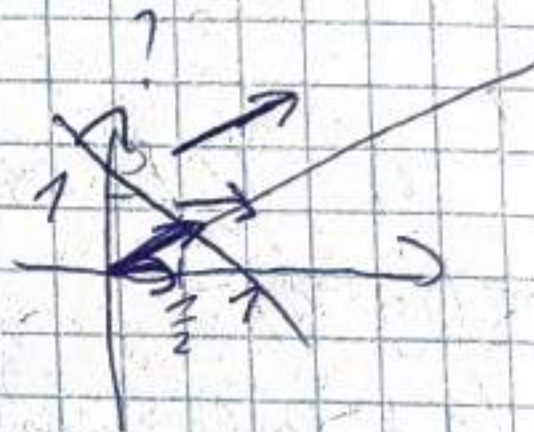
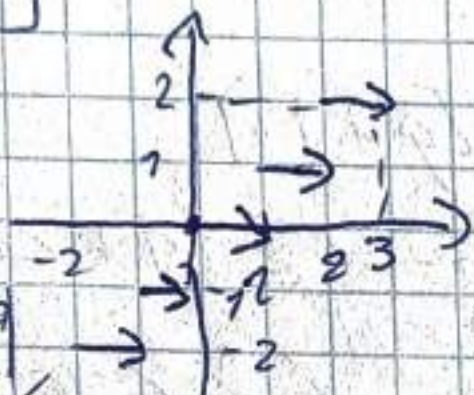
$$\sqrt{186} \quad y = \sqrt{x-x^2} \quad \text{or } y = \sqrt{\frac{1}{4} - (x - \frac{1}{2})^2} \quad y \in [0, \frac{1}{2}]$$

$$\sqrt{187} \quad y = \operatorname{ctg} \pi x \quad y \in (-\infty; +\infty)$$

$$\sqrt{188} \quad y = x + [2x]$$

$$[6.999] = 6$$

$$y \in (0, 2]$$



$$y \in (0; \frac{1}{2}] \cup [\frac{3}{2}; 2]$$

$$y = x$$

$$y = 1+x$$

$$x=0 \quad y = 0 + [2 \cdot 0] = 0$$

$$x = \frac{1}{8} \quad y = \frac{1}{8} + [\frac{1}{4}] = \frac{1}{8}$$

$$x = 0.499 \quad y = x$$

$$x = 1 \quad y = 3$$

$$\sqrt{193} \quad f(x) = \frac{1-x}{1+x}$$

$$f(0) = 1$$

$$f\left(\frac{1}{x}\right) = \frac{x-1}{x+1}$$

$$f(1-x) = \frac{1+x}{1-x}$$

$$\frac{1}{f(x)} = \frac{x+1}{1-x}$$

$$f(x+1) = -\frac{x}{x+2}$$

$$f(x)+1 = \frac{2}{1+x}$$

$$\sqrt{206} \quad \varphi(x) = x^2$$

$$\psi(x) = 2^x$$

$$\varphi(\varphi) = (x^2)^2 = x^4$$

$$\psi(\varphi) = 2^{x^2}$$

$$\varphi(\psi) = (2^x)^2 = 2^{2x}$$

$$\psi(\psi) = 2^{2x}$$

$\sqrt{207}$

$$\varphi = \text{sgn } x$$

$$\psi(x) = \frac{1}{x}$$

$$\varphi(\varphi) = \varphi$$

$$\psi(\varphi) = \varphi$$

$$\varphi(\psi) = \varphi$$

$$\psi(\psi) = x$$

$\sqrt{208}$

$$\varphi(x) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases}$$

$$\varphi(x) = \begin{cases} 0, & x \leq 0 \\ x^2, & x > 0 \end{cases}$$

$$\varphi(\varphi) = \varphi$$

$$\psi(\varphi) \equiv 0$$

$$\varphi(\psi) = \psi$$

$$\psi(\psi) \equiv 0$$

$$\sqrt{203} \quad f(x) = \frac{1}{1-x}$$

$$f(f(x)) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1}{\frac{1-x-1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}$$

$$f(f(f(x))) = \frac{1}{1 - \frac{x-1}{x}} = \frac{1}{1 - 1 + \frac{1}{x}} = x$$

$\sqrt{20}$

$$f(x) = \frac{x}{\sqrt{1+x^2}} \quad \underbrace{f(f(\dots f(x)))}_{n \text{ times}} = \frac{x}{\sqrt{1+x^2}}$$

$$f(f(x)) = \frac{\frac{x}{\sqrt{1+x^2}}}{\sqrt{1 + \frac{x^2}{1+x^2}}} = \frac{x(1+x^2)}{\sqrt{1+x^2} \cdot (1+x^2)} = \frac{x\sqrt{1+x^2}}{\sqrt{1+2x^2}}$$

$$f(f(f(x))) = \frac{\frac{x}{\sqrt{1+x^2}} \cdot \frac{1 + \frac{x^2}{1+x^2}}{1 + \frac{2x}{1+x^2}}}{\sqrt{1 + \frac{x^2}{1+x^2}}} = \frac{x}{\sqrt{1+2x^2}}$$

$$= \frac{x}{\sqrt{3x^2+1}}$$

$$\sqrt{211} \quad f(x+1) = x^2 - 3x + 1$$

$$f(x) = (x-1)^2 - 3(x-1) + 1 = x^2 - 5x + 6$$

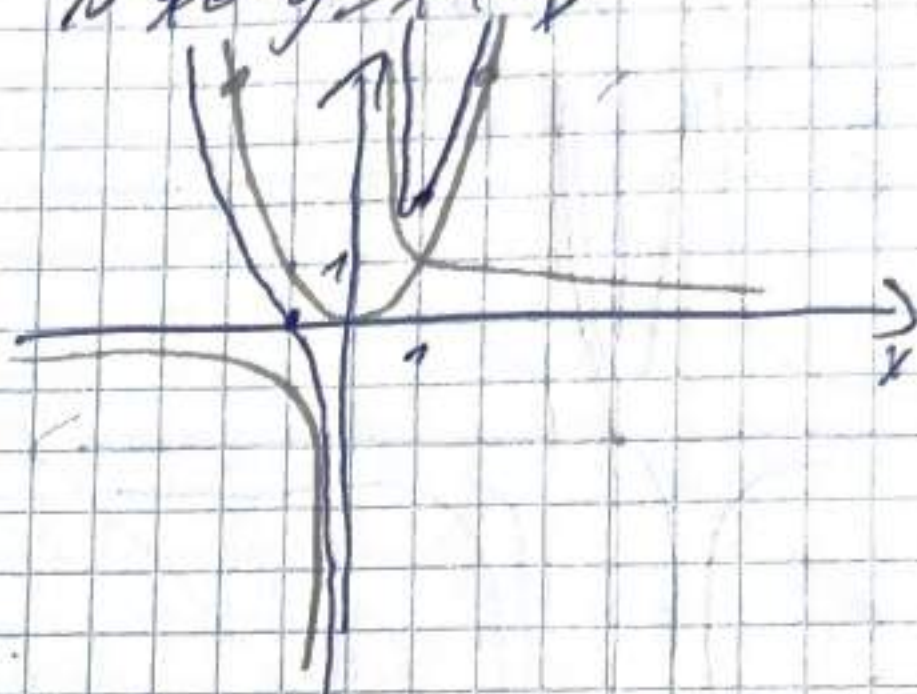
$$f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$$

$$f(x) = x^2 - 2$$

N/253 $y = x + \frac{1}{x}$



N/254 $y = x^2 + \frac{1}{x}$



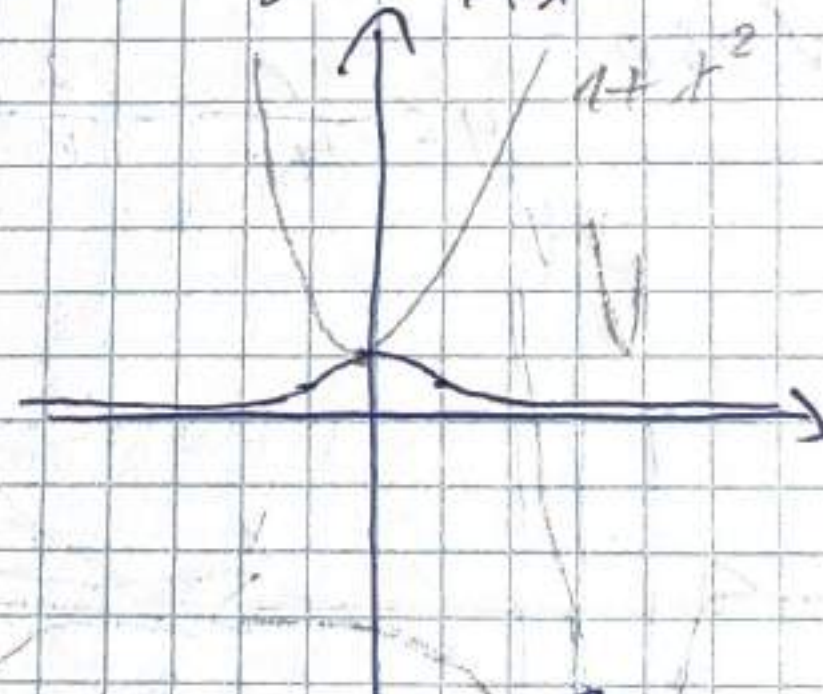
N/255

$y = x + \frac{1}{x^2}$

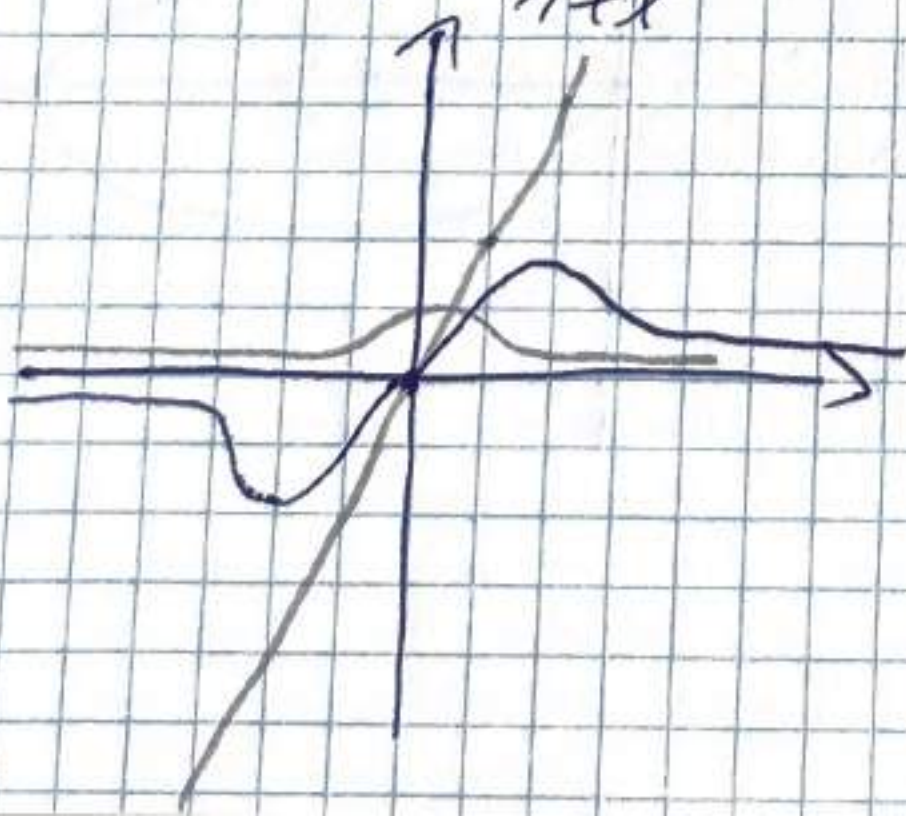


N/256

$y = \frac{1}{1+x^2}$

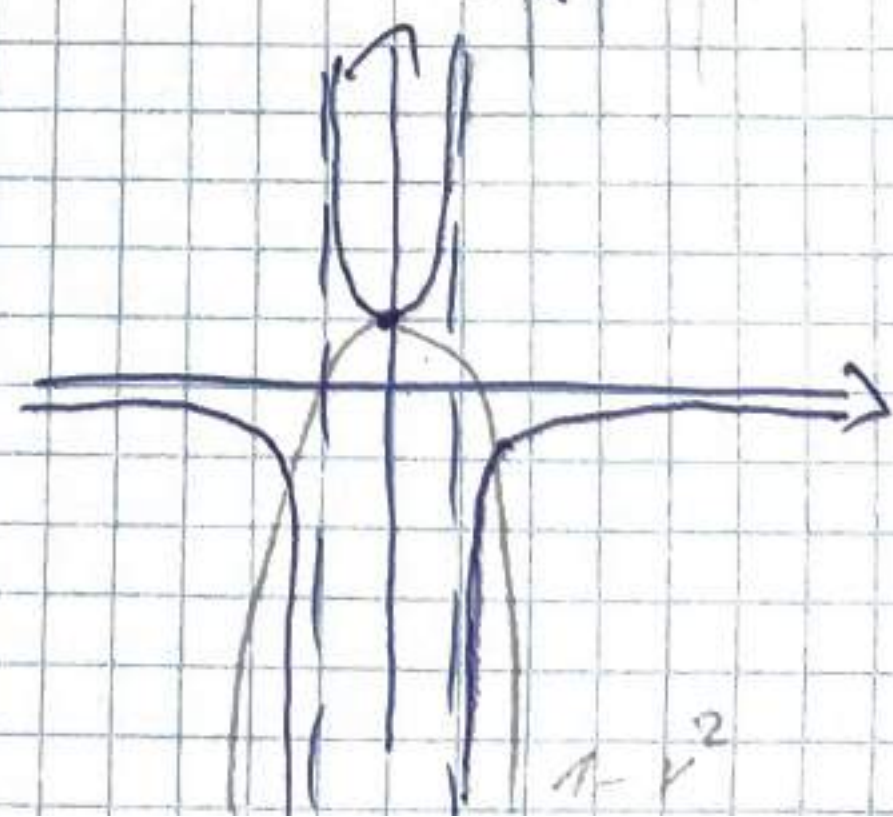


N/257 $y = \frac{2x}{1+x^2}$



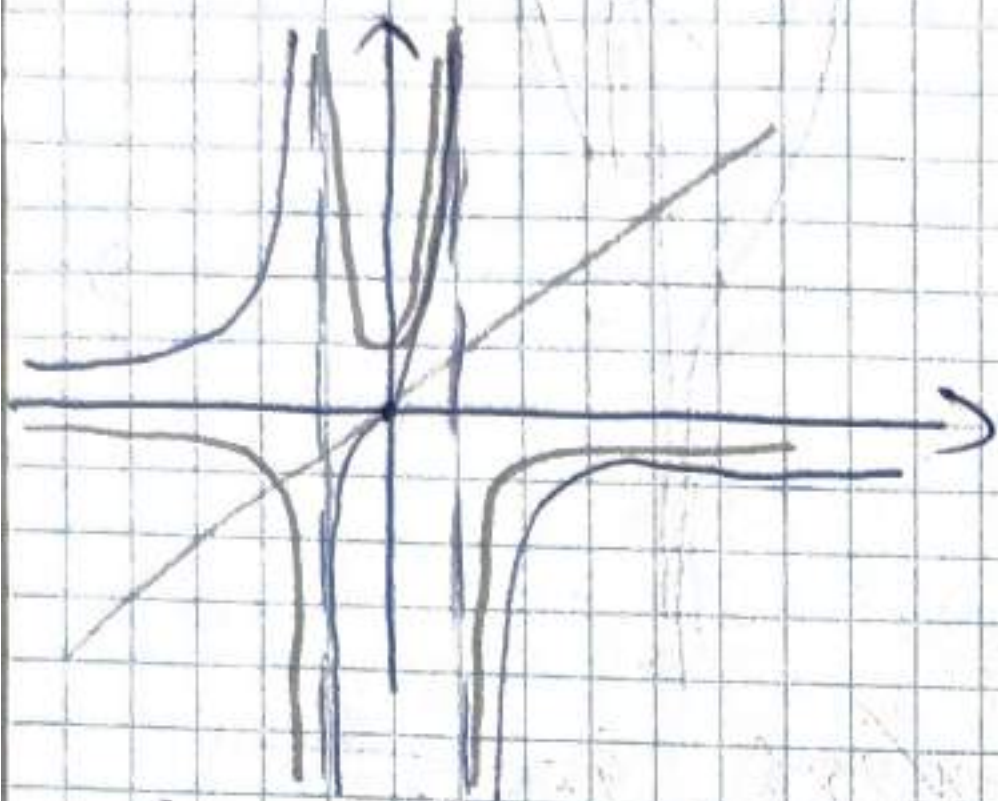
N/258

$y = \frac{1}{1-x^2}$



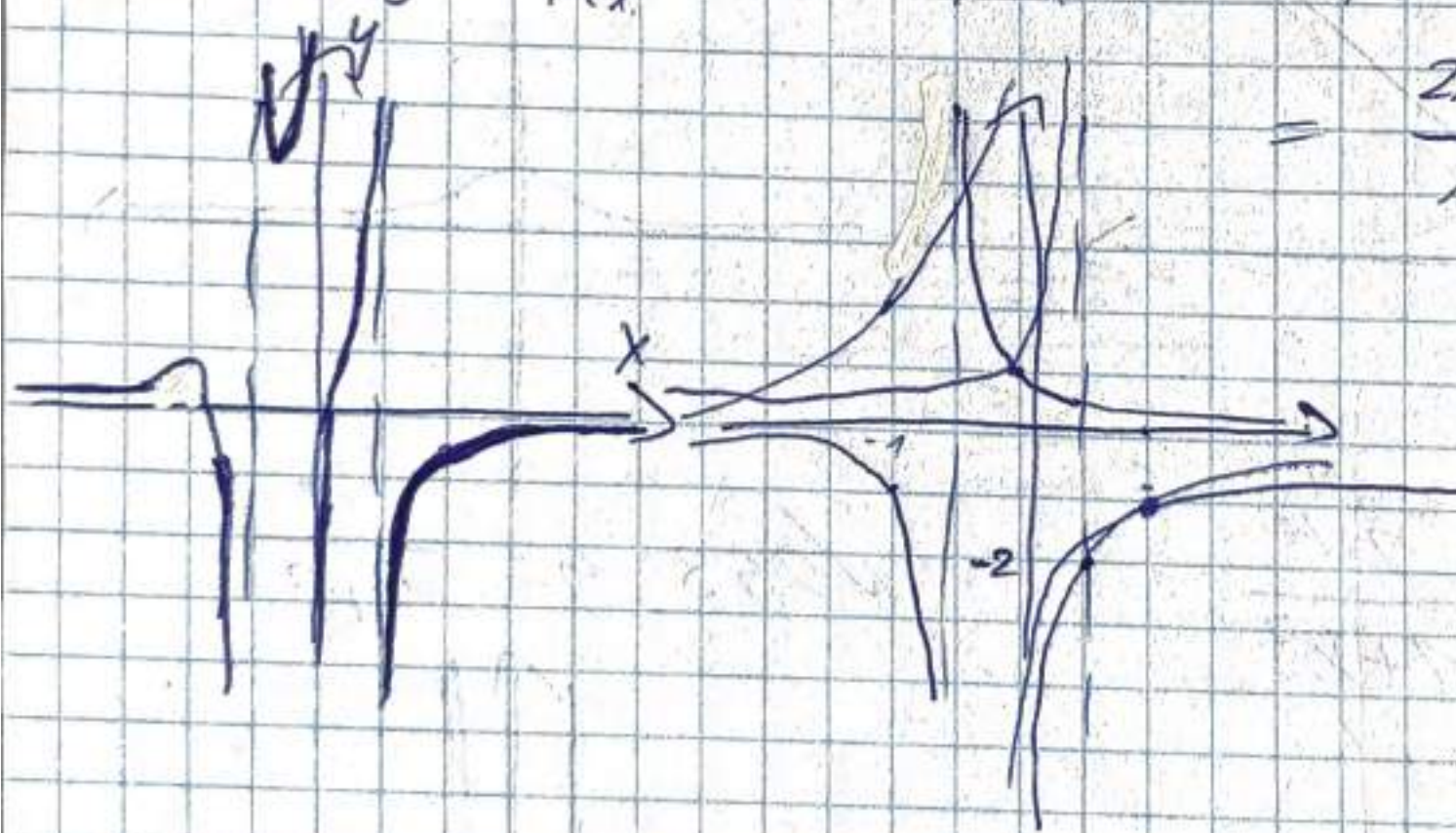
N 259 $y = \frac{x}{1-x^2}$

266-273



N 260

$$y = \frac{1}{1+x} - \frac{2}{x} + \frac{1}{1-x} = \frac{x - x^2 + 2(1-x^2+x^2+x)}{(1+x)(1-x)} = \frac{2x + 2x^2 - 2}{x(1+x)(1-x)}$$



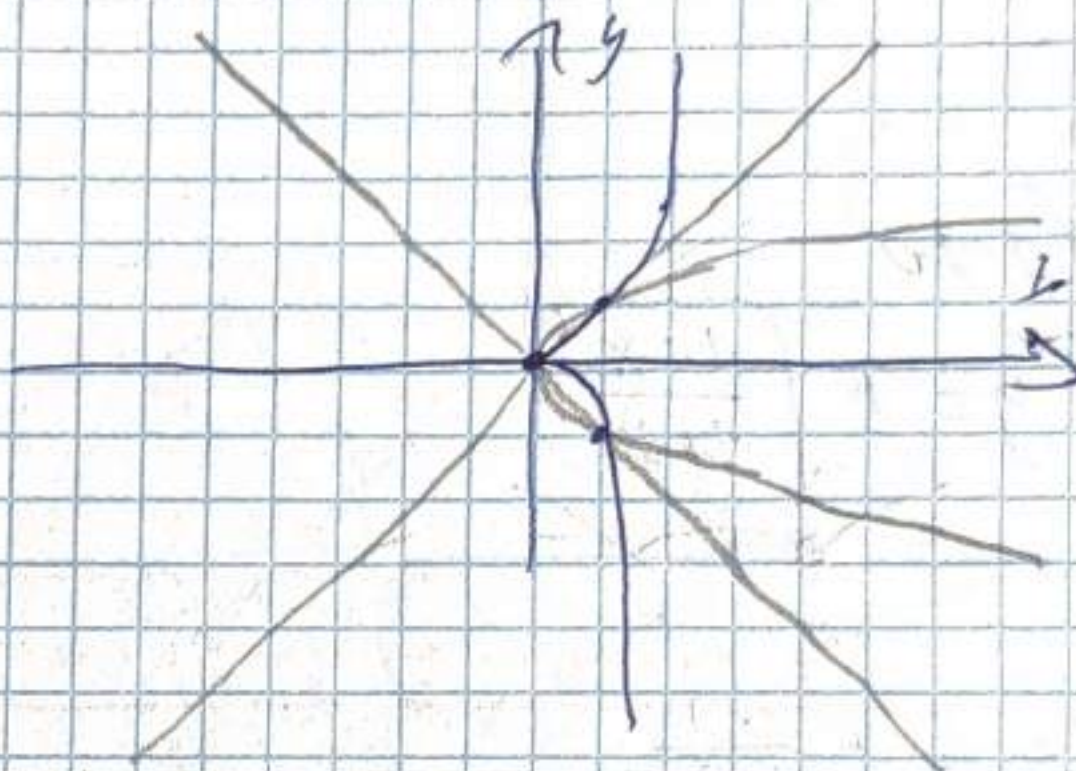
$$\sqrt{266} \quad y = \pm \sqrt{-x-2}$$



$\sqrt{267}$

$$y = \pm x\sqrt{x}$$

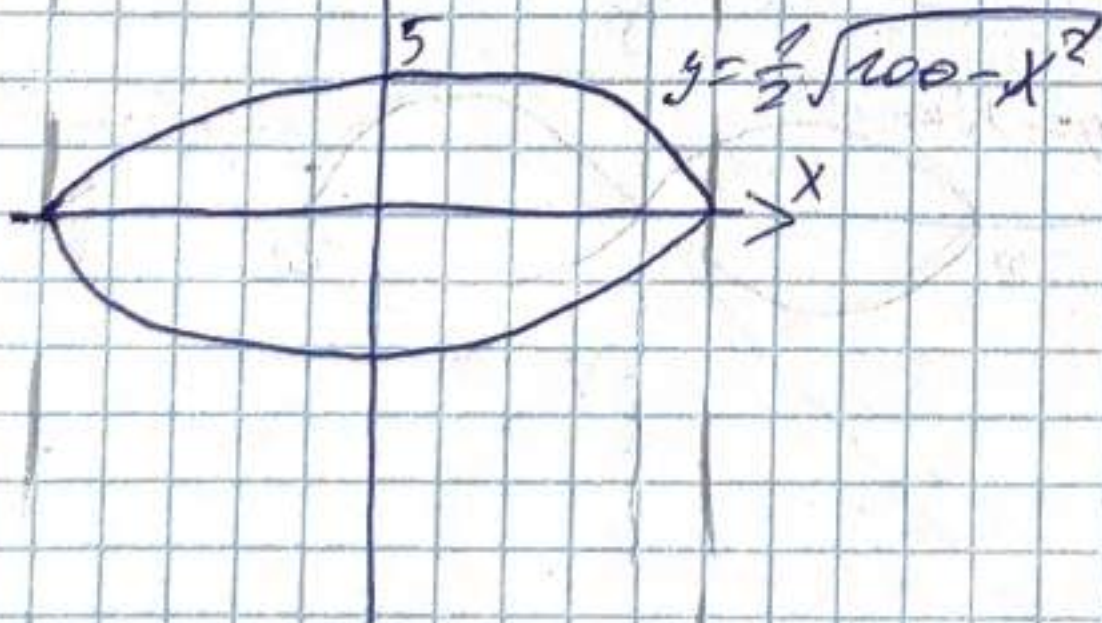
191



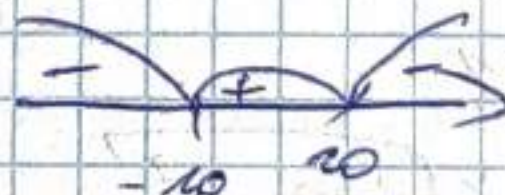
$\sqrt{268}$

$$y = \pm \frac{1}{2} \sqrt{100-x^2}$$

192

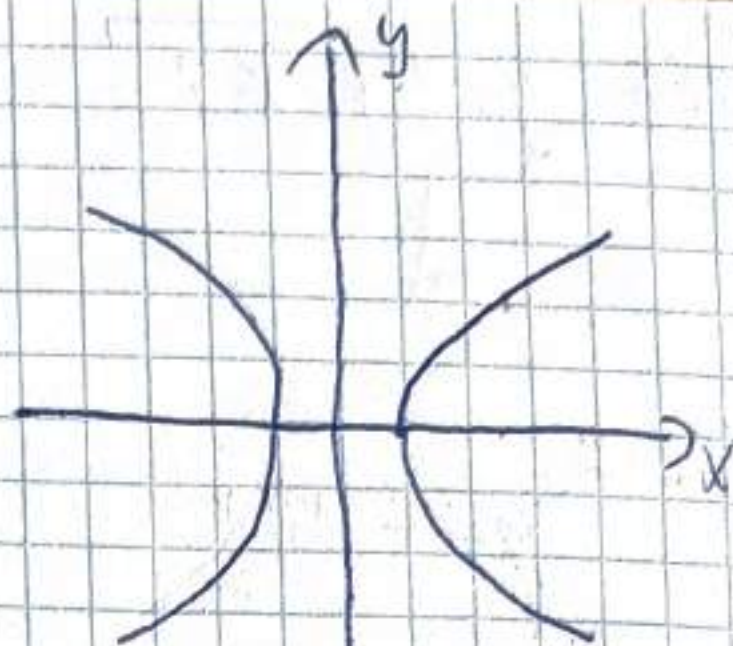


$$(10-x)(10+x) \geq 0$$



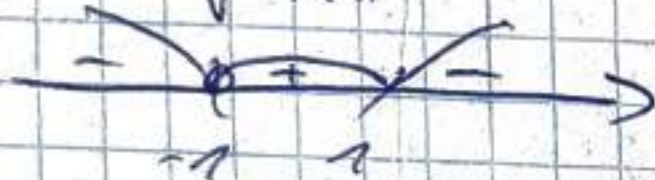
N 269

$$y = \pm \sqrt{x^2 - 1}$$



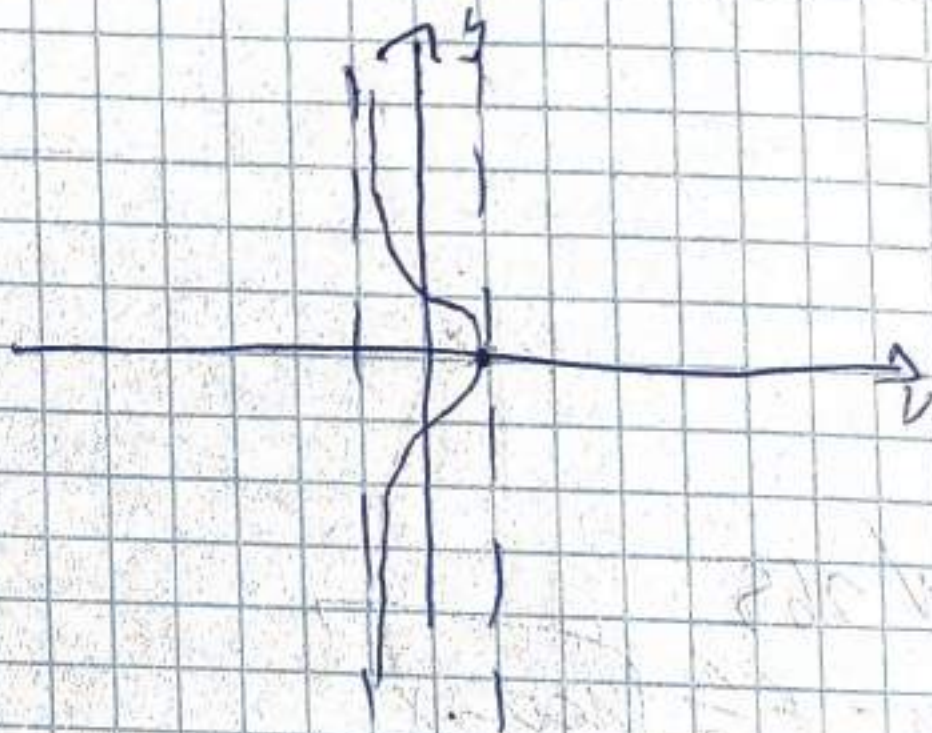
N 270

$$y = \pm \sqrt{\frac{1-x}{1+x}}$$



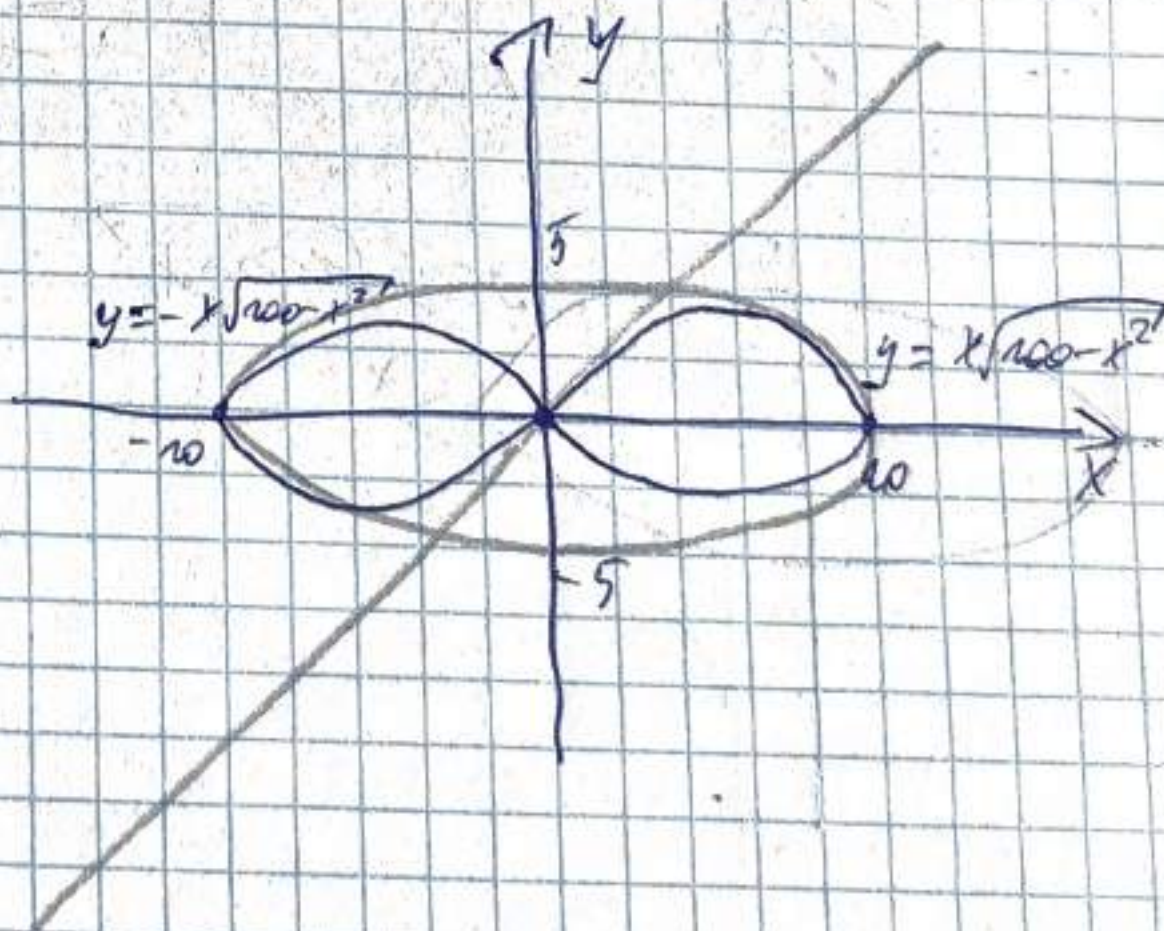
$$x = \frac{1}{2} \pm \frac{1 \cdot 3}{2 \cdot 2}$$

$$x = 0 \pm 1$$



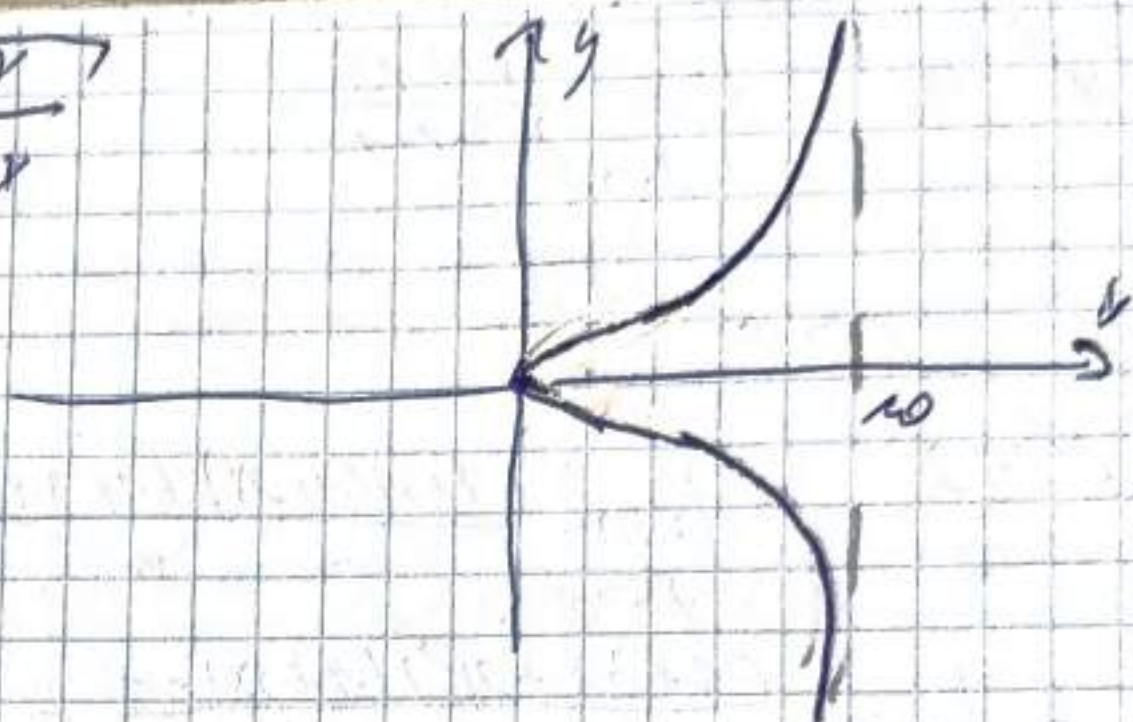
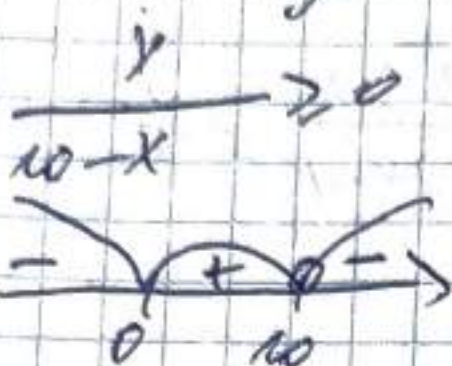
N 280 281

$$y = \pm x \sqrt{100 - x^2}$$



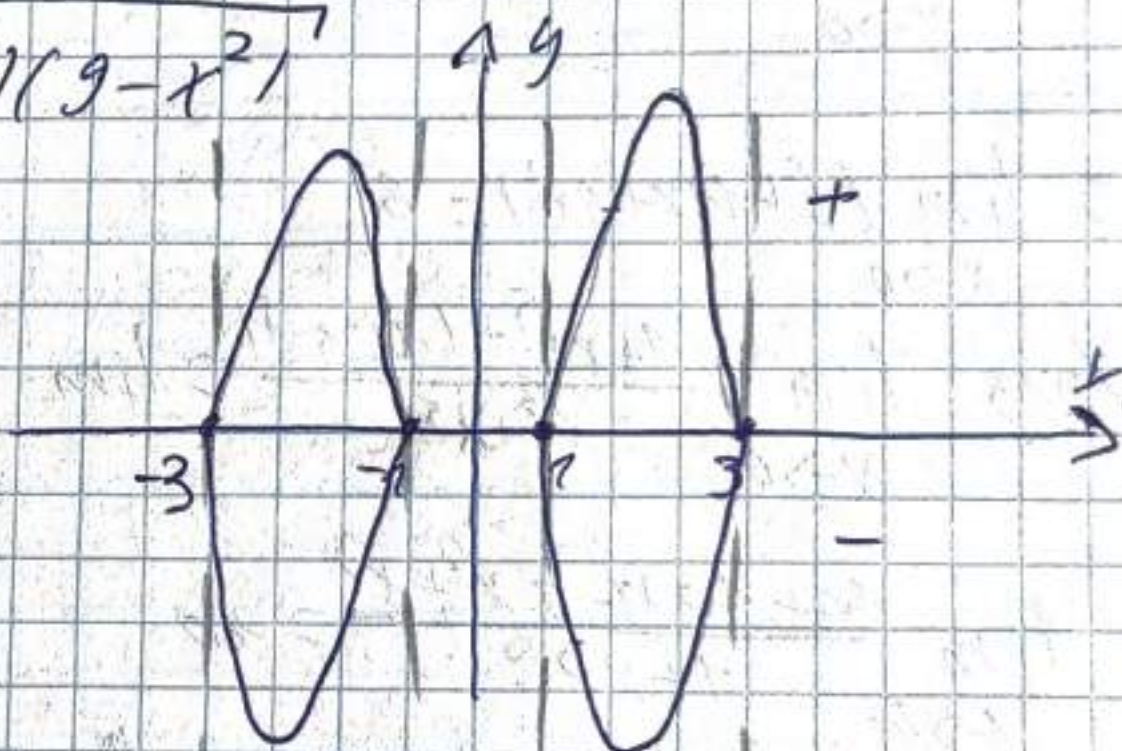
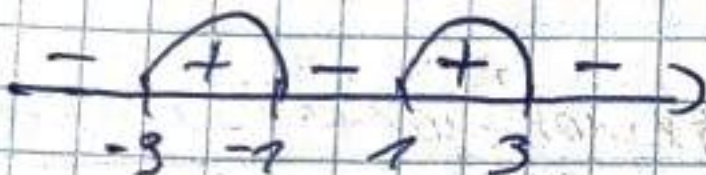
№172

$$y = \pm x \sqrt{\frac{y}{10-x}}$$



№173

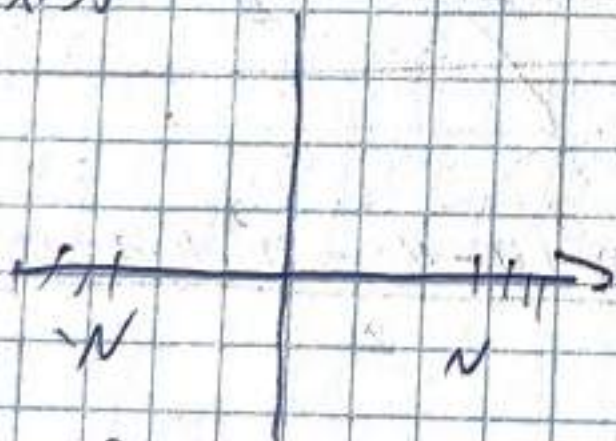
$$y = \pm \sqrt{(x^2-1)(9-x^2)}$$



411-433,

401, 403, 404, 405, 406

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad \forall M > 0 \quad \exists N > 0 \quad \forall x > N \quad |f(x)| > M$$



$$\lim_{x \rightarrow +\infty} f(x) = \infty \quad \forall M > 0 \quad \exists N > 0 \quad \forall x > N \quad |f(x)| > M$$

$$\text{Aufgabe 1} \quad \lim_{x \rightarrow 0} \frac{x^2 - 1}{2x^2 - x - 1} = 1$$

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{2x^2 - x - 1} = \frac{(x-1)(x+1)}{(x-1)(2x+1)} = \frac{2}{3}$$

$$x \rightarrow \infty \quad \frac{1}{2}$$

Aufgabe 2

$$\lim_{x \rightarrow 0} \frac{(1+x)(1+2x)(1+3x) - 1}{x} =$$

$$= \lim_{x \rightarrow 0} \frac{(1+3x+2x^2)(1+3x) - 1}{x} = \lim_{x \rightarrow 0} \frac{1 + 3x + 2x^2 + 3x + 9x^2 + 6x^3 - 1}{x}$$

$$= \lim_{x \rightarrow 0} (6 + 11x + 6x^2) = 6$$

$$\text{Aufgabe 3} \quad \lim_{x \rightarrow 0} \frac{(1+x)^5 - (1+5x)}{x^2 + x^5} = \lim_{x \rightarrow 0} \frac{1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5 - 1 - 5x}{x^2(1+x^3)}$$

$$= \lim_{x \rightarrow 0} \frac{10 + 10x + 5x^2 + x^3}{1+x^3} = 10$$

$$\text{Aufgabe 4} \quad \lim_{x \rightarrow 0} \frac{(1+mx)^n - (1+nx)}{x^2} = \lim_{x \rightarrow 0} \frac{1 + mnx + \frac{n(n-1)}{2} m^2 x^2 + \dots - 1 - nmx}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{mnx + \frac{n(n-1)}{2} m^2 x^2 + \dots - 1 - nmx - \frac{n(n-1)}{2} m^2 x^2 + \dots}{x^2} =$$

$$= \lim_{x \rightarrow 0} \frac{\frac{n(n-1)}{2} m^2}{x^2} - \frac{n(n-1)}{2} m^2 = \frac{nm^2(n-1) - n^2m(n-1)}{2} =$$

$$= \frac{m^2n^2 - m^2n - n^2m^2 + mn^2}{2} = \frac{mn(n-m)}{2}$$

$$\sqrt{419} \quad \lim_{x \rightarrow \infty} \frac{(x-1)(x-2)(x-3)(x-4)(x-5)}{(5x-1)^5} = \frac{1}{5^5}$$

$$\sqrt{416} \quad \lim_{x \rightarrow \infty} \frac{(2x-3)^{20}(3x+2)^{30}}{(2x+1)^{50}} = \left(\frac{3}{2}\right)^{30}$$

$$\sqrt{420} \quad \lim_{x \rightarrow 1} \frac{x^4 - 3x + 2}{x^5 - 4x + 3} = \left(\frac{0}{0}\right) = \lim_{x \rightarrow 1} \frac{(x-1)(x^3 + x^2 + x - 2)}{(x-1)(x^4 + x^3 + x^2 + x - 3)} = \frac{1}{1} = 1$$

$$\begin{array}{r} x^4 - 3x + 2 \quad | \quad x-1 \\ \underline{-(x^4 - x^3)} \\ x^3 - 3x + 2 \\ \underline{-(x^3 - x^2)} \\ x^2 - 3x + 2 \\ \underline{-(x^2 - x)} \\ -2x + 2 \\ \underline{-(-2x + 2)} \\ 0 \end{array}$$

$$\begin{array}{r} x^5 - 4x + 3 \quad | \quad x-1 \\ \underline{-(x^5 - x^4)} \\ x^4 - 4x + 3 \\ \underline{-(x^4 - x^3)} \\ x^3 - 4x + 3 \\ \underline{-(x^3 - x^2)} \\ -x^2 - 4x + 3 \\ \underline{-(x^2 - x)} \\ -3x + 3 \\ \underline{-(-3x + 3)} \\ 0 \end{array}$$

$$\sqrt{422} \quad \lim_{x \rightarrow -1} \frac{x^3 - 2x - 1}{x^5 - 2x - 1} = \lim_{x \rightarrow -1} \frac{x^2 - x - 1}{x^4 - x^3 + x^2 - x - 1} = \frac{+1}{3} = \frac{1}{3}$$

$$\begin{array}{c|c|c|c|c|c|c} x & -2 & 0 & 1 & -1 & & \\ \hline -1 & 1 & & & & & \end{array} \quad \begin{array}{c|c|c|c|c|c|c} 1 & 0 & -2 & -1 & & & \\ \hline -1 & 1 & -1 & -1 & 0 & & \end{array}$$

$$\begin{array}{c|c|c|c|c|c|c} 1 & 0 & 0 & 0 & -2 & -1 & \\ \hline -1 & 1 & -1 & 1 & -1 & -1 & 0 \end{array}$$

$$\sqrt{423} \quad \lim_{x \rightarrow 2} \frac{(x^2 - x - 2)^{20}}{(x^3 - 12x + 16)^{20}} = \lim_{x \rightarrow 2} \left(\frac{(x^2 - x - 2)^2}{x^3 - 12x + 16} \right)^{20} =$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{(x-2)^2(x+4)} = \frac{(x+1)^{20}}{(x+4)^{20}} = \left(\frac{3}{6}\right)^{20}$$

$$\sqrt{424} \quad \lim_{x \rightarrow 1} \frac{x^{100} - 2x + 1}{x^{99} - 2x + 1} = \lim_{x \rightarrow 1} \frac{x^{99} + \dots + x - 1}{x^{99} + x^{98} + \dots + x - 1} = \frac{99 - 1}{99 - 1} = \frac{98}{98}$$

$$\frac{x^{100} - 2x + 1}{x^{99} - 2x + 1} \Big| \frac{x-1}{x^{99} + x^{98} + \dots + x - 1} = \frac{49}{24}$$

$$\frac{x^{99} - 2x + 1}{x^{99} - 2x + 1}$$

$$\frac{x^2 - 2x}{x^2 - x} = \frac{-x + 1}{-x + 1} = 0$$

$$\sqrt{425} \quad \lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x^{m-1} + x^{m-2} + \dots + x + 1)}{(x-1)(x^{n-1} + x^{n-2} + \dots + x + 1)} = \frac{m}{n}$$

$$\sqrt{426} \quad \lim_{x \rightarrow a} \frac{(x^n - a^n) - na^{n-1}(x-a) - \frac{(x-a)(x^{n-1} + ax^{n-2} + \dots + a^{n-1})}{2}}{(x-a)^2}$$

$$\frac{x^n - a^n}{x^n - ax^{n-1}} \Big| \frac{x-a}{x^{n-1} + ax^{n-2} + \dots + a^{n-1}}$$

$$= \lim_{x \rightarrow a} \frac{(x-a)(x^{n-1} + ax^{n-2} + \dots + a^{n-2}x + a^{n-1}) - na^{n-1}(x-a)}{(x-a)^2}$$

$$= \lim_{x \rightarrow a} \frac{(x^{n-1} + ax^{n-2} + \dots + a^{n-2}x + a^{n-1}) - na^{n-1}}{x-a}$$

$$= \lim_{x \rightarrow a} \frac{(x^{n-1} - a^{n-1}) + a(x^{n-2} - a^{n-2}) + \dots + a^{n-2}(x - a) + a^{n-1}(x-a)}{x-a}$$

$$= \lim_{x \rightarrow a} (1x^{n-2} + ax^{n-3} + a^2x^{n-4} + \dots + a^{n-2}) + a(1x^{n-3} + ax^{n-4} + \dots + a^{n-3}) +$$

$$+ a^2(1x^{n-4} + \dots + a^{n-4}) + \dots + a^{n-2} = a^{n-2}((n-1) + (n-2) + (n-3) + \dots + 1) =$$

$$= \frac{n(n-1)}{2} a^{n-2}$$

✓ 401 $\lim_{x \rightarrow 2} x^2 = 4$

ε	0,1	0,01	0,001	0,0001
δ	0,02	0,002	0,0002	0,00002

$$\lim_{x \rightarrow c} f(x) = L \Leftrightarrow (\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D, 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon)$$

$$|x^2 - 4| < \varepsilon \quad |x^2 - 4| = |(x-2)(x+2)| =$$

$$= |x-2||x+2| = |x-2|(|x-2| + 4) \leq |x-2|(|x-2| + 4) =$$

$$= |x-2|^2 + 4|x-2|$$

$$|x^2 - 4| \leq |x^2 - 2| + 4|x-2|$$

$$\delta(0,1) = \sqrt{4,41} - 2 = 0,02$$

$$|x^2 - 2|^2 + 4|x-2| < \varepsilon$$

$$\delta(0,01) = \sqrt{4,01} - 2 = 0,002$$

$$|x-2| < \sqrt{\varepsilon + 4} - 2$$

$$\delta = \sqrt{\varepsilon + 4} - 2$$

✓ 403

a) $\lim_{x \rightarrow a} f(x) = b$

$$\forall \varepsilon > 0 \exists \delta > 0, 0 < |x - a| < \delta$$

$$\lim_{x \rightarrow 1} x^2 = 2$$

$$|f(x) - b| < \varepsilon$$

$$\delta) \lim_{x \rightarrow a-0} f(x) = b$$

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < a - x < \delta \Rightarrow |f(x) - b| < \varepsilon$$

$$\lim_{x \rightarrow 10} f(x) = 15$$

$$f(x) = \begin{cases} x+5, & x < 10 \\ x, & x \geq 10 \end{cases}$$

$$\delta) \lim_{x \rightarrow a+0} f(x) = b$$

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 \leq x - a < \delta \Rightarrow |f(x) - b| < \varepsilon$$

$$\lim_{x \rightarrow 10} f(x) = 10$$

$$f(x) = \begin{cases} x+5, & x < 10 \\ x, & x \geq 10 \end{cases}$$

hoy

$$a) \lim_{x \rightarrow \infty} f(x) = b$$

$$\forall \varepsilon > 0 \exists M = M(\varepsilon) > 0 : |x| > M \Rightarrow |f(x) - b| < \varepsilon$$

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x}{x^2 + x} = 3$$

$$\delta) \lim_{x \rightarrow -\infty} f(x) = b$$

$$\forall \varepsilon > 0 \exists M = M(\varepsilon) > 0 : x < -M \Rightarrow |f(x) - b| < \varepsilon$$

$$\lim_{x \rightarrow +\infty} f(x) = b$$

$$\forall \varepsilon > 0 \exists M = M(\varepsilon) > 0 : x > M \Rightarrow |f(x) - b| < \varepsilon$$

Notes a) $\lim_{x \rightarrow a} f(x) = \infty$

$\forall \varepsilon > 0, \exists \delta > 0 : 0 < x - a < \delta, |f(x)| > \varepsilon$

$$\lim_{x \rightarrow 0} \frac{1}{x} = \infty$$

b) $\lim_{x \rightarrow a} f(x) = -\infty$

$\forall \varepsilon > 0, \exists \delta > 0 : 0 < x - a < \delta, f(x) < -\varepsilon$

$$\lim_{x \rightarrow 0} \left(-\frac{1}{x}\right) = -\infty$$

c) $\lim_{x \rightarrow a} f(x) = +\infty$

$\forall \varepsilon > 0, \exists \delta > 0 : 0 < x - a < \delta, f(x) > \varepsilon$

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

d) $\lim_{x \rightarrow a-0} f(x) = \infty$

$\forall \varepsilon > 0, \exists \delta > 0 : 0 < a - x < \delta, |f(x)| > \varepsilon$

$$\lim_{x \rightarrow 0-0} \frac{1}{x} = \infty$$

e) $\lim_{x \rightarrow a-0} f(x) = -\infty$

$\forall \varepsilon > 0, \exists \delta > 0 : 0 < a - x < \delta, f(x) < -\varepsilon$

$$2) \lim_{x \rightarrow 0-0} f(x) = +\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < x - 0 < \delta, f(x) > \varepsilon$$

$$2) \lim_{x \rightarrow 0+0} f(x) = -\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < x - 0 < \delta, f(x) < -\varepsilon$$

$$3) \lim_{x \rightarrow 0+0} f(x) = +\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < x < \delta, f(x) > \varepsilon$$

$$4) \lim_{x \rightarrow 0-0} f(x) = -\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < x < \delta, f(x) < -\varepsilon$$

non

$$1) \lim_{x \rightarrow \infty} f(x) = \infty$$

$$\forall \varepsilon > 0, \exists \delta > 0 : x > \delta, f(x) > \varepsilon$$

$$2) \lim_{x \rightarrow \infty} f(x) = -\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0 : x > \delta, f(x) < -\varepsilon$$

$$3) \lim_{x \rightarrow -\infty} f(x) = +\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0 : x < -\delta, f(x) > \varepsilon$$

$$2) \lim_{x \rightarrow -\infty} f(x) = \infty$$

$$\forall \varepsilon > 0, \exists \delta > 0: x < -\delta, |f(x)| > \varepsilon$$

$$8) \lim_{x \rightarrow -\infty} f(x) = -\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0: x < -\delta, f(x) < -\varepsilon$$

$$9) \lim_{x \rightarrow -\infty} f(x) = +\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0: x < -\delta, f(x) > \varepsilon$$

$$10) \lim_{x \rightarrow +\infty} f(x) = \infty$$

$$\forall \varepsilon > 0, \exists \delta > 0: x > \delta, |f(x)| > \varepsilon$$

$$3) \lim_{x \rightarrow +\infty} f(x) = -\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0: x > \delta, f(x) < -\varepsilon$$

$$11) \lim_{x \rightarrow +\infty} f(x) = +\infty$$

$$\forall \varepsilon > 0, \exists \delta > 0: x > \delta, f(x) > \varepsilon$$

$$\lim_{n \rightarrow \infty} \frac{(1+x)(1+x^2) \dots (1+x^n)}{(1+x)^{\frac{n+1}{2}}} = \frac{1}{n^{\frac{n+1}{2}}} = n^{-\frac{n+1}{2}}$$

8v18

$$\lim_{x \rightarrow 3} \frac{x^2 + 5x + 6}{x^2 - 8x + 15} = \lim_{x \rightarrow 3} \frac{x-2}{x-5} = -\frac{1}{2}$$

$$\text{Aufg 1} \quad \lim_{x \rightarrow 1} \frac{x^3 - 3x + 2}{x^2 - 4x + 3} = \lim_{x \rightarrow 1} \frac{(x-1)^2(x+2)}{(x-1)^2(x^2+2x+3)} = \frac{3}{8} = \frac{1}{2}$$

$$\begin{array}{c|c|c|c|c} 1 & 0 & -3 & 2 & \\ \hline 1 & 1 & 1 & -2 & 0 \\ \hline 1 & 1 & 2 & 0 & \end{array} \quad \begin{array}{c|c|c|c|c|c} 1 & 0 & 0 & -4 & 3 & \\ \hline 1 & 1 & 1 & 1 & -3 & 0 \\ \hline 1 & 1 & 2 & 3 & 0 & \end{array}$$

$$\text{Aufg 2} \quad \lim_{x \rightarrow 2} \frac{x^3 - 2x^2 - 4x + 8}{x^4 - 8x^2 + 16} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2-4)}{(x-2)(x^3+2x^2-4x-8)} =$$

$$\begin{array}{c|c|c|c|c} 1 & -2 & -4 & 8 & \\ \hline 2 & 1 & 0 & -4 & 0 \\ \hline 2 & 1 & 0 & & \end{array} \quad \begin{array}{c|c|c|c|c|c} 1 & 0 & -8 & 0 & 16 & \\ \hline 2 & 1 & 2 & -4 & -8 & 0 \\ \hline 2 & 1 & 4 & -4 & 0 & 0 \end{array}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)(x^2+4x+4)} = \frac{4}{16} = \frac{1}{4}$$

$$\text{Aufg 3} \quad \lim_{x \rightarrow 1} \frac{x^{n+1} - (n+1)x + n}{(x-1)^2} =$$

$$y = x - 1 \quad \lim_{y \rightarrow 0} \frac{(y+1)^{n+1} - (n+1)y + 1 - n}{y^2} =$$

$$= \lim_{y \rightarrow 0} \frac{(1+y)^{n+1} - y(n+1) - 1}{y^2} = \lim_{y \rightarrow 0} \frac{1}{y^2} \left(y^{n+1} + (n+1)y^n + \dots + \frac{n(n+1)}{2}y^2 + \right.$$

$$\left. + (n+1)y + 1 - y(n+1) - 1 \right) = \lim_{y \rightarrow 0} \left(y^{n-1} + (n+1)y^{n-2} + \dots + \frac{n(n+1)}{2} \right) =$$

$$= \frac{n(n+1)}{2}$$

$$N429 \quad \lim_{n \rightarrow \infty} \frac{1}{n} \left[\left(x + \frac{a}{n}\right) + \left(x + \frac{2a}{n}\right) + \dots + \left(x + \frac{(n-1)a}{n}\right) \right] =$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(x(n-1) + \frac{a}{n} (1+2+\dots+n-1) \right) =$$

$$= \lim_{n \rightarrow \infty} \frac{xn - x}{n} + \frac{a}{n} \cdot \left(\frac{1+n-1}{2} \cdot (n-1) \right) = \lim_{n \rightarrow \infty} \frac{xn - x}{n} + \frac{an(n-1)}{2n} =$$

$$= \lim_{n \rightarrow \infty} \frac{n-1}{n} \left(x + \frac{a}{2} \right) = x + \frac{a}{2}$$

$$N430 \quad \lim_{n \rightarrow \infty} \frac{1}{n} \left[\left(x + \frac{a}{n}\right)^2 + \left(x + \frac{2a}{n}\right)^2 + \dots + \left(x + \frac{(n-1)a}{n}\right)^2 \right] =$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left[(n-1)x^2 + \frac{2a}{n} (1+2+\dots+n-1) + \left(\frac{a}{n}\right)^2 (1^2+2^2+\dots+(n-1)^2) \right] =$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left((n-1)x^2 + \frac{2a}{n} \cdot \frac{n(n-1)}{2} + \frac{a^2}{n^3} \cdot \frac{n(n-1)(2n-1)}{6} \right) =$$

$$= \lim_{n \rightarrow \infty} \frac{n-1}{n} \left(x^2 + a + \frac{a^2 (2 - \frac{1}{n})}{6} \right) = x^2 + a + \frac{a^2 (2+0)}{6} =$$

$$= x^2 + a + \frac{a^2}{3} \quad 28, 31-33$$

$$N433 \quad \lim_{n \rightarrow \infty} \frac{1^3 + 4^3 + 7^3 + \dots + (3n-2)^3}{[1+4+7+\dots+(3n-2)]^2} =$$

$$1^3 + 4^3 + 7^3 + \dots + (3n-2)^3 = (3 \cdot 1 - 2)^3 + (3 \cdot 2 - 2)^3 + \dots + (3n-2)^3 =$$

$$= 3^3 (1^3 + 2^3 + \dots + n^3) - 3 \cdot 9 \cdot 2 (1^2 + 2^2 + \dots + n^2) + 8n =$$

$$= 27(1+2+n)^2 - \frac{54n(n+1)(2n+1)}{6} + \frac{36n(n+1)}{2} + 84$$

$$= \lim_{n \rightarrow \infty} \frac{27 \frac{n^2(n+1)^2}{4} - 9n(n+1)(2n+1) + 18n(n+1) + 84}{\left(\frac{1+3n-2}{2} \cdot n\right)^2} =$$

$$= \lim_{n \rightarrow \infty} \frac{27 \cdot x}{x \cdot 9} = 3$$

$$\sqrt{435} \quad \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + \sqrt{x} + \sqrt{x}}{\sqrt{1+x}} = 1$$

$$\sqrt{437} \quad \lim_{x \rightarrow 4} \frac{\sqrt{1+2x} - 3}{\sqrt{x} - 2} = \lim_{x \rightarrow 4} \frac{2(x-4)(\sqrt{x}+2)}{(x-4)(\sqrt{1+2x}+3)} =$$

$$= \lim_{x \rightarrow 4} \frac{8}{5} = \frac{8}{5}$$

$$\sqrt{438} \quad \lim_{x \rightarrow -8} \frac{\sqrt{1-x} - 3}{2 + \sqrt[3]{x}} = \lim_{x \rightarrow -8} \frac{-1(x+8)(4 - 2\sqrt{x} + \sqrt{x^2})}{(\sqrt{1-x}+3)(8+x)} = \frac{12}{6} = 2$$

$$\sqrt{439} \quad = \lim_{x \rightarrow a} \frac{\sqrt{x} - \sqrt{a} + \sqrt{x-a}}{\sqrt{x^2 - a^2}} = \lim_{x \rightarrow a} \frac{x-a + \sqrt{x-a}(\sqrt{x} + \sqrt{a})}{\sqrt{x^2 - a^2}(\sqrt{x} + \sqrt{a})} =$$

$$= \lim_{x \rightarrow 7} \frac{\sqrt{x+2} - \sqrt[3]{x+20}}{\sqrt[4]{x+9} - 2} = \left| \frac{0}{0} \right| = \lim_{x \rightarrow 7} \left(\frac{\sqrt{x+2} - 3}{\sqrt[4]{x+9} - 2} - \frac{\sqrt[3]{x+20} - 3}{\sqrt[4]{x+9} - 2} \right) =$$

$$= \lim_{x \rightarrow 7} \left(\frac{(x-7)(\sqrt{x+9}+2)(\sqrt{x+9}+2)}{(\sqrt{x+2}+3)(x-7)} - \frac{(x-7)(\sqrt{x+9}+2)(\sqrt{x+9}+2)}{(3\sqrt{(x+2)^2}+3\sqrt{x+2}+9)(x-7)} \right) =$$

$$= \lim_{x \rightarrow 7} 32 \left| \frac{1}{6} - \frac{1}{27} \right| = \frac{32}{54} |9-2| = \frac{112}{27}$$

$$\text{N 451} \quad \lim_{x \rightarrow 0} \frac{x^2}{\sqrt[5]{1+5x} - (1+x)} = \lim_{x \rightarrow 0} \frac{x^2}{1+x + \frac{1}{5}(1-\frac{4}{5})25x^2 - 1 - x} =$$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \dots$$

$$(1+5x)^{\frac{1}{5}} = 1 + x + \frac{\frac{1}{5}(1-\frac{4}{5})}{2} 25x^2 + \dots$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{x + x + \frac{1-41}{2}x^2 - x - x + \dots} = -\frac{1}{2}$$

$$\text{N 452} \quad \lim_{x \rightarrow 0} \frac{\sqrt[m]{1+\alpha x} - \sqrt[n]{1+\beta x}}{x} = \frac{1 + \frac{\alpha x}{m} - (1 + \frac{\beta x}{n})}{x} =$$

$$(1+x)^\alpha - 1 \sim \alpha x \quad (1+x)^\beta \sim 1 + \beta x \quad = \frac{\alpha}{m} - \frac{\beta}{n}$$

$$\sqrt[m]{1+\alpha x} = (1+\alpha x)^{\frac{1}{m}} = 1 + \alpha x \cdot \frac{1}{m} + \dots - (1+\beta x)^{\frac{1}{n}}$$

$$\text{N 455} \quad \lim_{x \rightarrow 1} \frac{3}{1-\sqrt{x}} - \frac{3}{1-\sqrt[3]{x}}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt[m]{x} - 1}{\sqrt[n]{x} - 1} = \left| \begin{array}{l} x-1=t \\ x=t+1 \\ t \rightarrow 0 \end{array} \right| = \lim_{t \rightarrow 0} \frac{\sqrt[m]{t+1} - 1}{\sqrt[n]{t+1} - 1} = \frac{\frac{1}{m}}{\frac{1}{n}} = \frac{n}{m}$$

$$\text{Aufg. 5.2} \quad \lim_{x \rightarrow 1} \left(\frac{3}{1-\sqrt{x}} - \frac{2}{1+\sqrt{x}+\sqrt{x^2}} \right) = \lim_{x \rightarrow 1} \left(\frac{3(1+\sqrt{x})}{1-x} - \frac{2(1+\sqrt{x}+\sqrt{x^2})}{1-x} \right)$$

$$= \lim_{x \rightarrow 1} \frac{3+3\sqrt{x}+2-2\sqrt{x}-2\sqrt{x^2}}{1-x} = \frac{1+3\sqrt{x}-2\sqrt{x}-2\sqrt{x^2}}{1-x}$$

$$= \left| \begin{array}{l} x=t^6 \\ t=\sqrt[6]{x} \\ t \rightarrow 1 \end{array} \right| = \lim_{t \rightarrow 1} \frac{1+3t^3-2t^2-2t^4}{1-t^6} = \lim_{t \rightarrow 1} \frac{-2t^3+t^2-t-1}{1+t+\dots+t^5} = \frac{-3}{6} = -\frac{1}{2}$$

$$\text{Aufg. 58} \quad \lim_{x \rightarrow \infty} \sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x} =$$

$$= \lim_{x \rightarrow \infty} \frac{x + \sqrt{x + \sqrt{x}} - x}{\sqrt{x + \sqrt{x + \sqrt{x}}} + \sqrt{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{1}{\sqrt{x}}}}{\sqrt{x} \left(\sqrt{1 + \frac{1}{\sqrt{x}}} \sqrt{1 + \frac{1}{\sqrt{x}}} + 1 \right)} = \frac{1}{2}$$

$$\text{Aufg. 59} \quad \lim_{x \rightarrow +\infty} x(\sqrt{x^2+2x} - 2\sqrt{x^2+x} + x) = \lim_{x \rightarrow +\infty}$$

$$= \lim_{x \rightarrow +\infty} x \frac{x^2+2x-2x^2-2x}{\sqrt{x^2+2x}+2\sqrt{x^2+x}}$$

$$= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^2+2x}+x)^2 - 4(x^2+x)}{(\sqrt{x^2+2x}+x)+2\sqrt{x^2+x}} \cdot x =$$

$$= \lim_{x \rightarrow +\infty} x \frac{\sqrt{x^2+2x}+x}{x(\sqrt{1+\frac{2}{x}}+1)+2x\sqrt{1+\frac{1}{x}}} = \frac{1}{4} \lim_{x \rightarrow +\infty} (x^2+2x+2x\sqrt{x^2+2x} - 4x^2-4x) =$$

$$= \frac{1}{2} \lim_{x \rightarrow +\infty} (-x^2-x+x\sqrt{x^2+2x}) =$$

$$= \frac{1}{2} \lim_{x \rightarrow \infty} \frac{x \sqrt{x^2 + 2x} + x^2}{x^2 + 1}$$

$$= \frac{1}{2} \lim_{x \rightarrow \infty} \frac{x^4 + 2x^3 + x^2 - x^4 - 2x^3}{x \sqrt{x^2 + 2x} + x^2 + 1} = \frac{1}{2} \lim_{x \rightarrow \infty} \frac{-x^2}{x^2 \left(\sqrt{1 + \frac{2}{x}} + 1 + \frac{1}{x} \right)} =$$

$$= \frac{1}{4}$$

Sc 478

$$\lim_{x \rightarrow 1} \left(\frac{n}{1-x^n} - \frac{m}{1-x^m} \right) = \lim_{x \rightarrow 1} \left(\frac{n}{x^n-1} - \frac{m}{x^m-1} \right) =$$

$$= \lim_{\substack{x-1=t \\ x=t+1 \\ t \rightarrow 0}} \left(\frac{n}{(t+1)^n-1} - \frac{m}{(t+1)^m-1} \right) =$$

$$= \lim_{t \rightarrow 0} \frac{n(t+1)^m - m(t+1)^n}{((t+1)^m-1)((t+1)^n-1)}$$

$$(t+1)^n - 1 = 1 + nt + \frac{n(n-1)}{2} t^2 + \dots - 1$$

$$\lim_{t \rightarrow 0} \left(\frac{n \left(1 + nt + \frac{m(m-1)}{2} t^2 + \dots \right) - m \left(1 + nt - \frac{n(n-1)}{2} t^2 + \dots \right)}{\left(1 + nt + \frac{n(n-1)}{2} t^2 + \dots \right) \left(1 + mt - \frac{m(m-1)}{2} t^2 + \dots \right)} \right) =$$

$$= \lim_{t \rightarrow 0} \frac{t \left(\frac{nm(m-1)}{2} + \dots \right) - \left(\frac{mn(n-1)}{2} + \dots \right)}{nm \left(1 + \frac{n-1}{2} t + \dots \right) \left(1 + \frac{m-1}{2} t + \dots \right)} =$$

$$= \lim_{t \rightarrow 0} \frac{m-1}{2} - \frac{n-1}{2} = \frac{m-n}{2}$$

Nasib $y = \arcsin \frac{2x}{1+x^2}$

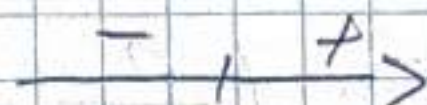
$\frac{2x}{1+x^2} \in [-1, 1] \quad \forall x$

$-1 \leq \frac{2x}{1+x^2} \leq 1$

$-1-x^2 \leq 2x \leq 1+x^2$

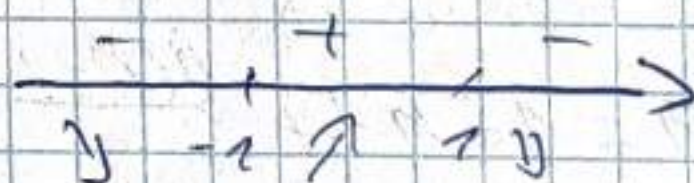
$D_y = \mathbb{R}$, representation at $y=0$

$x=0 \Leftrightarrow y=0$

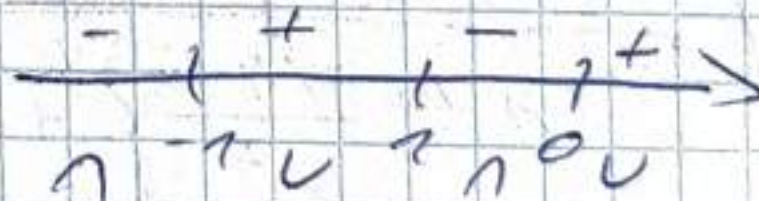


$y' = \frac{1}{\sqrt{1 - \frac{4x^2}{(1+x^2)^2}}} \cdot \frac{2(x^2+1) - 4x^2}{(1+x^2)^2} = \frac{2(1-x^2)}{\sqrt{(1-x^2)^2} (1+x^2)}$

$= \frac{2 \operatorname{sgn}(1-x^2)}{1+x^2}$

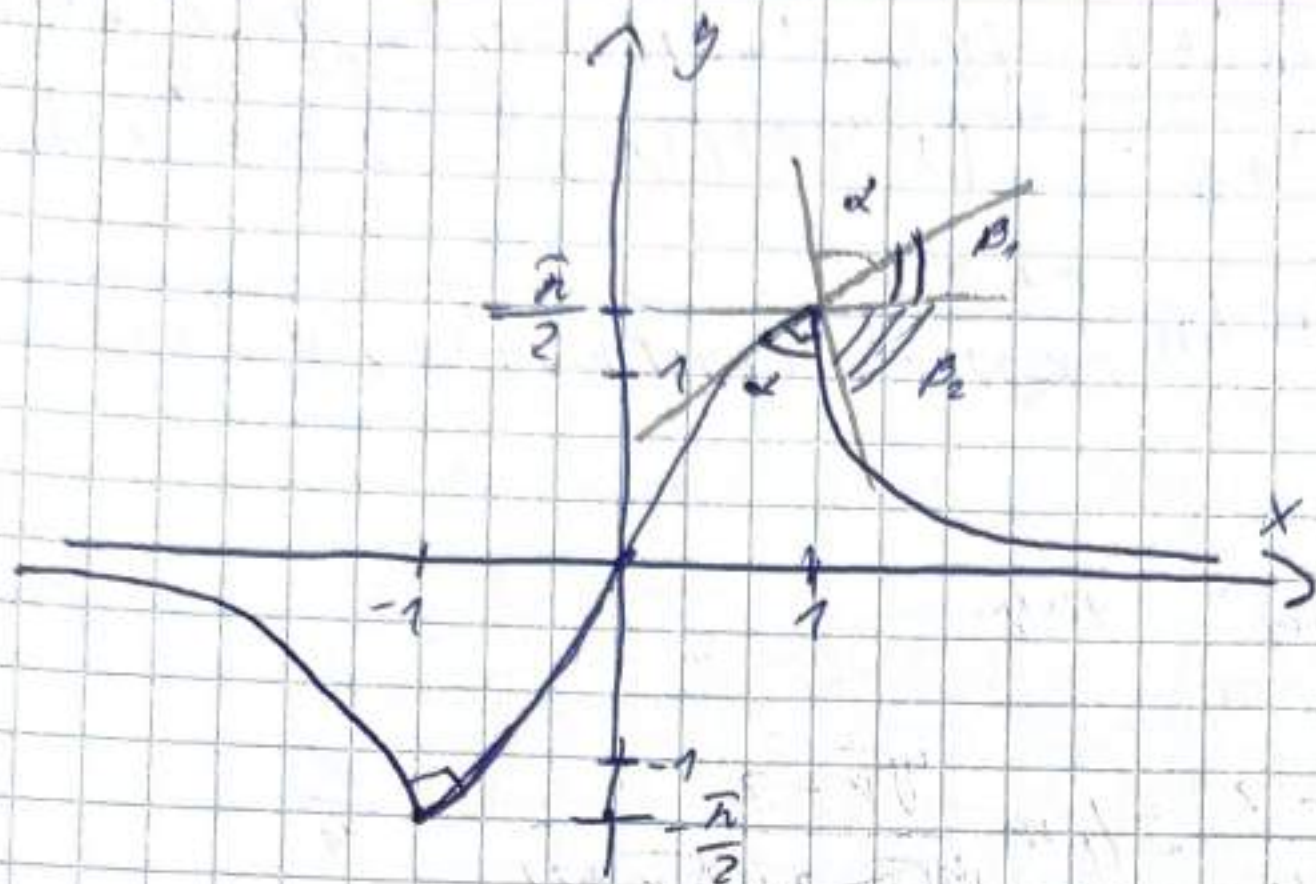


$y'' = \frac{2 \operatorname{sgn}(1-x^2) \cdot (-2x)}{(1+x^2)^2}$



x	$-\infty, -1$	-1	-1, 0	0	0, 1	1	$1, +\infty$
y	-	min	-	0	+	max	+
y'	$\searrow$	$-\frac{\pi}{2}$ never	$\nearrow$	0 never	$\nearrow$	$\frac{\pi}{2}$ never	$\searrow$
y''	$\cap$		$\cup$		$\cap$		$\cup$

$y \rightarrow$



$$B_1: y'(1-0) = \frac{2 \operatorname{sgn}(1-x^2)}{1+x^2} = 1 \quad \operatorname{tg} B_1 = 1 \Rightarrow B_1 = 45^\circ$$

$$B_2: y'(1+0) = -\frac{2}{2} = -1 \quad B_2 = -45^\circ$$

$$45 - (-45) = 90 = \alpha$$

$$\sqrt{436} \quad \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x}}{\sqrt{2x+1}} = \lim_{x \rightarrow +\infty} \frac{x^{\frac{1}{2}} + x^{\frac{1}{3}} + x^{\frac{1}{4}}}{\sqrt{2x+1}} =$$

$$= \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \left(1 + \frac{1}{x^{\frac{1}{6}}} + \frac{1}{x^{\frac{5}{12}}} \right)}{\sqrt{x} \sqrt{2 + \frac{1}{x}}} = \frac{1}{\sqrt{2}}$$

$$\sqrt{440} \quad \lim_{x \rightarrow 3} \frac{\sqrt{x+13} - 2\sqrt{x+1}}{x^2-9} = \lim_{x \rightarrow 3} \frac{(\sqrt{x+13} - 2\sqrt{x+1})(x^2+9)}{x^4-81}$$

$$= \lim_{x \rightarrow 3} \frac{(\sqrt{x+13} - 2\sqrt{x+1})(\sqrt{x+13} + 2\sqrt{x+1})}{(x+3)(x-3)(\sqrt{x+13} + 2\sqrt{x+1})} = \frac{x+13-2(x+1)}{(x+3)(x-3)(\sqrt{x+13} + 2\sqrt{x+1})} =$$

$$= \lim_{x \rightarrow 3} \frac{-3(\sqrt{x+13} + 2\sqrt{x+1})}{(x+3)(x-3)(\sqrt{x+13} + 2\sqrt{x+1})} = \frac{-3}{6(4+4)} = -\frac{1}{16}$$

$$\text{N447} \quad \lim_{x \rightarrow -2} \frac{\sqrt[3]{x-6}+2}{x^3+8} = \frac{(\sqrt[3]{x-6}+2)(x-6)^{\frac{2}{3}}-2\sqrt[3]{x-6}+4}{(x^3+8)(x-6)^{\frac{2}{3}}-2\sqrt[3]{x-6}+4}$$

$$= \lim_{x \rightarrow -2} \frac{x-6+8}{1 \cdot 1 \cdot 1} = \lim_{x \rightarrow -2} \frac{x+2}{(x+2)(x^2-2x+4)(x-6)^{\frac{2}{3}}-2\sqrt[3]{x-6}+4}$$

$$= \lim_{x \rightarrow -2} \frac{1}{12 \cdot 1 \cdot 4 + 4 + 4} = \frac{1}{164}$$

$$\text{N442} \quad \lim_{x \rightarrow 16} \frac{\sqrt[4]{x}-2}{\sqrt{x}-4} = \lim_{x \rightarrow 16} \frac{\sqrt[4]{x}-2}{(\sqrt{x}-2)(\sqrt{x}+2)} = \frac{1}{4}$$

$$\text{N443} \quad \lim_{x \rightarrow 8} \frac{\sqrt{9+2x}-5}{\sqrt[3]{x}-2} = \lim_{x \rightarrow 8} \frac{(\sqrt{9+2x}-5)(\sqrt{9+2x}+5)(x^{\frac{2}{3}}+2\sqrt[3]{x}+4)}{(x-8)(\sqrt{9+2x}+5)}$$

$$= \lim_{x \rightarrow 8} \frac{2(x-8)(x^{\frac{2}{3}}+2\sqrt[3]{x}+4)}{(x-8)(\sqrt{9+2x}+5)} = \frac{24}{20} = \frac{12}{5}$$

$$\text{N444} \quad \lim_{x \rightarrow 0} \frac{\sqrt[n]{x+1}-1}{x} = \left| \begin{array}{l} x+1=t \\ t \rightarrow 1 \end{array} \right| = \lim_{t \rightarrow 1} \frac{\sqrt[n]{t}-1}{\sqrt[n]{t^n}-1} =$$

$$= \lim_{t \rightarrow 1} \frac{\sqrt[n]{t}-1}{t-1} \cdot \frac{t^{\frac{n-1}{n}}+t^{\frac{n-2}{n}}+\dots+1}{t^{\frac{n-1}{n}}+t^{\frac{n-2}{n}}+\dots+1} = \frac{1}{n}$$

$$\text{N445} \quad \lim_{x \rightarrow 0} \frac{1-2x-x^2-(1+x)}{x} = \lim_{x \rightarrow 0} \frac{-x^2-2x+1-(1+x)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{-2x(x+1)}{x(\sqrt{1-2x-x^2}+(1+x))} = \lim_{x \rightarrow 0} \frac{-2x-4}{\sqrt{1-2x-x^2}+(1+x)} = \frac{-4}{2} = -2$$

√446

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{8+3x-x^2} - 2}{x+x^2} = \frac{8+3x-x^2-8}{x(x+1)(8+3x-x^2)^{\frac{2}{3}}+2(8-x^2)^{\frac{2}{3}}+4}$$

$$= \lim_{x \rightarrow 0} \frac{3-x}{(1+x)(8+3x-x^2)^{\frac{2}{3}}+2(8-x^2)^{\frac{2}{3}}+4} = \frac{3}{1(4+4+4)} = \frac{1}{4}$$

√447

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{x+27} - \sqrt[3]{27-x}}{x+2\sqrt[3]{x^3}} = \lim_{x \rightarrow 0} \frac{2x}{x(1+2\sqrt[3]{x})((27+x)^{\frac{2}{3}}+...+)}$$

$$= \lim_{x \rightarrow 0} \frac{2}{(1+2\sqrt[3]{x})((x+27)^{\frac{2}{3}}+(x+27)^{\frac{1}{3}}(27-x)^{\frac{1}{3}}+(27-x)^{\frac{2}{3}})}$$

$$= \frac{2}{1 \cdot (9+9+9)} = \frac{2}{27}$$

√448

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt[3]{1+x} - \sqrt[3]{1-x}} = \lim_{x \rightarrow 0} \frac{(1+x-\sqrt{1-x})(1+x)^{\frac{2}{3}}+...+1}{(1+x-\sqrt{1-x})(\sqrt{1+x}+\sqrt{1-x})}$$

$$= \frac{3}{2}$$

√450

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+\frac{x}{3}} - \sqrt{1+\frac{x}{4}}}{1 - \sqrt{1-\frac{x}{2}}} =$$

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+\frac{x}{3}} - \sqrt{1+\frac{x}{4}}}{1 - \sqrt{1-\frac{x}{2}}} \cdot \frac{1 + \sqrt{1-\frac{x}{2}}}{1 + \sqrt{1-\frac{x}{2}}} = \lim_{x \rightarrow 0} \frac{2(1 + \sqrt{1-\frac{x}{2}})}{1}$$

$$= \lim_{x \rightarrow 0} \frac{2(1 + \sqrt{1 - \frac{x}{2}})(\sqrt[6]{1 + \frac{x}{3}} - \sqrt[4]{1 + \frac{x}{4}})}{x} =$$

$$= \lim_{x \rightarrow 0} \frac{2(1 + \sqrt{1 - \frac{x}{2}})(\sqrt[6]{1 + \frac{x}{3}})^4 - \sqrt[2]{(1 + \frac{x}{4})^3}(\sqrt[2]{1 + \frac{x}{3}} + \sqrt[2]{1 + \frac{x}{4}})}{x(\sqrt[6]{1 + \frac{x}{3}})^4 + \sqrt[2]{1 + \frac{x}{4}}^3}$$

$$= \lim_{x \rightarrow 0} \frac{2(1 + \sqrt{1 - \frac{x}{2}})(\sqrt[6]{1 + \frac{x}{3}})^4 - \sqrt[2]{(1 + \frac{x}{4})^3}}{x(\sqrt[6]{1 + \frac{x}{3}})^4 + \sqrt[2]{1 + \frac{x}{4}}^3} =$$

$$= 2 \cdot \dots = \frac{17}{36}$$

№53 $\lim_{x \rightarrow 0} \frac{\sqrt[m]{1 + \alpha x} \cdot \sqrt[n]{1 + \beta x} - 1}{x} = A$

$$(1 + \alpha x)^{\frac{1}{m}} = 1 + \frac{\alpha x}{m} + \frac{1}{2} \left(\frac{1}{m} - 1 \right) \alpha^2 x^2 + \dots$$

$$(1 + \beta x)^{\frac{1}{n}} = 1 + \frac{\beta x}{n} + \frac{1}{2} \left(\frac{1}{n} - 1 \right) \beta^2 x^2 + \dots$$

$$A = \lim_{x \rightarrow 0} \frac{(1 + \frac{\alpha x}{m})(1 + \frac{\beta x}{n}) - 1}{x} =$$

$$= \lim_{x \rightarrow 0} \frac{x + \frac{\beta x}{n} + \frac{\alpha x}{m} + \frac{x^2 \alpha \beta}{m n} - 1}{x} = \frac{\alpha}{m} + \frac{\beta}{n}$$

№56 $\lim_{x \rightarrow 1} \frac{(1 - \sqrt{x})(1 - \sqrt[3]{x}) \dots (1 - \sqrt[n]{x})}{(1 - x)^{n-1}} =$

$$\lim_{x \rightarrow 1} \frac{(-1)^{n-1} (\sqrt{x}-1) \cdots (\sqrt[n]{x}-1)}{(-1)^{n-1} (x-1)^{n-1}} = \lim_{x \rightarrow 1} \frac{\sqrt{x}-1}{x-1} \cdots \frac{\sqrt[n]{x}-1}{x-1}$$

$$= \left| \begin{matrix} x-1=t \\ t \rightarrow 0 \end{matrix} \right| = \lim_{t \rightarrow 0} \frac{\sqrt{t+1}-1}{t} \cdots \frac{\sqrt[n]{t+1}-1}{t} = \frac{1}{2} \cdot \frac{1}{3} \cdots$$

$$= \frac{1}{n!}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt[n]{a+P(x)}-1}{x} = \frac{a_n}{n}$$

$$P(x) = a_1 x + a_2 x^2 + \cdots$$

N454

$$\lim_{x \rightarrow +\infty} \sqrt{(x+a)(x+b)} - x =$$

$$= \lim_{x \rightarrow +\infty} \frac{(x+a)(x+b) - x^2}{\sqrt{(x+a)(x+b)} + x} = \lim_{x \rightarrow +\infty} \frac{x(a+b) + ab}{x \left(\sqrt{\left(1+\frac{a}{x}\right)\left(1+\frac{b}{x}\right)} + 1 \right)}$$

$$= \lim_{x \rightarrow +\infty} \frac{a+b + \frac{ab}{x}}{\sqrt{\left(1+\frac{a}{x}\right)\left(1+\frac{b}{x}\right)} + 1} = \frac{a+b}{2}$$

N460

$$\lim_{x \rightarrow +0} \left(\sqrt{\frac{1}{x} + \sqrt{\frac{1}{x} + \sqrt{\frac{1}{x}}}} - \sqrt{\frac{1}{x} - \sqrt{\frac{1}{x} + \sqrt{\frac{1}{x}}}} \right) =$$

$$= \left| \begin{matrix} \frac{1}{x} = t \\ y \rightarrow +\infty \end{matrix} \right| = \lim_{y \rightarrow +\infty} \left(\sqrt{y + \sqrt{y + \sqrt{y}}} - \sqrt{y - \sqrt{y + \sqrt{y}}} \right) =$$

$$= \lim_{y \rightarrow +\infty} \frac{2\sqrt{y + \sqrt{y}}}{\sqrt{y + \sqrt{y + \sqrt{y}}} + \sqrt{y - \sqrt{y + \sqrt{y}}}} = \lim_{y \rightarrow +\infty} \frac{\sqrt{y} \cdot 2\sqrt{1 + \frac{1}{\sqrt{y}}}}{\sqrt{y} \left(\sqrt{1 + \frac{1}{\sqrt{y}} \sqrt{1 + \frac{1}{\sqrt{y}}}} + \sqrt{1 - \frac{1}{\sqrt{y}} \sqrt{1 + \frac{1}{\sqrt{y}}}} \right)}$$

$$\text{N461} \quad \lim_{x \rightarrow \infty} (\sqrt[3]{x^3 + x^2 + 1} - \sqrt[3]{x^3 - x^2 + 1}) =$$

$$= \lim_{x \rightarrow \infty} \frac{x^3 + x^2 + 1 - x^3 + x^2 - 1}{\sqrt[3]{(x^3 + x^2 + 1)^2} + \sqrt[3]{(x^3 - x^2 + 1)^2} + \sqrt[3]{(x^3 + x^2 + 1)(x^3 - x^2 + 1)}} = \lim_{x \rightarrow \infty} \frac{2x^2}{1} =$$

$$= \lim_{x \rightarrow \infty} \frac{2x^2}{x^2(1 + \frac{1}{x} + \frac{1}{x^3}) + x^2(\frac{1}{x} + \frac{1}{x^3} + 1) + x^2(1 - \frac{1}{x} + \frac{1}{x^3})} =$$

$$= \lim_{x \rightarrow \infty} \frac{2}{1+1+1} = \frac{2}{3}$$

$$\text{N462} \quad \lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 9x^2} - \sqrt[3]{x^3 - 2x}) = \lim_{x \rightarrow \infty} x \left(1 + \frac{9}{x} \right)^{\frac{1}{3}} - \left(1 - \frac{2}{x} \right)^{\frac{1}{3}} =$$

$$= \lim_{x \rightarrow \infty} \frac{x \left(1 + \frac{9}{x} \right)^{\frac{1}{3}} - \left(1 - \frac{2}{x} \right)^{\frac{1}{3}}}{1 + \frac{9}{x} + \dots + \left(1 - \frac{2}{x} \right)^{\frac{1}{3}}} = \lim_{x \rightarrow \infty} \frac{12 - \frac{3}{x} - \frac{8}{x^2}}{\left(1 + \frac{9}{x} \right)^{\frac{2}{3}} + \dots + \left(1 - \frac{2}{x} \right)^{\frac{2}{3}}} =$$

$$= \frac{12}{6} = 2$$

$$\text{N463} \quad \lim_{x \rightarrow \infty} x^{\frac{1}{3}} \left[(x+1)^{\frac{2}{3}} - (x-1)^{\frac{2}{3}} \right] =$$

$$= \lim_{x \rightarrow \infty} x^{\frac{1}{3}} \left[x^{\frac{2}{3}} \left[\left(1 + \frac{1}{x} \right)^{\frac{2}{3}} - \left(1 - \frac{1}{x} \right)^{\frac{2}{3}} \right] \right] =$$

$$= \lim_{x \rightarrow \infty} \frac{x \left[\left(1 + \frac{1}{x} \right)^2 - \left(1 - \frac{1}{x} \right)^2 \right]}{\left(1 + \frac{1}{x} \right)^{\frac{4}{3}} + \left(1 + \frac{1}{x} \right)^{\frac{2}{3}} \left(1 - \frac{1}{x} \right)^{\frac{2}{3}} + \left(1 - \frac{1}{x} \right)^{\frac{4}{3}}} =$$

$$= \lim_{x \rightarrow \infty} \frac{4}{1+1+1} = \frac{4}{3}$$

$$\sqrt{464} \quad \lim_{x \rightarrow \infty} x^{\frac{3}{2}} (\sqrt{x+2} - 2\sqrt{x+1} + \sqrt{x}) =$$

$$= \lim_{x \rightarrow +\infty} x^{\frac{3}{2}} (x^{\frac{1}{2}} (\sqrt{1+\frac{2}{x}} - 2\sqrt{1+\frac{1}{x}} + 1)) =$$

$$= \lim_{x \rightarrow \infty} x^2 (\sqrt{1+\frac{2}{x}} - 2\sqrt{1+\frac{1}{x}} + 1) = \lim_{x \rightarrow \infty} x^2 ((\sqrt{1+\frac{2}{x}} + 1) - 2\sqrt{1+\frac{1}{x}}) =$$

$$= \lim_{x \rightarrow +\infty} \frac{x^2 (\sqrt{1+\frac{2}{x}} + 1)^2 - 4(1+\frac{1}{x})}{(\sqrt{1+\frac{2}{x}} + 1) + 2\sqrt{1+\frac{1}{x}}} =$$

$$= \lim_{x \rightarrow \infty} \frac{x^2 (1 + \frac{2}{x} + 2\sqrt{1+\frac{2}{x}} + 1 - 4 - \frac{4}{x})}{4} =$$

$$= \dots \quad \frac{-2}{4} = -2$$

$$\sqrt{466} \quad \lim_{x \rightarrow +\infty} \frac{(x - \sqrt{x^2 - 1})^n + (x + \sqrt{x^2 - 1})^n}{x^n} =$$

$$= \lim_{x \rightarrow +\infty} \frac{x^n (1 - \sqrt{1 - \frac{1}{x^2}})^n + x^n (1 + \sqrt{1 + \frac{1}{x^2}})^n}{x^n} = 2$$

$$\sqrt{467} \quad \lim_{x \rightarrow \infty} \frac{(\sqrt{1+x^2} + x)^n + (\sqrt{1+x^2} - x)^n}{x^n} =$$

Ans 85 $\lim_{x \rightarrow +\infty} [\sqrt[n]{(x+q_1) \dots (x+q_n)} - x] =$

$$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + ab^{n-2} + b^{n-1})$$

$$= \lim_{x \rightarrow +\infty} \left(\sqrt[n]{(x+q_1) \dots (x+q_n)} - x \right) = \lim_{x \rightarrow +\infty} \left(\sqrt[n]{(1 + \frac{q_1}{x}) \dots (1 + \frac{q_n}{x})} - 1 \right)$$

$$= \lim_{x \rightarrow +\infty} \frac{x^n \left(1 + \frac{q_1}{x} + \frac{q_2}{x} + \dots + \frac{q_n}{x} - 1 \right)}{x^{n-1} \left(1 + \frac{q_1}{x} + \dots + 1 \right)} = \frac{1}{n} \lim_{x \rightarrow +\infty} x \left(1 + \frac{q_1}{x} \right)$$

$$\left(1 + \frac{q_1}{x} + \dots + \frac{q_n}{x} - 1 \right) = \frac{q_1 + q_2 + \dots + q_n}{n}$$

Ans 83 $\lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx} = \left| \frac{t = \pi - x}{t \rightarrow 0} \right| = \lim_{t \rightarrow 0} \frac{\sin m(\pi - t)}{\sin n(\pi - t)} = (-1)^{m+n} \frac{\sin mt}{\sin nt}$

$$= \frac{(-1)^{m+n} \sin mt}{(-1)^{n+m} \sin nt} = (-1)^{m-n} \frac{m}{n}$$

$$1 - \cos x = \frac{x^2}{2}$$

Ans 85 $\lim_{x \rightarrow 0} \frac{(9x - \sin x)}{x^3} = \lim_{x \rightarrow 0} \frac{9 - \frac{1}{\cos x}}{3x^2} =$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} - 1}{x^2} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2 \cos x} = \lim_{x \rightarrow 0} \frac{x^2}{2x^2 \cos x} = \frac{1}{2}$$

$$\text{Aufg 5} \quad \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 + \sin px - \cos px} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \sin^2 \frac{px}{2} + 2 \sin \frac{px}{2} \cos \frac{px}{2}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2} (\sin \frac{x}{2} + \cos \frac{x}{2})}{\sin \frac{px}{2} (\sin \frac{px}{2} + \cos \frac{px}{2})} \quad \lim_{x \rightarrow 0} \frac{\frac{x}{2}}{\frac{px}{2}} = \frac{1}{p}$$

$$\text{Aufg 9} \quad \lim_{x \rightarrow \frac{\pi}{4}} \lg 2x \lg \left(\frac{\pi}{4} - x \right) = \left| \begin{matrix} t = \frac{\pi}{4} - x \\ t \rightarrow 0 \end{matrix} \right| = \lim_{t \rightarrow 0} \lg t \cdot \lg \left(\frac{\pi}{2} - 2t \right)$$

$$= \frac{t}{2t} = \frac{1}{2} = \frac{\lg t}{\lg 2t}$$

$$\text{Aufg 8} \quad \lim_{x \rightarrow 0} \frac{\sin(a+2x) - 2 \sin(a+x) + \sin a}{x^2} =$$

$$= \lim_{x \rightarrow 0} \frac{(\sin(a+2x) - \sin(a+x)) + (\sin a - \sin(a+x))}{x^2} =$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2} \cos \left(a + \frac{3x}{2} \right) + 2 \sin \frac{x}{2} \cos \left(a + \frac{x}{2} \right)}{x^2} =$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2} \left(\cos \left(a + \frac{3x}{2} \right) - \cos \left(a + \frac{x}{2} \right) \right)}{x^2} =$$

$$= \lim_{x \rightarrow 0} \frac{-2 \sin \left(a + \frac{x}{2} \right) \sin \frac{x}{2}}{x} \stackrel{x \approx \frac{x}{2}}{=} -\sin a$$